

Ahmedabad Institute of Technology

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DEPARTMENT OF MASTER OF COMPUTER APPLICATION

Research work Project Phase-II Report On

“E-NGO WEB APPLICATION”

Submitted as Partial Fulfillment of Master of Computer Application Sem-IV

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“E-NGO WEB APPLICATION”

As a partial fulfilment of the study of

Master of Computer Application

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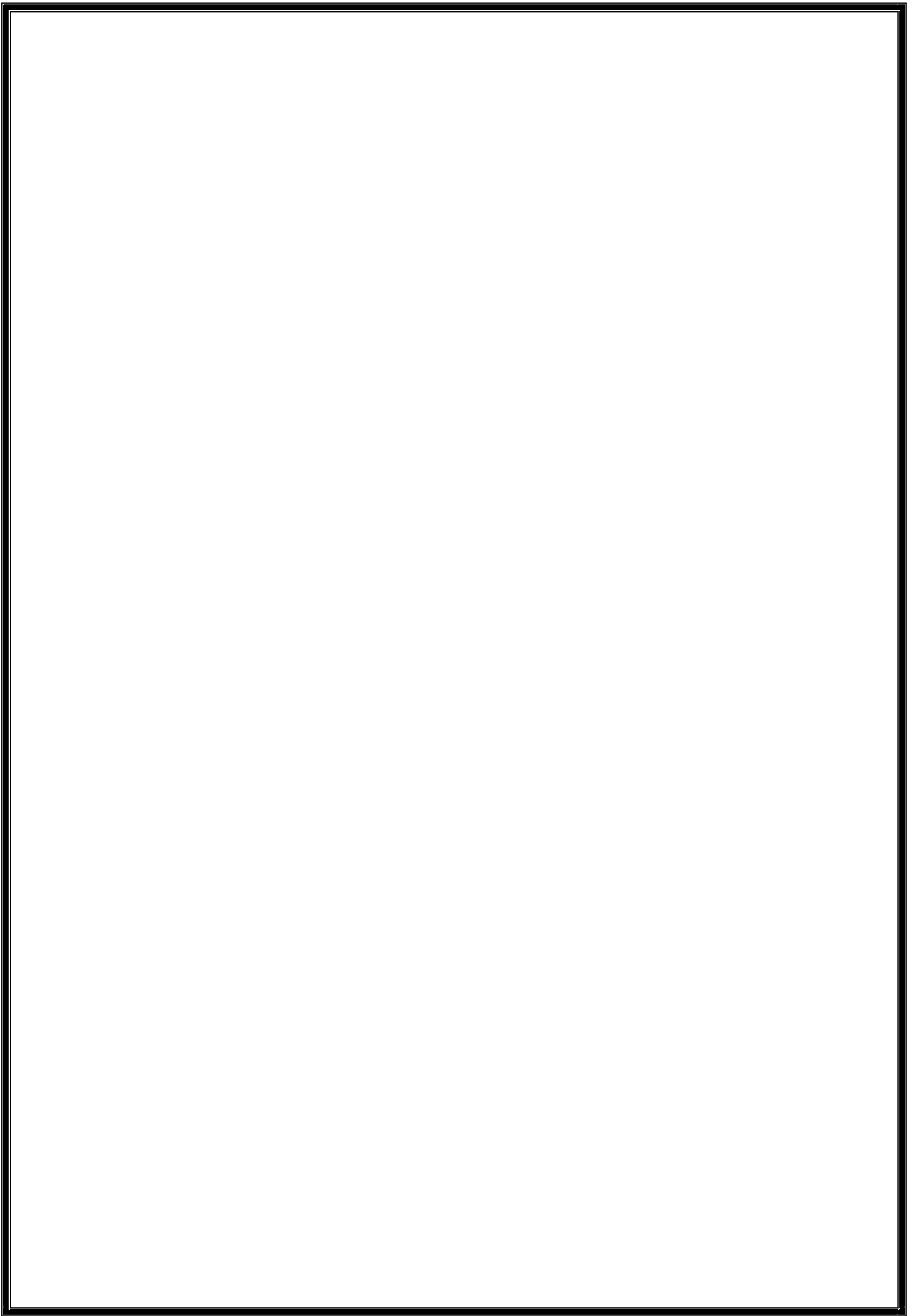
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ABSTRACT

Non-Governmental Organizations (NGOs) require modern digital solutions to improve transparency, scalability, and engagement. This project presents an advanced **E-NGO web application**, designed as a **multi-tenant and modular platform** that connects donors, volunteers, NGOs, and corporate partners within a unified system.

The platform supports **multiple contribution models**, including one-time, recurring, pledge-based, and wallet-based donations, along with volunteer management and goods/services contributions. A **unified user module with role-based access control (RBAC)** enables a single user to perform multiple roles seamlessly.

To enhance efficiency and user engagement, the system integrates **Artificial Intelligence (AI)** for campaign recommendations, donor behavior analysis, and volunteer matching. Additionally, **block chain technology** is incorporated to provide **secure, transparent, and immutable donation records** with fund traceability.

The system is built using **Django, REST APIs, Web Sockets, and Celery with Redis**, enabling real-time updates, asynchronous processing, and scalable architecture. Corporate partners are supported through **CSR dashboards** for compliance tracking and impact analysis.

Overall, the proposed system transforms traditional NGO platforms into a **smart, transparent, and future-ready ecosystem**, ensuring trust, accountability, and improved collaboration among all stakeholders.

Project Profile

Field	Details
Project Title	E-NGO (Electronic Non-Governmental Organization) Web Application
Project Category	Web Application
Objectives	- To design a multi-tenant, modular E-NGO platform. - Supports donations, pledges, goods, and services. - Enables volunteer management and NGO collaboration. - Ensures transparency, accountability, and engagement. - Provides real-time updates and AI-driven insights. - Integrates CSR functionalities. - Scalable and future-ready system using Django.
Front-End	Django Templates, HTML5, CSS3, Bootstrap
Back-End	Python Django
Tools	REST Framework, Celery, Redis, Stripe
Server	Django Built-in Development Server
Browser	Google Chrome
Documentation Tools	MS Office
Developed By	Sangani Mitali Jayeshbhai (245850694028) Katoch Rishita Ravindrabhai (245850694021)

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1. Introduction

The **ENGO Web Application (EcoNexus NGO)** is designed to provide a modern digital platform for Non-Governmental Organizations (NGOs) to efficiently manage donations, volunteers, and social impact activities. NGOs play a vital role in addressing social and environmental challenges, but traditional systems often lack transparency, scalability, and user engagement.

Many existing NGO platforms are limited to basic donation management and do not support advanced features such as real-time tracking, personalized recommendations, or multi-organization collaboration. This creates challenges for donors, volunteers, and organizations in terms of trust, participation, and operational efficiency.

To overcome these limitations, the EcoNexus NGO system introduces a **multi-tenant and modular architecture**, allowing multiple NGOs to operate within a single platform while maintaining data isolation. The system provides a **unified user account with role-based access control (RBAC)**, enabling users to act as donors, volunteers, or corporate partners seamlessly.

The platform supports various functionalities, including **monetary donations (one-time, recurring, pledge-based, and wallet-based), volunteer management, and goods/services contributions**. It also integrates **Artificial Intelligence (AI)** for campaign recommendations, donor behavior analysis, and volunteer matching to enhance user engagement.

Additionally, the system incorporates **Blockchain technology** to ensure transparency and trust by maintaining secure and immutable records of donations and fund usage. Built using **Django, REST APIs, WebSockets, and Celery with Redis**, the application provides real-time updates and scalable performance.

Overall, the ENGO Web Application (EcoNexus NGO) aims to create a **smart, transparent, and future-ready NGO ecosystem**, improving collaboration, accountability, and impact across all stakeholders.

1.1 Background of the Study

In today's digital era, technology plays a vital role in improving the efficiency and transparency of various sectors, including Non-Governmental Organizations (NGOs). NGOs work towards social welfare by managing donations, volunteers, and community services. However, many NGOs still rely on manual or semi-digital systems, which lead to problems such as lack of transparency, inefficient management, and limited reach.

Traditional NGO systems often face challenges in handling donations, tracking volunteer activities, and managing campaigns effectively. Donors may not have clear visibility of how their contributions are used, which reduces trust. Similarly, volunteers may find it difficult to connect with suitable opportunities, and NGOs struggle to manage resources efficiently.

With the growth of web technologies, there is a need for a centralized digital platform that can streamline NGO operations. An online system can provide better accessibility, real-time updates, and improved

communication between donors, volunteers, and organizations. It can also enhance transparency and accountability through proper tracking and reporting mechanisms.

The ENGO Web Application (EcoNexus NGO) is developed to address these challenges by providing an integrated platform for donation management, volunteer coordination, and campaign monitoring. It leverages modern technologies such as web development frameworks, real-time systems, and advanced features like AI and Blockchain (optional) to create a smart and efficient NGO ecosystem.

Thus, this study focuses on designing and developing a digital solution that improves the overall functioning, transparency, and impact of NGO activities.

1.2 Problem Statement

Non-Governmental Organizations (NGOs) are essential for social welfare, working in areas such as education, healthcare, environment, and poverty reduction. Despite their importance, many NGOs still rely on traditional or semi-digital systems to manage their operations. These systems are often inefficient, unorganized, and unable to meet the growing demands of modern digital environments.

One of the primary challenges faced by NGOs is the absence of a centralized and integrated system. Different activities such as donation management, volunteer coordination, and campaign tracking are handled separately, often using manual records or basic tools like spreadsheets. This leads to data inconsistency, duplication of records, and increased chances of human error. Managing large volumes of data becomes difficult, resulting in poor organization and reduced operational efficiency.

Transparency and trust are major concerns in NGO operations. Donors contribute funds with the expectation that their money will be used effectively. However, in many cases, there is no proper system to track how donations are utilized. The lack of real-time updates and clear reporting reduces donor confidence and may discourage future contributions. Without transparency, it becomes difficult for NGOs to build long-term relationships with donors.

Volunteer management is another significant issue. Volunteers play a crucial role in the success of NGO activities, but there is often no structured system to manage them efficiently. Volunteers face difficulties in finding suitable opportunities that match their skills, interests, and availability. At the same time, NGOs struggle to assign the right tasks to the right people, leading to underutilization of human resources and reduced effectiveness of campaigns.

Campaign management also presents several challenges. NGOs frequently run multiple campaigns simultaneously, but without proper tools, it becomes difficult to monitor progress, track goals, and evaluate outcomes. The absence of dashboards and reporting systems limits the ability to analyze performance and make data-driven decisions. This affects the overall success and impact of campaigns.

Security and data protection are critical issues in handling sensitive information such as user data and financial transactions. Traditional systems often lack proper security measures, making them vulnerable to data breaches, unauthorized access, and misuse of information. This can lead to serious consequences, including loss of trust and legal complications.

Additionally, the lack of real-time communication and automation reduces system responsiveness. Users do not receive instant updates about donations, campaign progress, or notifications. This creates a gap in user engagement and reduces the overall efficiency of the platform.

Therefore, there is a strong need for a modern, digital solution that can address these challenges effectively. The ENGO Web Application (EcoNexus NGO) is developed to provide a centralized, secure, and user-friendly platform that integrates all NGO activities. It aims to enhance transparency, improve efficiency, and provide better coordination between donors, volunteers, and organizations.

In conclusion, this project focuses on solving key issues such as lack of integration, poor transparency, inefficient resource management, weak security, and limited user engagement in traditional NGO systems, thereby providing a comprehensive and scalable solution.

1.3 Objectives of the Study

The primary objective of the ENGO Web Application (EcoNexus NGO) is to design and develop a modern, efficient, and transparent digital platform that improves NGO operations. The objectives are categorized into functional and technical aspects for better clarity.

A. Functional Objectives

1. Centralized System Development

- To develop a unified platform for managing all NGO activities.
- To integrate donations, volunteers, campaigns, and CSR in one system.
- To eliminate manual and fragmented processes.

2. Efficient Donation Management

- To provide a simple and secure donation process.
- To support multiple donation types such as one-time, recurring, and pledge-based donations.
- To maintain complete donation history and generate receipts.
- To ensure transparency in fund usage.

3. Volunteer Management System

- To enable volunteer registration and profile creation.
- To store skills, interests, and availability of volunteers.
- To allow volunteers to apply for campaigns easily.
- To track volunteer participation and working hours.

4. Campaign Management

- To allow NGOs to create, update, and manage campaigns.
- To track campaign goals, progress, and outcomes.
- To support multiple campaigns simultaneously.
- To improve planning and execution of social activities.

5. CSR (Corporate Social Responsibility) Support

- To provide features for corporate funding of campaigns.
- To track CSR activities and contributions.
- To generate compliance and performance reports.
- To support employee volunteering programs.

6. User Experience and Accessibility

- To design a user-friendly and responsive interface.
- To provide easy navigation for all types of users.
- To ensure accessibility across different devices.
- To improve overall user satisfaction.

B. Technical Objectives

7. Secure System Design

- To implement secure authentication and authorization mechanisms.
- To protect sensitive user data and financial transactions.
- To ensure data privacy and integrity.

8. Real-Time System Implementation

- To provide live updates for donations and campaign progress.
- To implement instant notifications using real-time technologies.
- To enhance system responsiveness and user engagement.

9. Scalable and Modular Architecture

- To design a modular system for easy maintenance and upgrades.
- To support multi-tenant architecture for multiple NGOs.
- To ensure scalability for future expansion.

10. Efficient Data Management

- To use a relational database (SQLite3) for structured data storage.
- To maintain data consistency using proper relationships.
- To enable fast data retrieval and reporting.

C. Advanced Technology Objectives

11. Artificial Intelligence (AI) Integration

- To develop a campaign recommendation system for donors.
- To analyze donor behavior and preferences.
- To implement volunteer skill-matching algorithms.
- To provide predictive analytics for campaign success.
- To generate AI-based impact reports.

12. Blockchain Integration

- To ensure secure and immutable donation records.
- To provide transparency through traceable transactions.
- To implement hash-based verification for donations.
- To support smart contracts for conditional fund release (future scope).
- To build trust among users through transparent systems.

D. Overall Objective

- To create a smart, transparent, secure, and scalable NGO management system.
- To improve efficiency, accountability, and collaboration among stakeholders.
- To provide a future-ready platform using modern technologies.

1.4 Scope of the Study

The scope of this project includes the development of a web-based platform that manages donations, volunteer activities, goods and services contributions, and NGO collaboration. It supports multiple stakeholders such as donors, volunteers, NGOs, and corporate partners within a single system. The platform focuses on real-time tracking, transparency, and scalability using modern technologies. Future enhancements may include mobile applications, advanced AI analytics, and decentralized governance systems

1. Functional Scope

- The system provides a platform for **donation management**, including one-time, recurring, and pledge-based donations.
- It supports **volunteer management**, allowing users to register, apply for campaigns, and track participation.
- NGOs can **create and manage campaigns**, monitor progress, and handle multiple campaigns simultaneously.
- The system includes **CSR (Corporate Social Responsibility) features** for corporate funding and reporting.
- Provides **dashboard and reporting tools** for better monitoring and decision-making.

2. User Scope

- The system is designed for multiple users:
 - **Donors** – to contribute funds and track donations.
 - **Volunteers** – to participate in social activities.
 - **NGO Admins** – to manage campaigns and operations.
 - **Corporate Users** – for CSR funding and compliance.
- A single user can perform multiple roles using the system.

3. Technical Scope

- The system is developed using **Django framework** with a modular architecture.
- Uses **SQLite3 database** for structured data management.
- Supports **real-time features** using WebSockets (Django Channels).
- Implements **asynchronous processing** using Celery and Redis.

- Designed to be **scalable and maintainable** for future expansion.

4. Advanced Technology Scope

- Includes **Artificial Intelligence (AI)** features such as:
 - Campaign recommendation system
 - Donor behavior analysis
 - Volunteer matching
- Includes **Blockchain (optional)** features such as:
 - Secure and immutable donation records
 - Transparent transaction tracking

5. Operational Scope

- The system can be used by NGOs to manage daily operations digitally.
- Supports online interaction between donors, volunteers, and organizations.
- Improves communication and coordination among stakeholders.
- Suitable for small to medium-scale NGO operations.

6. Limitations of the Study

- The system uses **SQLite3**, which may not be suitable for very large-scale deployment.
- Full **blockchain implementation is optional** and not fully deployed.
- Limited to web-based platform (mobile app is future scope).
- Requires internet connectivity for real-time features.

7. Future Expansion Scope

- Can be extended to **mobile applications (Android/iOS)**.
- Can support **global users and multi-currency transactions**.
- Can integrate with **government and external systems**.
- Can include more advanced AI and analytics features.

1.5 Significance of the Study

The ENGO Web Application (EcoNexus NGO) plays an important role in improving the efficiency, transparency, and effectiveness of NGO operations. The significance of this study is explained as follows:

1. Improvement in NGO Operations

- Provides a centralized platform to manage donations, volunteers, and campaigns.
- Reduces manual work and paperwork.
- Improves overall efficiency and organization.

2. Enhanced Transparency and Trust

- Enables proper tracking of donations and fund usage.
- Provides clear reports and real-time updates.
- Builds trust between donors and NGOs.

3. Better Donor Engagement

- Simplifies the donation process.
- Allows donors to track their contributions.
- Encourages repeated donations through transparency and ease of use.

4. Efficient Volunteer Management

- Helps volunteers find suitable opportunities easily.
- Enables NGOs to match volunteers based on skills.
- Improves utilization of human resources.

5. Improved Decision Making

- Provides dashboards and reports for analysis.
- Helps NGOs monitor campaign performance.
- Supports data-driven decision-making.

6. Integration of Modern Technologies

- Introduces Artificial Intelligence (AI) for recommendations and predictions.
- Uses Blockchain concepts for transparency and security.
- Makes the system modern and future-ready.

7. Increased Accessibility and Reach

- Web-based platform accessible from anywhere.
- Connects donors, volunteers, and NGOs globally.
- Expands the reach of social initiatives.

8. Support for Corporate Social Responsibility (CSR)

- Enables corporate organizations to fund social campaigns.
- Provides CSR tracking and reporting features.
- Ensures compliance and transparency.

9. Contribution to Digital Transformation

- Promotes digitization in the NGO sector.
- Replaces traditional manual systems with automated solutions.
- Encourages adoption of technology in social services.

10. Academic and Practical Importance

- Provides practical knowledge of web development and system design.
- Demonstrates integration of AI, Blockchain, and real-time systems.
- Useful for future research and development in similar domains.

2. Literature Review

NGO web applications have evolved from simple static websites to dynamic platforms supporting donations, volunteer management, and campaign coordination. Existing systems provide basic functionalities such as fundraising, user registration, and reporting, but they often lack scalability, transparency, and user engagement. Most platforms operate independently for each NGO, resulting in fragmented experiences and limited cross-organization collaboration.

Modern web technologies have introduced multi-tenant architectures, enabling multiple NGOs to operate on a single platform while maintaining data isolation. Tools like Django, REST APIs, and cloud computing improve scalability and performance, while asynchronous processing ensures efficient handling of tasks. However, many NGO systems still rely on traditional designs and do not fully utilize these advancements.

Artificial Intelligence (AI) is increasingly used for donor analysis, campaign recommendations, and volunteer matching, enhancing decision-making and engagement. Blockchain technology improves transparency by providing secure and immutable donation records, ensuring trust among donors.

Despite these advancements, existing systems do not combine multi-tenancy, AI, Blockchain, and real-time features in a unified platform. The ENGO Web Application (EcoNexus NGO) addresses this gap by providing a scalable, intelligent, and transparent solution for modern NGO operations.

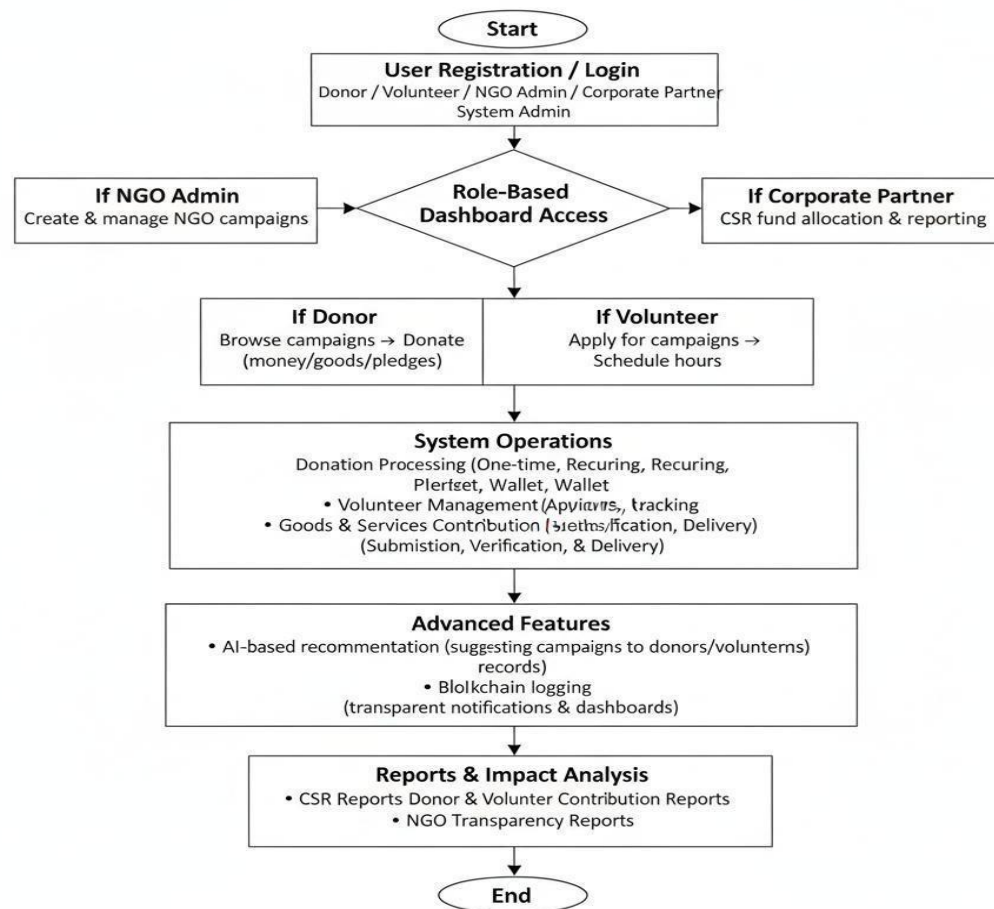
2.1 Overview of NGO Web Applications

Non-Governmental Organizations (NGOs) play a significant role in addressing global challenges such as poverty, education, healthcare, and environmental sustainability. With the increasing adoption of digital technologies, NGOs have shifted from traditional manual processes to web-based platforms for managing their operations. Early NGO web applications were primarily static websites that provided basic information about the organization, its mission, and ongoing activities. These systems lacked interactivity and did not support functionalities such as online donations or volunteer management.

Over time, NGO platforms evolved into dynamic web applications that enabled online donations, event management, and volunteer registration. These systems improved accessibility and allowed NGOs to reach a wider audience. However, most existing applications still focus on limited functionalities and do not provide a comprehensive solution that integrates all aspects of NGO operations. Additionally, issues such as lack of transparency, limited scalability, and poor user engagement remain significant challenges.

Modern NGO platforms require advanced capabilities such as real-time updates, personalized user experiences, and integration with emerging technologies. The need for a unified, intelligent, and scalable system has led to the development of next-generation solutions like the ENGO Web Application (EcoNexus NGO), which aims to overcome the limitations of traditional systems.

Flowchart – E-NGO Web Application



2.2 Existing Donation and Volunteer Management Systems

Several platforms exist for managing donations and volunteer activities, including online fundraising websites and volunteer matching systems. These platforms provide essential features such as donation processing, campaign creation, and volunteer registration. Some systems also offer basic reporting tools that allow NGOs to track contributions and generate summaries.

Despite these functionalities, existing systems have several limitations. Most platforms operate independently for each NGO, resulting in a fragmented user experience. Donors and volunteers are required to create separate accounts for different organizations, which reduces engagement and convenience. Additionally, these systems typically lack advanced features such as predictive analytics, intelligent recommendations, and automated decision-making.

Another major limitation is the lack of transparency in fund utilization. Donors often do not have clear visibility into how their contributions are being used, which can lead to trust issues. Similarly, volunteer management systems do not always provide efficient matching between volunteers and opportunities, resulting in underutilization of resources.

Furthermore, many existing platforms rely on synchronous processing, which can affect performance and scalability during high traffic periods. They also lack integration with modern technologies such as Artificial Intelligence (AI) and Blockchain, which are essential for improving efficiency, transparency, and user engagement in today's digital ecosystem.

2.3 Multi-Tenant and Scalable Web Platforms

Multi-tenant architecture has become a key design approach in modern web applications, especially for platforms that need to serve multiple organizations within a single system. In a multi-tenant system, multiple users or organizations share the same application infrastructure while maintaining separate and secure data environments. This approach reduces operational costs, improves resource utilization, and simplifies maintenance.

Frameworks such as Django support multi-tenancy through schema-based or shared-database approaches, enabling developers to build scalable and flexible applications. In addition, the use of REST APIs allows seamless integration with external systems and mobile applications. Asynchronous processing tools like Celery and Redis play a crucial role in handling background tasks such as sending notifications, generating reports, and processing large datasets.

Scalability is another critical factor in modern web applications. With the increasing number of users and transactions, systems must be able to handle high loads efficiently. Cloud computing platforms provide infrastructure that supports horizontal scaling, ensuring that the system can accommodate growth without performance degradation.

However, many existing NGO platforms still use single-tenant architectures, which limit their ability to scale and support multiple organizations. These systems also lack efficient resource management and fail to provide real-time updates, which are essential for enhancing user experience. The ENGO Web Application addresses these challenges by implementing a multi-tenant architecture combined with real-time and asynchronous processing capabilities.

2.4 AI and Blockchain in NGO Transparency

Artificial Intelligence (AI) and Blockchain are emerging technologies that have the potential to transform NGO operations significantly. AI enables systems to analyze large volumes of data, identify patterns, and make intelligent predictions. In the context of NGO platforms, AI can be used for donor segmentation, campaign recommendation, volunteer matching, and impact analysis.

Machine learning algorithms can analyze donor behavior, including donation frequency, amount, and preferences, to predict future contributions and recommend suitable campaigns. Similarly, AI can match volunteers with opportunities based on their skills, availability, and interests, improving overall efficiency and engagement. Natural Language Processing (NLP) techniques can be used to analyze feedback from donors and volunteers, providing valuable insights into user satisfaction and system performance.

Blockchain technology, on the other hand, provides a decentralized and tamper-proof mechanism for recording transactions. In NGO systems, Blockchain can be used to maintain immutable records of donations, ensuring transparency and accountability. Each transaction can be stored as a block with a unique hash, making it impossible to alter or manipulate the data. This builds trust among donors, as they can verify how their contributions are being utilized.

Smart contracts further enhance Blockchain capabilities by enabling automated and conditional execution of transactions. For example, funds can be released only when specific milestones are achieved, ensuring that donations are used effectively. Despite these advantages, the adoption of AI and Blockchain in NGO platforms is still limited due to technical complexity and implementation challenges.

The ENGO Web Application integrates both AI and Blockchain technologies to provide a more intelligent and transparent system, addressing the shortcomings of traditional NGO platforms.



2.5 Research Gap

Based on the analysis of existing literature and systems, it is evident that current NGO platforms lack a unified approach that combines advanced technologies and comprehensive functionalities. Most systems focus on either donation management or volunteer coordination, but fail to integrate both effectively within a single platform. Additionally, the absence of multi-tenancy limits scalability and increases operational costs.

Another significant gap is the lack of transparency and accountability in fund utilization. While some systems provide basic reporting features, they do not offer real-time tracking or immutable records that can ensure trust among stakeholders. The integration of Blockchain technology in NGO systems is still at an early stage and has not been widely implemented.

Furthermore, existing platforms do not fully leverage the potential of Artificial Intelligence for enhancing user engagement and decision-making. Features such as predictive analytics, personalized recommendations, and automated impact analysis are either missing or not adequately developed. Real-time communication and asynchronous processing are also lacking in many systems, leading to delays and reduced user satisfaction.

There is also limited support for cross-NGO collaboration, where multiple organizations can share resources, volunteers, and campaigns within a unified system. Corporate participation through CSR initiatives is another area that is not fully explored in existing platforms.

To address these gaps, the ENGO Web Application (EcoNexus NGO) proposes a comprehensive solution that integrates multi-tenancy, AI-driven analytics, Blockchain-based transparency, real-time updates, and modular architecture. This approach ensures scalability, efficiency, and trust, making the system a future-ready platform for NGO operations.

3. System Design & Methodology

The ENGO Web Application (EcoNexus NGO) is designed using a **multi-tier and modular architecture** to ensure scalability, flexibility, and performance. The system follows a structured methodology that integrates modern web technologies with AI and Blockchain components.

The architecture consists of three main layers: **Presentation Layer, Application Layer, and Data Layer**. The Presentation Layer is developed using Django Templates, HTML, CSS, and Bootstrap, providing a user-friendly interface for donors, volunteers, NGOs, and corporate users.

The Application Layer is built using the Django framework and Django REST APIs, handling business logic, authentication, and role-based access control (RBAC). The Data Layer uses SQLite3 with multi-tenancy support to ensure secure and isolated data for multiple NGOs.

To enhance performance, the system uses **Celery with Redis** for asynchronous task processing, such as sending notifications, generating reports, and handling background AI operations. **Django Channels (WebSockets)** are used for real-time updates, including live donation tracking and instant notifications.

The methodology follows a **modular approach**, dividing the system into key modules such as User Management, Donation Management, Volunteer Management, NGO Administration, CSR Module, and Reporting Dashboard. Each module is independently developed but interconnected through APIs for seamless communication.

Additionally, **Artificial Intelligence (AI)** is integrated for campaign recommendations, donor behavior analysis, and volunteer matching. **Blockchain technology** is optionally used to maintain immutable donation records and ensure transparency.

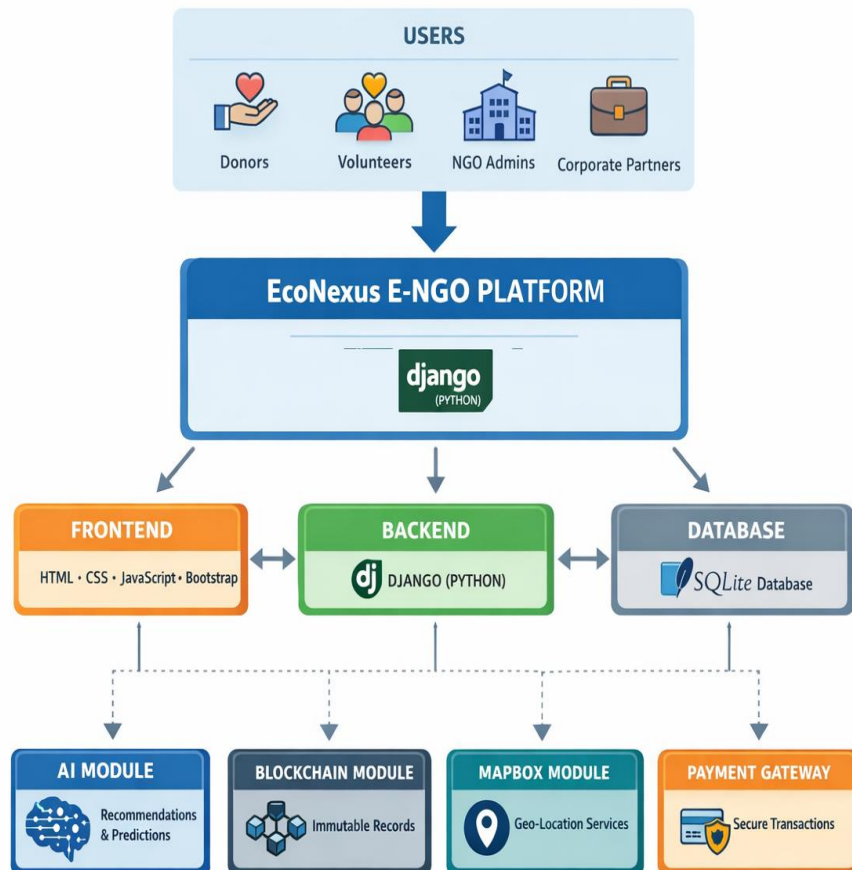
Overall, the system design focuses on creating a **scalable, secure, and intelligent platform** that supports efficient NGO operations and enhances user engagement.

3.1 System Architecture

The ENGO Web Application (EcoNexus NGO) follows a **multi-tier architecture** consisting of Presentation Layer, Application Layer, and Data Layer. The Presentation Layer provides a user-friendly interface using HTML, CSS, Bootstrap, and Django Templates.

The Application Layer is built with Django and REST APIs, handling business logic, authentication, and role-based access control (RBAC).

The Data Layer uses SQLite3 for structured data storage. Real-time updates are implemented using Django Channels (WebSockets), and background tasks are managed using Celery with Redis, ensuring scalability and performance.



3.2 Database Design

- The system uses **SQLite3**, a lightweight and serverless relational database.
- It follows a **relational database model** with tables, rows, and relationships.
- Data is organized using **primary keys and foreign keys** to maintain integrity.

Main Tables:

- **User Table:** Stores user details like ID, name, email, password, and role.
- **Role Table:** Defines roles such as Donor, Volunteer, NGO Admin, and Corporate.
- **Donor Table:** Contains donor-specific details and links to User.
- **Volunteer Table:** Stores skills, availability, and experience of volunteers.
- **NGO Table:** Maintains NGO details such as name, description, and contact info.
- **Campaign Table:** Stores campaign data like title, goal amount, and duration.
- **Donation Table:** Records all donations (amount, type, status, date).
- **Volunteer Application Table:** Tracks volunteer participation in campaigns.
- **Goods/Services Table:** Handles non-monetary contributions.
- **CSR Table:** Stores corporate funding and compliance details.

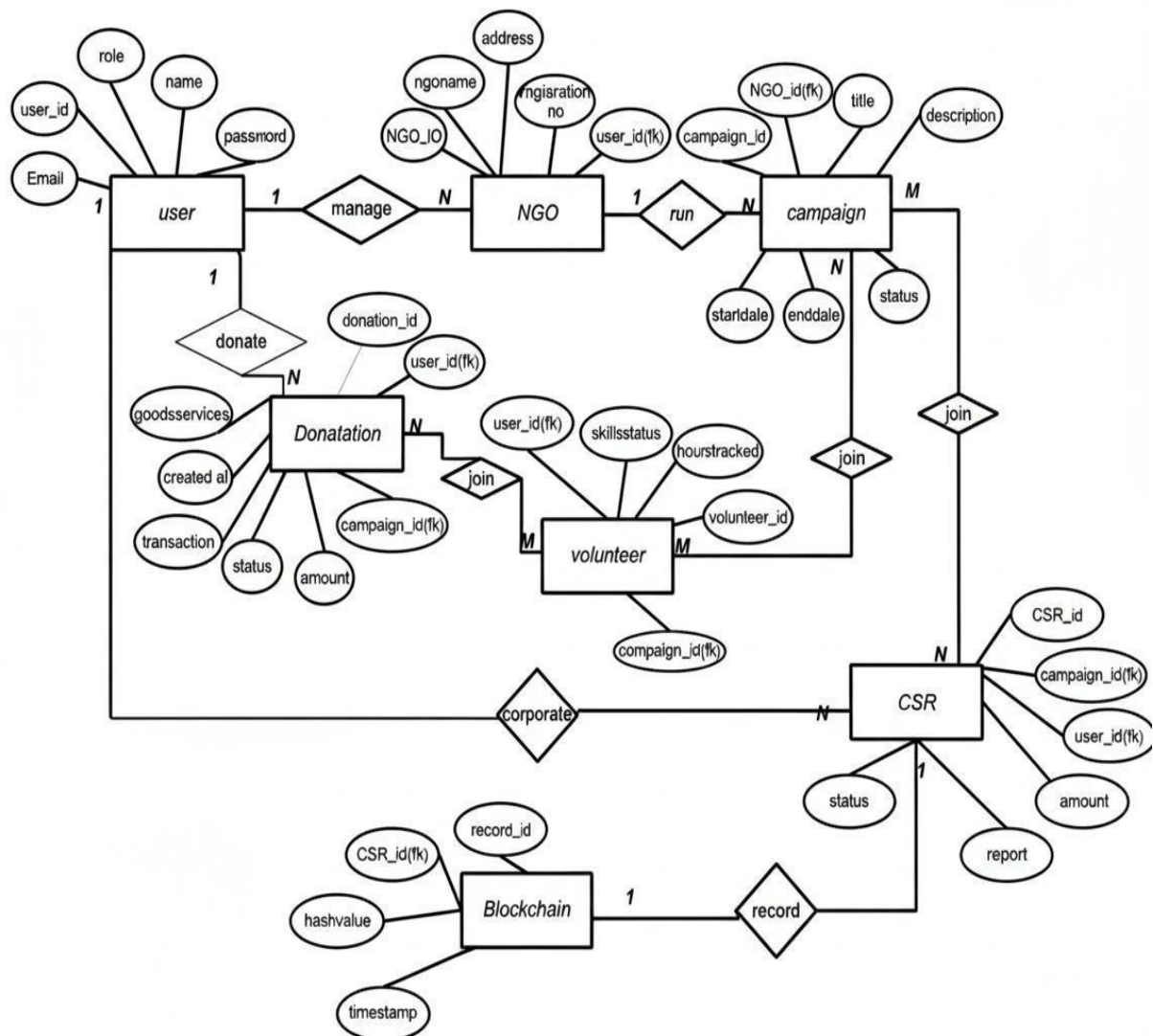
Relationships:

- One **User** can have multiple roles (Donor/Volunteer).
- One **NGO** can create multiple campaigns.
- One **Campaign** can receive multiple donations.
- One **User** can donate to multiple campaigns.
- Volunteers can apply to multiple campaigns.

Key Features:

- Supports multiple donation types (one-time, recurring, pledge).
- Ensures **data consistency and normalization**.
- Enables efficient data retrieval and reporting.

Overall, the database design is simple, efficient, and well-structured for NGO operations.



3.3 Module Description

The ENGO Web Application (EcoNexus NGO) is divided into several functional modules to ensure smooth operation and scalability.

- **User Module:**

- Handles user registration and login.
- Supports multiple roles (Donor, Volunteer, NGO Admin, and Corporate).
- Implements Role-Based Access Control (RBAC).
- Maintains user profiles and authentication.

- **Donor Module:**

- Enables one-time, recurring, and pledge-based donations.
- Maintains donation history and receipts.
- Supports wallet-based micro-donations.
- Provides campaign browsing and selection.

- **Volunteer Module:**

- Allows volunteer registration and profile creation.
- Stores skills, availability, and interests.
- Enables application to campaigns.
- Tracks volunteer hours and participation.

- **NGO Admin Module:**

- Allows NGOs to create and manage campaigns.
- Monitors donations and volunteer activities.
- Approves volunteer applications.
- Generates reports and dashboards.

- **CSR (Corporate) Module:**

- Supports corporate funding of campaigns.
- Tracks CSR activities and compliance.
- Enables employee volunteering programs.
- Provides CSR reporting dashboards.

- **Admin/System Module:**

- Manages overall system operations.
- Handles user roles and permissions.
- Maintains audit logs and system monitoring.
- Ensures security and data integrity.

- **AI Module (Advanced):**

- Recommends campaigns to donors.
- Matches volunteers based on skills.
- Analyzes donor behavior and trends.

- **Blockchain Module (Optional):**

- Maintains secure and immutable donation records.
- Provides transparency and traceability.

3.4 Tools & Technologies

The ENGO Web Application (EcoNexus NGO) is developed using a combination of modern technologies that ensure scalability, performance, and efficient system management. These technologies are categorized as follows:

1. Frontend Technologies:

- **HTML (HyperText Markup Language):**

- Used to structure the content of web pages.
- Defines elements such as forms, buttons, tables, and navigation menus.

- **CSS (Cascading Style Sheets):**

- Used for styling and layout design.
- Enhances visual appearance with colors, fonts, spacing, and responsiveness.

- **Bootstrap:**

- A frontend framework used for responsive and mobile-friendly design.
- Provides pre-built components such as buttons, forms, cards, and navigation bars.

- **JavaScript:**

- Adds interactivity and dynamic behavior to the application.
- Used for form validation, real-time updates, and user interaction handling.

2. Backend Technologies:

- **Python:**

- Primary programming language used for backend development.
- Known for simplicity, readability, and strong library support.

- **Django Framework:**

- A high-level Python web framework that follows the MVT (Model-View-Template) architecture.
- Handles business logic, authentication, URL routing, and security features.
- Provides built-in admin panel and ORM for database operations.

- **Django REST Framework (DRF):**

- Used to build RESTful APIs.
- Enables communication between frontend and backend.
- Supports JSON data exchange and integration with external systems.

3. Database:

- **SQLite3:**
 - Lightweight, serverless relational database.
 - Easy to set up and integrate with Django.
 - Stores structured data such as user details, campaigns, donations, and volunteer records.
 - Uses primary keys and foreign keys to maintain relationships and data integrity.

4. Libraries & Frameworks:

- **Django ORM (Object Relational Mapper):**
 - Allows database operations using Python code instead of SQL.
 - Simplifies data handling and improves development speed.
- **Celery:**
 - An asynchronous task queue used to process background jobs.
 - Handles tasks like sending emails, generating reports, and running AI computations.
- **Redis:**
 - In-memory data store used as a message broker for Celery.
 - Ensures fast communication between application components.
- **Django Channels:**
 - Extends Django to handle WebSockets.
 - Enables real-time features like live donation tracking and instant notifications.

5. Models (Database Models):

- **User Model:** Stores user credentials and profile data.
- **Role Model:** Defines access roles (Donor, Volunteer, Admin, Corporate).
- **Donor Model:** Contains donor-specific information.
- **Volunteer Model:** Stores skills, availability, and experience.
- **NGO Model:** Maintains NGO details and management data.
- **Campaign Model:** Stores campaign information such as goals and duration.
- **Donation Model:** Records all donation transactions.
- **CSR Model:** Handles corporate funding and compliance.
- **Goods/Services Model:** Manages non-monetary contributions.

6. AI Technologies (Optional Advanced Feature):

- Machine Learning techniques for analyzing donor behavior.
- Recommendation systems for suggesting relevant campaigns.
- Volunteer skill-matching algorithms.
- Predictive analytics for donation trends and campaign success.

7. Blockchain Technologies (Optional Advanced Feature):

- Hash generation for each donation transaction.
- Immutable and tamper-proof record storage.
- Transparent tracking of fund usage.
- Enhances trust and accountability in the system.

8. Other Tools & Development Utilities:

- **Git & GitHub:**
 - Used for version control and collaboration.
 - Tracks code changes and manages project versions.
- **VS Code:**
 - Code editor used for writing and debugging code.
 - Supports extensions for Django and Python development.
- **Postman:**
 - API testing tool used to test REST APIs.
 - Helps verify request-response cycles.
- **Web Browser (Chrome):**
 - Used for testing and debugging frontend features.
 - Developer tools help inspect UI and fix errors.
- **Django Admin Panel:**
 - Built-in interface for managing database records.
 - Allows easy monitoring and control of system data.

3.5 Advanced Features

1. Artificial Intelligence (AI) Features:

- AI-based **campaign recommendation system** for donors.
- **Donor behavior analysis** based on past donations.
- **Donor churn prediction** to identify inactive donors.
- **Donor Lifetime Value (LTV)** prediction.
- **Volunteer skill-matching** with suitable campaigns.
- **Sentiment analysis** from user feedback.

2. Blockchain Features:

- **Immutable donation records** using blockchain.
- Unique **hash generation** for each donation.
- **Transparent fund tracking** and verification.
- **Donation traceability** from donor to usage.
- Support for **smart contracts** (future scope).

3. Real-Time Features:

- **Live donation updates** using WebSockets.
- **Real-time campaign progress tracking.**
- **Instant notifications** for users.
- Improved user engagement and interaction.

4. Asynchronous Processing:

- Background task execution using **Celery**.
- **Redis** used as a message broker.
- Handles email sending, report generation, and AI tasks efficiently.

5. Multi-Tenant Architecture:

- Supports **multiple NGOs on a single platform**.
- Ensures **data isolation and security**.
- Reduces operational cost and improves scalability.

6. Advanced Donation Features:

- Supports **one-time, recurring, and pledge-based donations**.
- **Wallet-based micro-donations**.
- Automated **receipt generation**.

7. Volunteer Enhancement Features:

- **Smart volunteer matching** based on skills.
- **Volunteer hour tracking**.
- Future support for **gamification (badges, leaderboards)**.
- Auto-generated **volunteer certificates**.

8. CSR & Corporate Features:

- Corporate **CSR campaign funding**.
- **Impact tracking and reporting**.
- Employee volunteering support.
- CSR compliance dashboards.

9. Future-Ready Features:

- **AI-generated impact reports**.
- **NFT-based donation certificates**.
- **Geo-based volunteer matching**.
- Cross-NGO collaboration.
- DAO-based governance (future concept).

Overall, these advanced features enhance the system by making it intelligent, transparent, scalable, and user-friendly.

3.6 Ethical Considerations

1. Data Privacy:

- User personal information (name, email, donation details) must be protected.
- The system ensures that sensitive data is not shared without user consent.
- Follows proper data handling and privacy policies.

2. Data Security:

- Secure authentication and authorization mechanisms are implemented.
- Passwords are encrypted to prevent unauthorized access.
- Protection against common threats such as hacking and data breaches.

3. Transparency:

- All donation transactions are recorded clearly.
- Donors can track how their funds are used.
- Blockchain (optional) ensures immutable and verifiable records.

4. Accountability:

- NGOs are responsible for proper utilization of funds.
- System maintains logs and reports for monitoring activities.
- Ensures trust between donors, volunteers, and organizations.

5. Fairness and Non-Discrimination:

- AI features are designed to avoid bias in recommendations.
- Equal opportunities are provided to all users and volunteers.
- No discrimination based on gender, region, or background.

6. User Consent:

- Users must agree to terms and conditions before using the platform.
- Clear communication about data usage and permissions.
- Option to update or delete personal data.

7. Ethical Use of AI:

- AI is used responsibly for recommendations and analysis.
- Avoids manipulation or misleading users.
- Ensures explainability and fairness in decision-making.

8. Responsible Use of Technology:

- Technology is used for social good and community development.
- Avoid misuse of the platform for fraudulent activities.
- Promotes honesty and integrity in all operations.

4. Implementation & Results

Implementation:

The ENGO Web Application (EcoNexus NGO) is implemented using the Django framework with a modular and multi-tier architecture. The backend is developed in Python using Django and Django REST Framework, while the frontend is built using HTML, CSS, Bootstrap, and JavaScript to provide a responsive and user-friendly interface.

The database is implemented using SQLite3, which stores all system data including users, campaigns, donations, and volunteer records. Django ORM is used for efficient database operations without writing complex SQL queries.

Core modules such as User Management, Donor, Volunteer, NGO Admin, and CSR are developed and integrated successfully. Role-Based Access Control (RBAC) is implemented to allow users to perform multiple roles securely.

Real-time features such as live donation updates and notifications are implemented using Django Channels (WebSockets). Background tasks like email notifications and report generation are handled using Celery with Redis, ensuring smooth system performance.

Advanced features such as AI-based campaign recommendations and optional Blockchain-based donation tracking are conceptually implemented to enhance system intelligence and transparency.

Results:

The system was tested with different user roles, including donors, volunteers, and NGO administrators. The results show that the platform successfully handles multiple functionalities such as user registration, campaign management, donation processing, and volunteer coordination.

The donation module efficiently processes different types of donations and maintains accurate transaction records. The volunteer module effectively matches users with campaigns and tracks participation. The NGO admin module provides clear dashboards and reports for monitoring activities.

Real-time updates improve user engagement by providing instant feedback on campaign progress. Asynchronous processing ensures that background tasks do not affect system performance.

Overall, the system demonstrates improved **efficiency, transparency, and user experience** compared to traditional NGO management methods. The integration of modern technologies makes the platform scalable and future-ready for real-world applications.

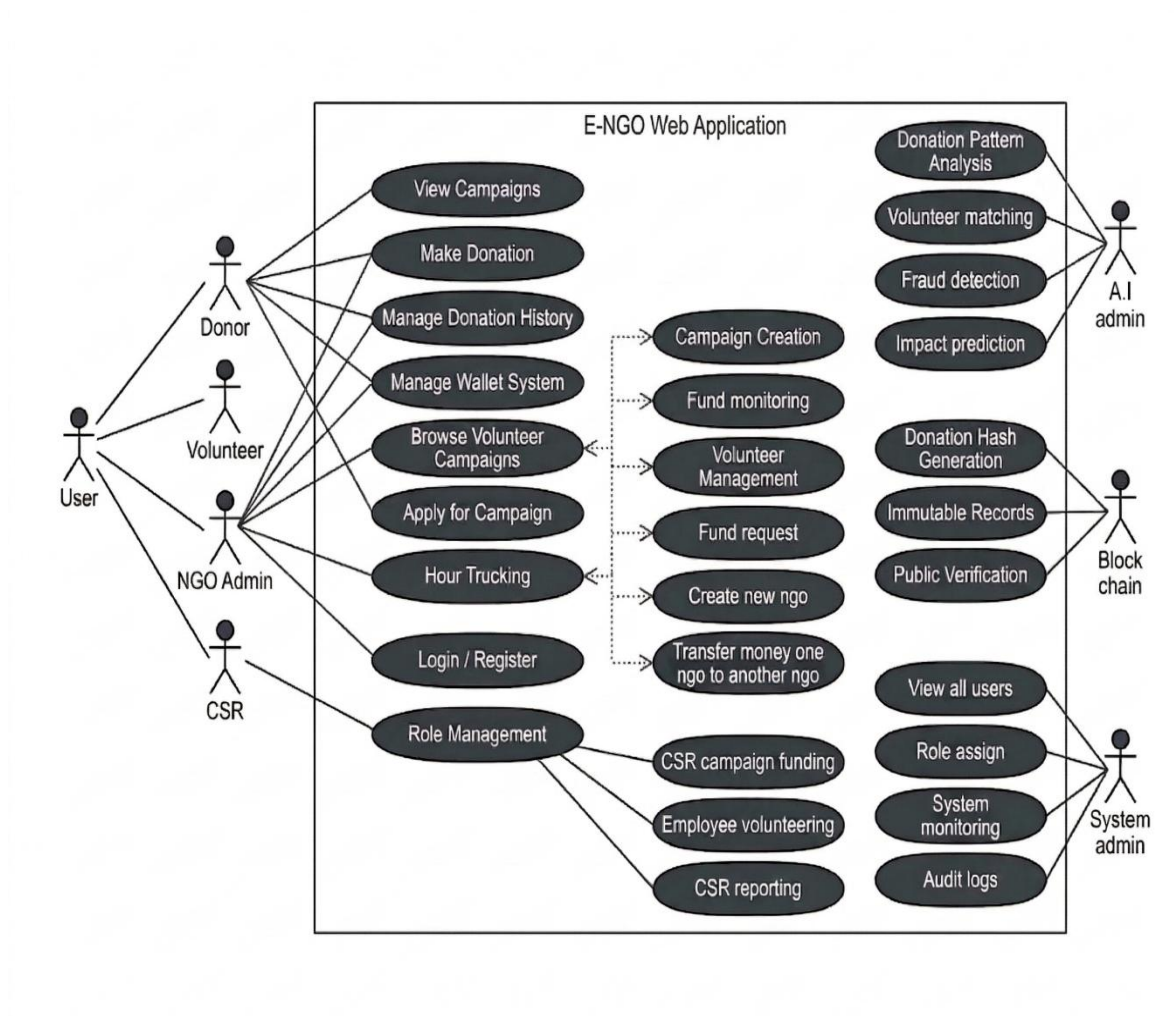
4.1 UML Diagrams

Unified Modeling Language (UML) diagrams are an essential part of system design, used to visually represent the structure, functionality, and behavior of the ENGO Web Application (EcoNexus NGO). These diagrams help developers and stakeholders clearly understand how different components of the system interact with each other. UML diagrams also improve communication, reduce development errors, and provide a blueprint for implementation.

1. Use Case Diagram:

The Use Case Diagram represents the interaction between different users (actors) and the system. In this application, the main actors include Donor, Volunteer, NGO Admin, Corporate User, and System Admin. Each actor performs specific actions such as registration, login, making donations, applying for volunteering, creating campaigns, and managing reports.

For example, a donor can browse campaigns, donate funds, and view donation history, while a volunteer can apply for campaigns and track their participation. NGO Admins manage campaigns and monitor activities. This diagram helps in identifying system requirements and user functionalities.

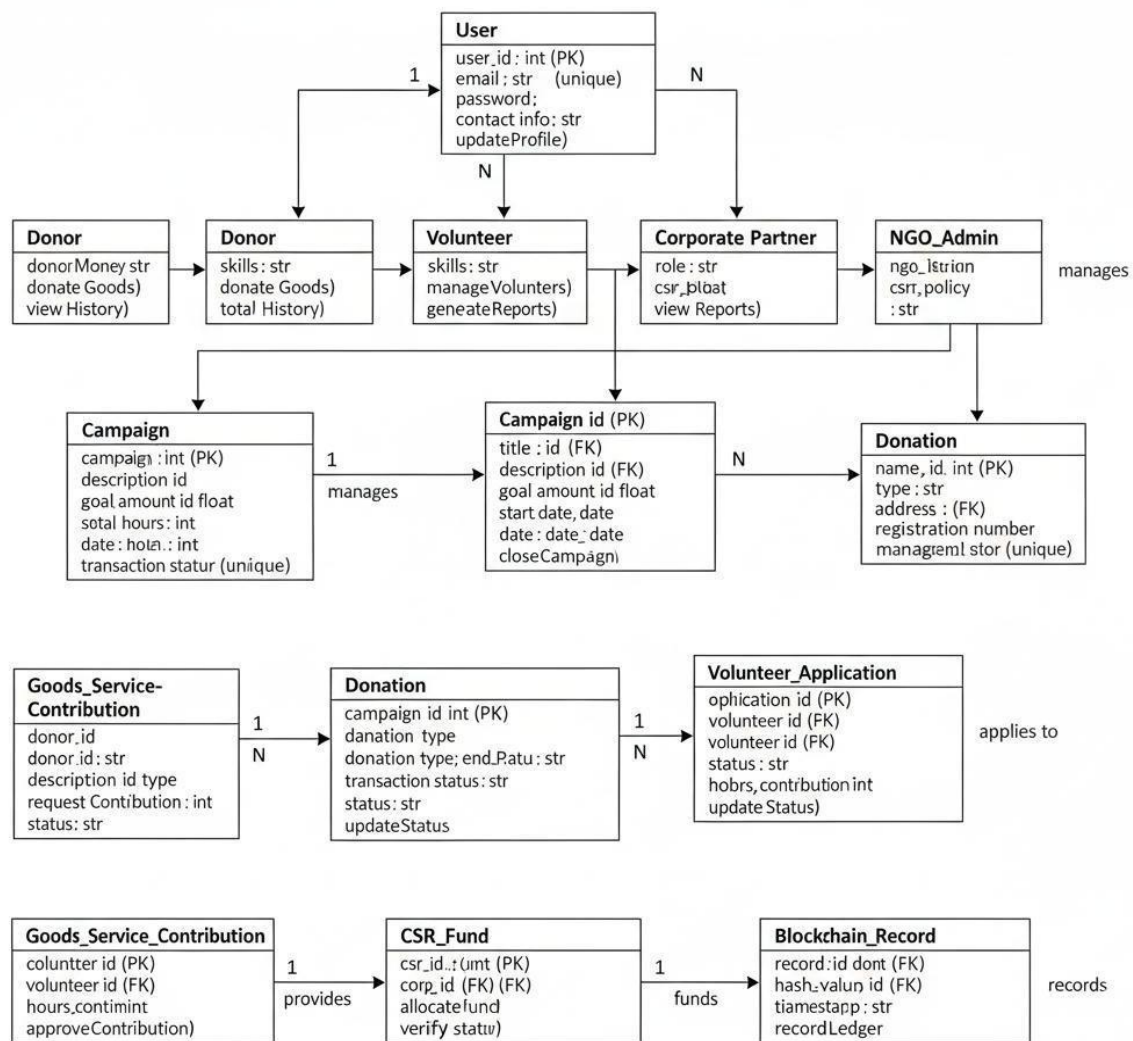


2. Class Diagram:

The Class Diagram shows the static structure of the system by representing classes, their attributes, methods, and relationships. Important classes in the system include User, Role, Donor, Volunteer, NGO, Campaign, Donation, and CSR.

Each class contains attributes (e.g., user name, campaign goal) and methods (e.g., login, donate, create campaign). Relationships such as inheritance (User → Donor/Volunteer), association (Campaign → Donation), and aggregation are defined. This diagram is useful for backend development and database design.

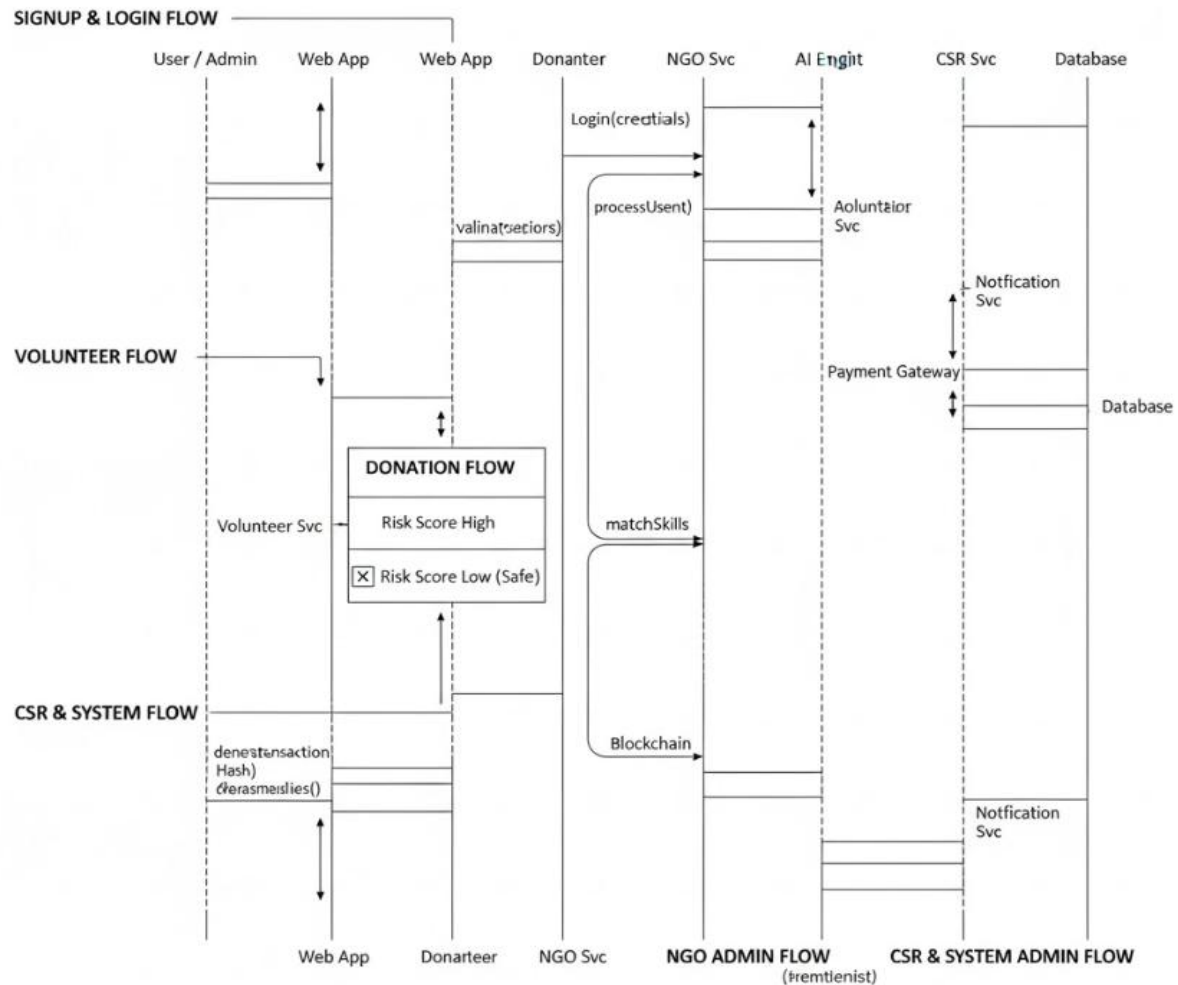
Class Diagram – E-NGO Web Application



3. Sequence Diagram:

The Sequence Diagram illustrates the dynamic behavior of the system by showing how different components interact in a sequence over time. It represents the flow of messages between objects.

For instance, in the donation process, the sequence starts with the user selecting a campaign, entering donation details, processing payment, and receiving confirmation. Each step is shown in order, helping developers understand the execution flow and system logic.



4. Activity Diagram:

The Activity Diagram represents the workflow of different processes in the system. It shows the sequence of activities, decision points, and parallel processes. For example, the donation process includes steps such as login, campaign selection, payment Processing and confirmation. This diagram helps in analyzing business logic and improving process efficiency.

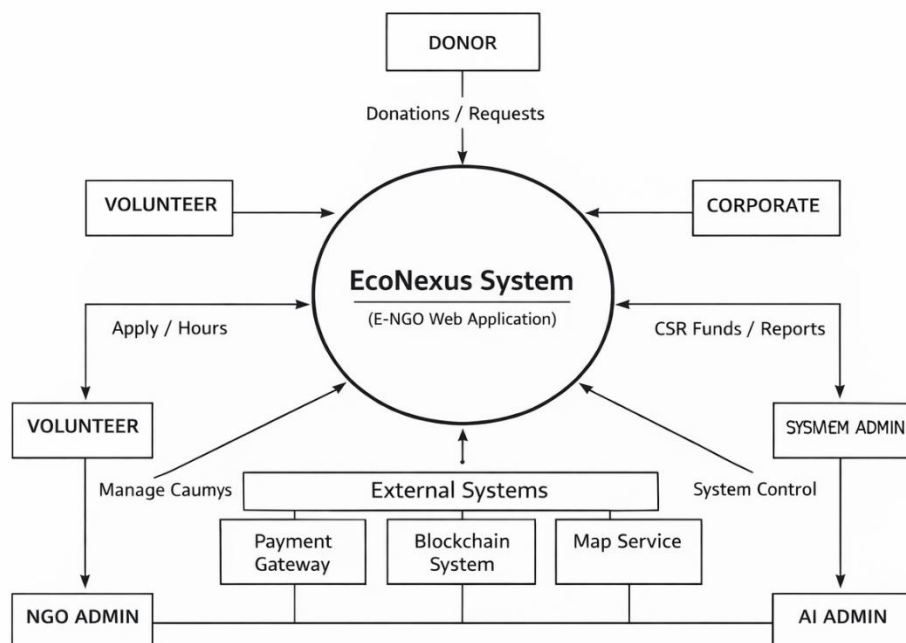


5. DFD Diagram:

5.1 0LEVEL:

The Level 0 DFD, also referred to as a context diagram, represents the entire ENGO Web Application as a single high-level process. It defines the system boundaries and illustrates the primary data exchanges between the application and external entities.

- **Central Process:** The core of the diagram is the **EcoNexus System (E-NGO Web Application)**.
- **External Entities:** The system interacts with four primary actors: **Donors**, **Volunteers**, **NGO Admins**, and **Corporate Partners**.
- **Primary Data Flows:**
 - ❖ **Donors** provide donations and requests.
 - ❖ **Volunteers** submit applications and track hours.
 - ❖ **NGO Admins** manage campaigns.
 - ❖ **Corporate Partners** provide CSR funds and receive impact reports.
- **External System Integrations:** The diagram also maps interactions with external services including **Payment Gateways**, **Block chain Systems**, and **Map Services**.

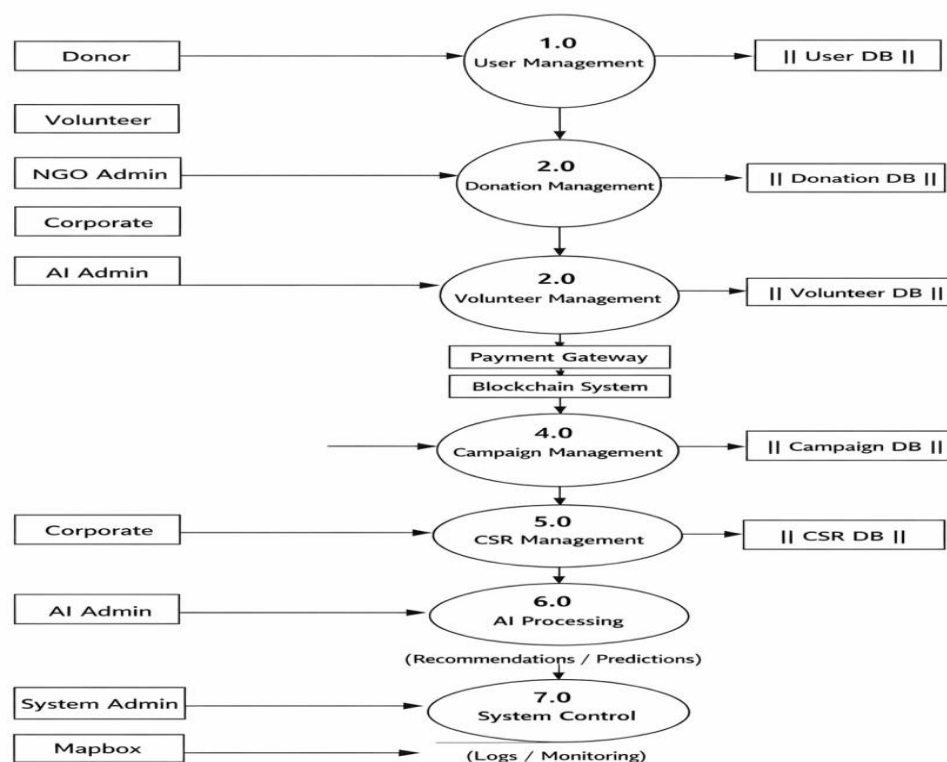


5.2 1 LEVEL:

The Level 1 DFD provides a more granular view of internal operations by breaking the system down into distinct functional processes. It shows how data is transformed and routed between these processes and the various databases.

- **1.0 User Management:** This process handles user authentication and profile data, interacting directly with the **User DB**.
- **2.0 Donation Management:** Manages the flow of contributions from donors, processing them through the **Payment Gateway** and recording entries in the **Donation DB** and **Block chain System**.
- **3.0 Volunteer Management:** Facilitates registration and application tracking, storing all volunteer-related metrics in the **Volunteer DB**.
- **4.0 Campaign Management:** NGOs utilize this process to create and monitor social initiatives, with data stored in the **Campaign DB**.
- **5.0 CSR Management:** Handles corporate-specific activities, such as fund allocation and compliance tracking, utilizing the **CSR DB**.
- **6.0 AI Processing:** An advanced layer that generates intelligent campaign recommendations and predictive analytics.
- **7.0 System Control:** This administrative process oversees logs, monitoring, and overall system integrity

EcoNexus – Level 1 DFD



6. DATA DICTIONARY:

1. USER & AUTHENTICATION MODULE

Table: auth_user Description: Stores all registered system users.

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique user identifier
username	varchar(150)	Unique	Login username
password	varchar(128)	-	Encrypted password
first_name	varchar(150)	-	User first name
last_name	varchar(150)	-	User last name
email	varchar(254)	-	Email address
is_superuser	bool	-	Admin access flag
is_staff	bool	-	Staff access flag
is_active	bool	-	Active status
last_login	datetime	-	Last login time
date_joined	datetime	-	Account creation date

Table: accounts_profile Description: Stores user role and verification details.

Column	Type	Key	Description
id	INTEGER	PK	Profile ID
role	varchar(20)	-	User role (Donor/NGO/Volunteer/CSR/Admin)
verified	bool	-	Verification status
user_id	INTEGER	FK	Linked to auth_user

2. DONOR MODULE

Table: donor_campaign Description: Stores donation campaigns created for fundraising.

Column	Type	Key	Description
id	INTEGER	PK	Campaign ID
title	varchar(200)	-	Campaign title
description	TEXT	-	Campaign details
goal_amount	decimal	-	Target donation amount
raised_amount	decimal	-	Amount collected
is_active	bool	-	Campaign status
created_at	datetime	-	Creation date

Table: donor_donation Description: Stores individual donor transactions

Column	Type	Key	Description
id	INTEGER	PK	Donation ID
amount	decimal	-	Donation amount
donation_type	varchar(20)	-	Type (Online/Wallet/etc.)
message	TEXT	-	Donor message
blockchain_hash	varchar(256)	-	Blockchain transaction reference
created_at	datetime	-	Donation timestamp

Table: donor_wallet Description: Stores donor wallet balance.

Column	Type	Key	Description
id	INTEGER	PK	Wallet ID
balance	decimal	-	Current wallet balance
user_id	INTEGER	FK	Linked donor

3. NGO ADMIN MODULE

Table: ngo_admin_ngo Description: Stores NGO details.

Column	Type	Key	Description
id	INTEGER	PK	NGO ID
name	varchar(200)	-	NGO name
balance	REAL	-	Available funds
created_at	datetime	-	Registration date

Table: ngo_admin_project Description: NGO projects for funding.

Column	Type	Key	Description
id	INTEGER	PK	Project ID
title	varchar(200)	-	Project title
goal_amount	REAL	-	Target fund
created_at	datetime	-	Created date
ngo_id	bigint	FK	Linked NGO

Table: ngo_admin_fundrequest Description: NGO fund request records.

Column	Type	Key	Description
id	INTEGER	PK	Request ID
required_amount	REAL	-	Requested amount
description	TEXT	-	Purpose
fulfilled	bool	-	Status
created_at	datetime	-	Request date
ngo_id	bigint	FK	Linked NGO

Table: ngo_admin_transaction Description: Blockchain-based NGO transactions.

Column	Type	Key	Description
id	INTEGER	PK	Transaction ID
type	varchar(20)	-	Credit/Debit
amount	REAL	-	Transaction amount
previous_hash	varchar(64)	-	Previous block hash
hash	varchar(64)	-	Current block hash
timestamp	datetime	-	Transaction time
ngo_id	bigint	FK	Linked NGO

4. VOLUNTEER MODULE

Table: volunteer_volunteer

Description: Stores volunteer profiles.

Column	Type	Key	Description
id	INTEGER	PK	Volunteer ID
availability	varchar(100)	-	Availability details
languages	varchar(100)	-	Known languages
total_hours	integer	-	Total contributed hours
impact_score	integer	-	Volunteer impact score
is_verified	bool	-	Verification status
leaderboard_rank	integer	-	Rank position
user_id	INTEGER	FK	Linked user

Table: volunteer_campaign

Description: Volunteer campaigns managed by NGOs.

Column	Type	Key	Description
id	INTEGER	PK	Campaign ID
title	varchar(200)	-	Campaign title
category	varchar(100)	-	Campaign category
description	TEXT	-	Details
location_name	varchar(200)	-	Location
latitude	REAL	-	Geo coordinate
longitude	REAL	-	Geo coordinate
start_date	date	-	Start date
end_date	date	-	End date
is_emergency	bool	-	Emergency flag
is_active	bool	-	Status
created_at	datetime	-	Created date
ngo_id	bigint	FK	Linked NGO

Table: volunteer_skill

Description: Master table storing all available skills.

Column Name	Data Type	Key	Description
id	INTEGER	PK	Unique skill ID
name	varchar(100)	-	Skill name
description	TEXT	-	Skill details

Table: volunteer_campaign_required_skills

Description: Maps required skills to campaigns.

Column Name	Data Type	Key	Description
id	INTEGER	PK	Mapping ID
campaign_id	INTEGER	FK	Linked campaign
skill_id	INTEGER	FK	Required skill

Table: volunteer_volunteer_skills Description: Maps volunteers with their skills.

Column Name	Data Type	Key	Description
id	INTEGER	PK	Mapping ID
volunteer_id	INTEGER	FK	Linked volunteer
skill_id	INTEGER	FK	Skill possessed

5. CSR MODULE

Description:

Manages Corporate Social Responsibility campaigns, ESG scoring, compliance, blockchain ledger, and employee volunteering.

Table: csr_corporatecsrprofile

Column	Type	Key	Description
id	INTEGER	PK	Corporate profile ID
company_name	varchar(200)	-	Company name
registration_no	varchar(100)	-	Registration number
total_budget	REAL	-	CSR allocated budget
created_at	datetime	-	Registration date

Table: csr_csrcampaign

Column	Type	Key	Description
id	INTEGER	PK	Campaign ID
title	varchar(200)	-	Campaign name
budget	REAL	-	Allocated budget
start_date	date	-	Start date
end_date	date	-	End date
corporate_id	INTEGER	FK	Linked corporate

Table: csr_blockchainledger

Column	Type	Key	Description
id	INTEGER	PK	Ledger ID
transaction_hash	varchar(256)	-	Blockchain hash
previous_hash	varchar(256)	-	Previous hash
timestamp	datetime	-	Record time
campaign_id	INTEGER	FK	Linked campaign

Table: csr_csrcertificate

Column	Type	Key	Description
id	INTEGER	PK	Certificate ID
certificate_no	varchar(100)	-	Certificate number
issued_date	date	-	Issued date
corporate_id	INTEGER	FK	Linked corporate

6. AI ENGINE MODULE

Description:

Handles AI-based fraud detection, donation analysis, sentiment analysis, and NGO performance scoring.

Table: ai_engine_ngo

Column	Type	Key	Description
id	INTEGER	PK	NGO AI record
risk_score	REAL	-	Fraud risk score
performance_score	REAL	-	AI performance score
ngo_id	INTEGER	FK	Linked NGO

Table: ai_engine_donation

Column	Type	Key	Description
id	INTEGER	PK	Donation AI record
anomaly_flag	bool	-	Fraud detection flag
sentiment_score	REAL	-	Donor sentiment
donation_id	INTEGER	FK	Linked donation

Table: ai_engine_impactreport

Column	Type	Key	Description
id	INTEGER	PK	Impact report ID
impact_score	REAL	-	AI calculated impact
report_summary	TEXT	-	AI generated summary
ngo_id	INTEGER	FK	Linked NGO

7. SYSTEM ADMIN MODULE

Description:

Manages roles, multi-tenant system, audit logs, anomaly events, and system metrics.

Table: system_admin_role

Column	Type	Key	Description
id	INTEGER	PK	Role ID
role_name	varchar(100)	-	Role name
description	TEXT	-	Role details

Table: system_admin_userrole

Column	Type	Key	Description
id	INTEGER	PK	Mapping ID
user_id	INTEGER	FK	Linked user

8. DJANGO SYSTEM TABLES

Description:

Internal tables automatically created by Django framework.

Table Name	Purpose
django_admin_log	Admin activity logs
django_content_type	Model type registry
django_migrations	Migration history
django_session	User session storage
auth_group	User groups
auth_permission	Permissions
auth_group_permissions	Group-Permission mapping
auth_user_groups	User-Group mapping
auth_user_user_permissions	Direct user permissions

4.2 Module Implementation

The ENGO system is designed using a modular architecture where each functional component is implemented independently to ensure high performance, security, and maintainability.

1. User & Authentication Module

This module serves as the entry point for all stakeholders and manages identity and access.

- **Authentication Engine:** Implements secure registration and login functionalities using encrypted passwords to protect user data.
- **Role-Based Access Control (RBAC):** A unified user account system that allows a single individual to seamlessly switch between multiple roles, such as Donor, Volunteer, or NGO Admin.
- **Profile Management:** Maintains detailed user profiles, including contact information and verification status.

2. Donor Module

The Donor Module focuses on facilitating transparent and efficient social contributions.

- **Diverse Contribution Models:** Supports multiple donation types, including one-time, recurring, and pledge-based contributions.
- **Interactive Exploration:** Provides a comprehensive campaign browsing interface where donors can select social causes based on their interests.
- **Financial Tracking:** Implements a personal donation history portal and a wallet-based system for micro-donations.
- **Automated Verification:** Automatically generates donation receipts and 80G tax certificates for eligible contributions.

3. Volunteer Module

This module manages human capital by connecting passionate individuals with NGOs in need of specific expertise.

- **Profile & Skill Inventory:** Enables volunteers to create professional profiles, listing their specific skills, languages, and availability.
- **Smart Application System:** Allows users to browse active volunteer campaigns and apply based on skill requirements.
- **Participation Metrics:** Tracks contributed working hours and generates "Proof-of-Work" records to verify community impact.
- **Certification:** Issues automated volunteer certificates upon successful completion of campaign duties.

4. NGO Admin Module

This administrative hub provides NGOs with the tools necessary to manage their daily digital operations.

- **Campaign Lifecycle Management:** Empowers NGO admins to create, update, and monitor multiple social campaigns simultaneously.
- **Resource Coordination:** Includes interfaces for reviewing and approving volunteer applications and managing fund requests.
- **Real-time Monitoring:** Provides live dashboards to track donation progress and volunteer activities through instant feedback loops.

5. CSR (Corporate Social Responsibility) Module

Specifically designed for corporate partners, this module facilitates large-scale social investments.

- **Strategic Campaign Funding:** Enables corporations to allocate budgets and fund specific NGO social projects directly.
- **Compliance & ESG Reporting:** Generates automated compliance reports and ESG (Environmental, Social, and Governance) scores to meet statutory requirements.
- **Employee Engagement:** Supports specialized programs for employee volunteering and corporate-level social impact tracking.

6. Advanced Intelligence & Security Modules

- **AI Module:** Handles intelligent volunteer skill-matching, campaign recommendations for donors, and predictive analytics for donation trends.
- **Blockchain Module:** Generates unique SHA-256 hashes for each transaction to ensure that donation records remain immutable and transparent.

4.3 Payment Gateway Integration

The Payment Gateway Module is a critical component of the EcoNexus system, designed to facilitate secure, transparent, and seamless financial contributions from donors to NGO campaigns. It ensures that every transaction is processed through encrypted channels and recorded accurately in the system's ledger.

Key Functional Features:

- **Support for Multiple Donation Models:** To cater to different donor preferences, the system supports:
 - **One-time Donations:** Immediate contributions for urgent causes.
 - **Recurring Donations:** Automated monthly or periodic contributions to ensure sustained support for long-term projects.
 - **Pledge-Based Donations:** Commitments made by donors to pay at a future date or in installments.
- **Integrated Payment Workflow:** The implementation follows a streamlined 4-step process:
 1. **Campaign Selection:** Donors choose a specific verified NGO campaign from the marketplace.
 2. **Amount Specification:** Users enter the desired contribution amount with options for predefined impact tiers (e.g., "Feed a child for a month").
 3. **Secure Processing:** The system initiates a secure handshake with the payment processor to handle sensitive card or UPI details.
 4. **Instant Confirmation:** Upon successful authorization, the system displays a real-time confirmation message and triggers the donation success protocol.
- **Automated Document Generation:** Immediately after a successful transaction, the module generates a digital receipt containing a unique transaction ID, campaign details, and the NGO's tax-exemption information (e.g., 80G status).

Security and Technical Architecture:

- **End-to-End Encryption:** All financial data is transmitted using **SSL/TLS encryption** to prevent man-in-the-middle attacks and ensure donor privacy.
- **Immutable Transaction Logs:** Every payment event is logged with a unique SHA-256 hash, preparing the data for the system's blockchain-simulated ledger to prevent record tampering.
- **Third-Party Integration (Future Scope):** While the current prototype handles secure transaction simulation, the architecture is designed for seamless API integration with industry-leading gateways like **Razorpay** or **Stripe** for real-world deployment.
- **Webhook Implementation:** The system is designed to use webhooks to listen for payment status updates (Success, Pending, or Failed) to ensure the database remains synchronized with the actual bank status.

Technical Impact:

By digitizing the donation process, this module eliminates manual reconciliation errors for NGOs and builds high levels of trust with donors through instant feedback and transparent documentation.

4.4 Real-Time Features

Real-time capabilities are integrated into the EcoNexus platform to enhance user engagement, provide immediate feedback, and optimize system responsiveness. By eliminating the need for manual page refreshes, the system ensures a fluid and dynamic experience for donors, volunteers, and administrators alike.

Key Functional Features:

- **Live Donation Feed:** Whenever a contribution is made, the system broadcasts a live update to the campaign page. This social proofing encourages further community participation and provides donors with immediate recognition.
- **Dynamic Progress Tracking:** Each campaign features a real-time progress bar that visually represents the journey toward the funding goal. As donations are processed, the bar updates instantly across all active user sessions.
- **Instant Notification System:**
 - **Donors** receive immediate "Success" alerts upon payment.
 - **Volunteers** get instant updates on their application status (Approved/Pending).
 - **Admins** are notified in real-time of high-value transactions or new volunteer registrations.
- **Live Statistical Syncing:** The dashboard's global metrics—such as total social impact percentage and total blocks added to the ledger—are updated live as activities occur across the network.

Technical Architecture:

- **WebSockets & Django Channels:** Unlike traditional HTTP requests where the client must ask for data, the system utilizes **WebSockets** via **Django Channels**. This allows for a persistent, full-duplex communication channel between the server and the browser.
- **Asynchronous Event Handling:** The system uses an asynchronous layer to push data packets to specific "Groups" (e.g., all users viewing a specific campaign) the moment a database change is detected.
- **Redis Backplane:** **Redis** is utilized as a channel layer to manage high-speed message passing, ensuring that real-time updates are delivered with sub-second latency even during high traffic.

Impact on User Experience:

- **Increased Transparency:** Real-time updates confirm that the system is active and that funds are being tracked at the very moment they are given.
- **Enhanced Performance:** By pushing only small data packets instead of reloading entire HTML pages, the system reduces server load and bandwidth consumption.
- **Engagement & Trust:** The "Live Node" status and instant notifications create a sense of a living, breathing social network, which is vital for building long-term trust in digital NGO management.

4.5 Dashboard & Reporting

The Dashboard and Reporting module is designed to transform raw system data into actionable insights. By providing customized interfaces for different user roles, the system ensures that complex financial and operational data is presented in a simplified, visual, and user-friendly format.

Role-Specific Dashboards:

- **Donor Dashboard (Individual Impact Portal):**
 - **Financial Overview:** Allows donors to view their complete donation history, including pending pledges and successful transactions.
 - **Impact Tracking:** Features visual metrics that show how their contributions have directly impacted social causes (e.g., "Number of lives touched" or "Projects completed").
 - **Certificate Access:** A dedicated section to download automated tax-exemption receipts and participation certificates.
- **NGO Admin Dashboard (Operations Control Center):**
 - **Campaign Monitoring:** Provides real-time tracking of active campaigns, including fund collection speed and target-to-achievement ratios.
 - **Volunteer Analytics:** Displays active volunteer counts, pending approvals, and total service hours contributed to various projects.
 - **Engagement Metrics:** Visualizes donor retention and new acquisition trends to help NGOs plan future outreach.
- **CSR Dashboard (Corporate Governance Suite):**
 - **Fund Allocation Tracking:** Enables corporate users to monitor how their CSR budgets are distributed across various NGO partners and specific projects.
 - **Compliance & ESG Reporting:** A specialized tool to generate automated reports required for statutory compliance and Environmental, Social, and Governance (ESG) audits.
 - **Impact Benchmarking:** Compares project outcomes against corporate social goals using standardized data points.

Comprehensive Reporting Engine:

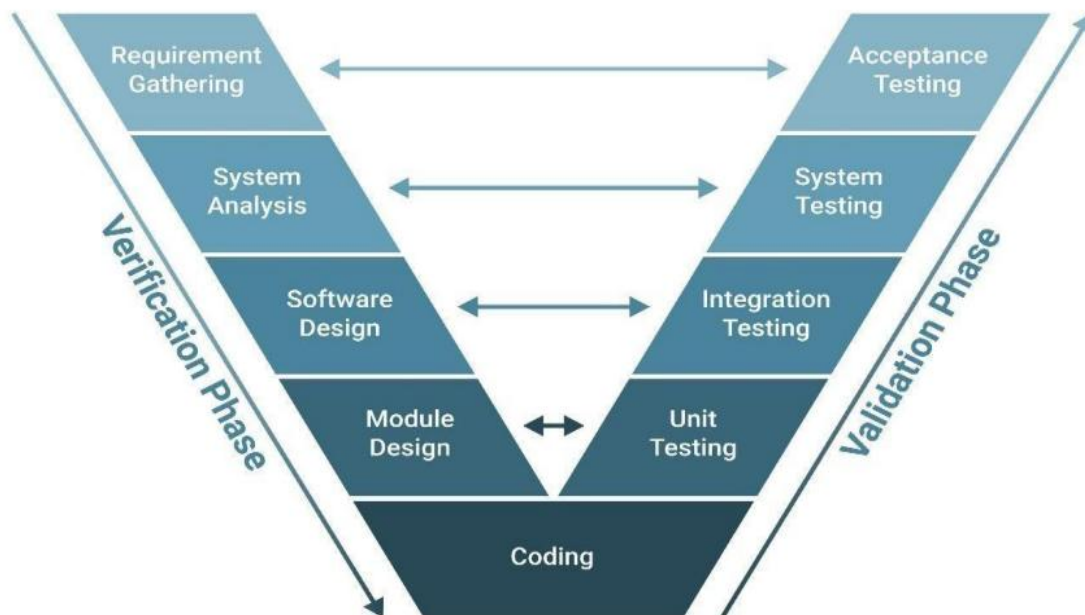
The system includes a robust reporting engine capable of generating detailed documents in multiple formats (PDF/Excel):

- **Donation Reports:** Detailed audit trails of all financial inflows, categorized by donor type, date, and payment method.
- **Campaign Performance Reports:** In-depth analysis of campaign success, identifying high-performing initiatives and areas needing more support.
- **Volunteer Impact Reports:** Documentation of human capital contribution, including total hours, skill utilization, and project milestones achieved.
- **Blockchain Audit Logs:** Technical reports showing the cryptographic integrity of the ledger, ensuring all recorded data is transparent and untampered.

Technical Implementation:

- **Data Visualization:** Implemented using charting libraries (such as Chart.js or D3.js) to convert database queries into interactive graphs and pie charts.
- **Dynamic Filtering:** Users can filter reports by date range, NGO category, or specific geographic locations.

4.6 Testing & Validation



Testing is a fundamental phase of the EcoNexus development lifecycle, ensuring that the platform is robust, secure, and capable of handling complex NGO operations. A multi-layered testing strategy was implemented to verify that every component functions correctly under various conditions.

Core Testing Methodologies:

- **Unit Testing (Module-Level Verification):**
 - Focuses on isolating and testing individual functions and components.
 - **Implementation:** Specific tests were written for the **Login System**, **Donation Processing Logic**, and **Hash Generation**. Each unit was checked to ensure it returns the expected output for a given input, preventing logic errors at the foundational level.
- **Integration Testing (Interface Verification):**
 - Verifies that different modules interact seamlessly with one another.
 - **Implementation:** This ensured that when a donor completes a payment, the **Donation Module** correctly triggers the **Blockchain Module** to hash the data and the **Notification Module** to alert the NGO Admin.
- **System Testing (End-to-End Verification):**
 - Validates the complete, integrated system against the initial requirements.
 - **Implementation:** The entire workflow—from NGO registration and campaign creation to fund disbursement and report generation—was tested as a single continuous process to ensure no "broken links" exist in the user journey.
- **User Acceptance Testing (UAT):**
 - Conducted with real-world scenarios to evaluate the system's usability and effectiveness.
 - **Implementation:** Feedback was gathered from simulated "Donors" and "NGO Admins" to refine the UI/UX, ensuring that the dashboards are intuitive and the navigation is straightforward for non-technical users.

Validation & Error Handling:

- **Data Integrity Validation:** Implemented strict input validation to ensure that only correct data formats (e.g., valid email addresses, positive currency amounts) are accepted. This prevents SQL injection and data corruption.
- **Edge Case Testing:** The system was stressed with "boundary" data, such as zero-value donations or duplicate registrations, to verify that error messages are triggered correctly without crashing the system.

Quality Benchmarks:

The testing phase was deemed successful by achieving three primary benchmarks:

1. **Security:** Verified that **Role-Based Access Control (RBAC)** prevents unauthorized users from accessing sensitive financial data.
2. **Reliability:** Ensured that **WebSockets** and **Django Channels** maintain persistent connections without data loss during high-frequency updates.
3. **Accuracy:** Cross-verified that the **Blockchain Ledger** hashes match the database entries exactly, providing a 100% accurate audit trail.

5. Discussion

5.1 System Functionality

The ENGO Web Application (EcoNexus NGO) is developed to provide a seamless, efficient, and user-friendly platform that connects donors, volunteers, NGOs, and corporate users in a unified ecosystem. The system is designed with a focus on accessibility, usability, and performance to ensure that all users can interact with the platform effortlessly.

One of the key features of the system is its support for multiple user roles, including donors, volunteers, NGO administrators, and corporate users. Each role is equipped with specific functionalities tailored to their needs. Users can easily register on the platform, securely log in, and switch between roles when required, making the system highly flexible and convenient.

The donation process is designed to be simple, transparent, and quick. Users can browse campaigns, select causes they care about, and complete donations through a clear step-by-step process. This ensures a smooth experience while encouraging more user participation and trust in the platform.

For volunteers, the system provides an efficient way to explore and engage with various campaigns. Volunteers can search for opportunities based on their skills, interests, and availability. The application process is straightforward, allowing users to apply quickly and track their participation.

NGO administrators are provided with powerful tools to manage their activities effectively. They can create and update campaigns, monitor donations, and manage volunteer participation through an organized dashboard. This helps NGOs streamline their operations and maintain better control over their initiatives.

The user interface of the ENGO Web Application is clean, responsive, and adaptable across different devices such as desktops, tablets, and smartphones. This ensures that users can access the platform anytime and anywhere without any usability issues.

Additionally, the system incorporates real-time updates and notifications, which keep users informed about important activities such as donation confirmations, campaign updates, and application statuses. Interactive dashboards further enhance the user experience by presenting information in a structured and easy-to-understand manner.

Overall, the ENGO Web Application ensures a smooth, reliable, and engaging user experience by combining robust functionality with an intuitive design, making it an effective platform for social impact and collaboration.

5.2 Benefits for Users

The ENGO Web Application (EcoNexus NGO) provides a centralized and integrated platform that improves efficiency, transparency, and communication among all stakeholders involved in social initiatives.

- The system reduces dependency on manual processes by digitizing donation management, volunteer coordination, and campaign tracking.
- **For Donors:**
 - The platform offers a simple, secure, and user-friendly donation process with clear steps.
 - Donors can easily browse different campaigns and choose causes that match their interests.
 - It allows tracking of donation history, including details of past contributions.
 - Transparency in fund utilization ensures donors can see how their money is being used.
 - Builds trust and encourages long-term engagement and repeated donations.
- **For Volunteers:**
 - Volunteers can register quickly and explore a wide range of campaigns.
 - Skill-based matching helps volunteers find opportunities that suit their abilities and preferences.
 - The platform simplifies the application process for participating in campaigns.
 - Volunteers can track their participation, performance, and working hours.
 - Enhances engagement by providing meaningful opportunities to contribute to society.
- **For NGOs:**
 - NGOs can efficiently create, update, and manage multiple campaigns from a single platform.
 - The system helps in tracking donations in real time, improving financial management.
 - Volunteer management becomes easier with proper tracking and communication tools.
 - Dashboards and reports provide insights that support better planning and decision-making.
 - Reduces administrative workload and improves overall operational efficiency.
- **For Corporate Users (CSR):**
 - Corporate users can easily invest in social initiatives and support NGOs.
 - The system enables monitoring of CSR activities and fund allocation.
 - Provides structured reports that help in compliance with corporate and legal requirements.
 - Enhances corporate reputation by showcasing social responsibility efforts.
- **Overall Benefits:**
 - Improves transparency and accountability across the entire system.
 - Saves time and reduces manual errors through automation.
 - Strengthens collaboration between donors, volunteers, NGOs, and corporates.
 - Provides real-time updates and better communication among users.
 - Offers a scalable and flexible solution that can grow with increasing users and campaigns.
 - Promotes social impact by making participation in social causes easier and more accessible.

5.3GUI Screenshots:

- A.Client Side

HOMEPAGE





Donate

Turning Crisis into Action.

From climate resilience to community empowerment, see how we are making a measurable impact on the ground.

[View Active Projects](#)[Donate to Intervention](#)

 TOTAL LIVES IMPACTED: 2,450,120+

Our Work

Region Sector Status 

DISASTER RELIEF

Project Phoenix: Flood Response

Completed

Goal: 15,000 People

100%

SUSTAINABILITY

Project Azure: Coral Reefs

Ongoing

Goal: 1MM Corals

65%

EDUCATION

Project Literacy: Global Schools

Active

Goal: 12 Schools

65%

EDUCATION

Project Literacy: Tech Outreach

Active

Goal: 50 Labs

30%

OUR MODEL OF CHANGE



Assessment

Identifying core community needs.



Mobilization

Deploying resources and experts.



Sustainability

Long-term local leadership.

FEATURED INTERVENTION

Crisis Response: The 2025 Flood Relief



Extreme rainfall displaced 50,000 families. We deployed mobile clinics and clean water stations within 48 hours.
[\[Read the Full Report \]](#)

GET_INVOLVED



INTERVENTIONSGET INVOLVEDSTORIES

Join Us

JOIN THE MOVEMENT

Your hands. Our mission.

Real change.

Whether you have five minutes or five days a month, there are a dozen ways you can help us protect the planet and empower communities.



Volunteer

Join our field teams or help remotely with translation, design, or data.

[Apply to Team →](#)



Donate

Fuel our interventions. 92% of every dollar goes directly to project costs.

[Give Now](#)



Partner

For corporations and NGOs looking to scale impact through collaboration.

[Get in Touch →](#)



Advocate

Spread the word. Use your voice to influence policy and awareness.

[Download Kit →](#)

Not sure where to start?

Tell us your background, and we'll match you with a project that needs your specific expertise.

Your Name

My Skillset 

[FIND MY PLACE](#)



"I started as a remote translator. Three months later, I was on the ground in Kenya helping with the flood relief. It changed my life."

— Sarah J., Volunteer

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ABOUT US

EcoNexus

A multipurpose foundation fueling a global spectrum of kindness, technology, and transparent giving.

Why EcoNexus Exists

We bridge the gap between resources and results across four vital sectors.

Environment

Restoring Earth's natural resources and biodiversity through innovative conservation programs.

Education

Empowering the global workforce through digital literacy and technical training.

Healthcare

Providing essential medical supplies and services to underserved communities.

CSR / Social

A transparent pipeline for corporate social responsibility and ethical sourcing.

How We Work

Verified NGOs

A strict 5-star authentication process for every campaign.

Smart Giving

Blockchain integrated secure gateways for instant action.

Full Clarity

Track fund movement from your dashboard to the field.

Automation

Instant NGO tax receipts and PDF project reports.

Our Global Mission

To create a unified digital ecosystem where generosity meets accountability. We aim to use technology to prove that every donation, no matter how small, can trigger a massive, colorful wave of change.

© 2025 EcoNexus Foundation | Impacting 5M+ Lives Globally

INTERVENTIONS



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Turning Crisis into Action.

From climate resilience to community empowerment, see how we are making a measurable impact on the ground.

[View Active Projects](#)[Donate to Intervention](#)

TOTAL LIVES IMPACTED: 2,450,120+

Our Work

Region

Sector

Status

GLOBAL IMPACT
2,450,120+ Lives Saved

DISASTER RELIEF

Project Phoenix: Flood Response

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Goal: 15,000 People **100%**

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Project Azure: Coral Reefs

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Goal: 1MM Corals **65%**

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RESOURCES

ENGO

Donate

Turning Crisis into Action.

From climate resilience to community empowerment, see how we are making a measurable impact on the ground.

[View Active Projects](#)[Donate to Intervention](#)

TOTAL LIVES IMPACTED: 2,450,120+

Our Work

Region | Sector | Status

DISASTER RELIEF

Project Phoenix: Flood Response

Completed

Goal: 15,000 People

100%

SUSTAINABILITY

Project Azure: Coral Reefs

Ongoing

Goal: 1MM Corals

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Project Literacy: Global Schools

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DONATE NOW

"Giving is not just about making a donation. It's about making a difference."

Your smallest contribution can be the answer to someone's biggest prayer. Join our mission to create a world where no one walks alone.

100% Transparency

Track your funds via AI Audit Logs.

Verified Impact

Donations reach pre-verified NGOs only.

Start Your Legacy

Fill the form below to share your blessings.

FULL NAME

John Doe

EMAIL ADDRESS

john@example.com

GIFT TYPE

Choose...

AMOUNT (₹)

₹ 500

NOTES OF HOPE (OPTIONAL)

Tell us more about the items or leave a message for the recipients...

CONFIRM MY CONTRIBUTION

Secure & AI-Audited Transaction

SSL ENCRYPTED

"We make a living by what we get, but we make a life by what we give."

© 2026 E-NGO Platform | Building a Trusted Ecosystem for Social Good

"Giving is not just about making a donation. It's about making a difference."

Your smallest contribution can be the answer to someone's biggest prayer. Join our mission to create a world where no one walks alone.

**100% Transparency**
Track your funds via AI Audit Logs.

**Verified Impact**
Donations reach pre-verified NGOs only.

Start Your Legacy

Fill the form below to share your blessings.

FULL NAME
virat kohli

EMAIL ADDRESS
kohli@gmail.com

GIFT TYPE
Nutrition/Food

AMOUNT (₹)
₹ 40000

NOTES OF HOPE (OPTIONAL)
A "Two-Handed" Life: You have two hands—one for helping yourself and one for helping others. The Ripple Effect: Good deeds are like stones cast into water; the stone may sink, but the ripples will last.

CONFIRM MY CONTRIBUTION

🎉 Donation Successful! Thank you 🎉

🔒 Secure & AI-Audited Transaction

SSL ENCRYPTED

"We make a living by what we get, but we make a life by what we give."

© 2026 E-NGO Platform | Building a Trusted Ecosystem for Social Good

JOIN THE MISSION



"The best time to act was yesterday. The second best time is now."

By joining our mission, you aren't just a donor or a volunteer—you are a pillar of resilience for communities in crisis.

#ACTNOW

#ENGO2026

#SUSTAINABLEFUTURE

BACK TO HOME →

MEMBERSHIP

What it means to join.

01

A Seat at the Table
Members receive quarterly strategic updates and have the opportunity to vote on regional project priorities.

02

Unfiltered Transparency
Gain access to the 'Ground-Truth' portal—a live feed of raw data, field photos, and expense reports directly from our interventions.

03

Collective Power
Join a network of 50,000+ activists and professionals who leverage their unique skills to solve complex environmental challenges.

Ready to take the next step?
Official registration and membership inquiries are handled via our secure portal or through our regional offices.
econexusfoundation.org

* Membership is voluntary and open to individuals committed to our non-partisan, science-led charter.

SIGNUP

START YOUR LEGACY

Plant a Seed of Hope.

"Do not wait for extraordinary circumstances to do good; try to use ordinary situations."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

ravi12

ravi@gmail.com

.....

Perfect strength!

.....

Perfect match!

YOUR MISSION ROLE

❤️ Donor

👤 Volunteer

🏢 NGO

🏢 Corporate

🚀 LAUNCH MY IMPACT

Already registered? [Sign In](#)

LOGIN

🌟 YOUR QUEST CONTINUES

Welcome Home.

"Every morning we are born again. What we do today matters most."

YOUR DREAMS ARE WAITING INSIDE.

Login

Enter your credentials to access the portal.

IDENTITY / USERNAME

ravi12

SECRET KEY / PASSWORD

.....

UNLOCK PORTAL 🗝️

New to the dream? [Sign up here](#)

1.NGO_ADMIN

User choose to ngo_admin role:

START YOUR LEGACY

Plant a Seed of Hope.

"Seeds are not just what you
sow, but what you wish others
to grow."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

Getting better!

Perfect match!
YOUR MISSION ROLE

Donor

Volunteer

NGO

Corporate

LAUNCH MY IMPACT

Already registered? [Sign In](#)

Dashboard.html:

NGO Blockchain Admin

System Live: 2026

Control Panel - 15 Features

Create NGO
Add new NGO to Ledger

Volunteer Approval
Manage applications

Fund Transfer
NGO to NGO transaction

Blockchain Ledger
View all blocks

AI Fraud Scan
Security analysis

AI Forecast
Predict future funds

Impact Reports
Generate NGO results

Smart Contracts
Manage release rules

Notifications
Real-time alerts

Payment Gateway
Simulated donations

Blockchain: Active

AI Engine: Ready

Active Volunteers:

Campaigns
Track fundraising

NGO Directory
View all registered NGOs

System Health
Node & Database status

Messaging
Talk to contributors

Settings
Account & Security

Volunteer_list.html:

Back to Control Panel

Volunteer Network

Manage and track your professional contributors

Total: 20

Back to Home

ACTIVE NOW20

All Registered Volunteers

Search Skills

VOLUNTEER PROFILE	EXPERTISE / SKILLS	CONTRIBUTION	STATUS	ACTION
A Aarav Patel Qarav_Patel	Content Writer, Data Analyst	122 Hrs	PENDING	Approve Now
V Vikash Sharma Qvish_Sharma	Social Worker, UI/UX Designer	46 Hrs	PENDING	Approve Now
I Ishan Mehta Qshan_Mehta	Project Coordinator, Community Manager	156 Hrs	PENDING	Approve Now
A Ananya Iyer Qanya_Iyer	Public Speaker, Social Worker	165 Hrs	PENDING	Approve Now
A Aditya Joshi Qditya_Joshi	Community Manager, Data Analyst	200 Hrs	PENDING	Approve Now
S Siddharth Shah Qsddharth_Shah	Social Worker, Data Analyst	143 Hrs	PENDING	Approve Now
D Diya Trivedi Qdiya_Trivedi	Blockchain Developer, Data Analyst	166 Hrs	PENDING	Approve Now
K Kabir Mathotra Qabir_Mathotra	Content Writer, UI/UX Designer	166 Hrs	PENDING	Approve Now
R Rohan Gupta Qrohan_Gupta	Public Speaker, Public Speaker	121 Hrs	PENDING	Approve Now
K Klara Desai Qklara_Desai	Project Coordinator, Content Writer	144 Hrs	PENDING	Approve Now
A Arijit Kukarni Qarjit_Kukarni	Content Writer, Community Manager	50 Hrs	PENDING	Approve Now
M Meha Rajput Qmeha_Rajput	UI/UX Designer, Content Writer	46 Hrs	PENDING	Approve Now
A Aryan Varma Qaryan_Varma	Blockchain Developer, Blockchain Developer	155 Hrs	PENDING	Approve Now
S Saanvi Reddy Qsaanvi_Reddy	UI/UX Designer, Blockchain Developer	56 Hrs	PENDING	Approve Now
R Rishi Pandey Qrishi_Pandey	Project Coordinator, Data Analyst	166 Hrs	PENDING	Approve Now
T Tanisha Choudhury Qtanisha_Choudhury	Community Manager, Project Coordinator	176 Hrs	PENDING	Approve Now
Z Zoya Khan Qzoia_Khan	Blockchain Developer, Blockchain Developer	112 Hrs	PENDING	Approve Now
K Karan Mathotra Qkaran_Mathotra	Project Coordinator, Community Manager	136 Hrs	PENDING	Approve Now
N Navya Singh Qnavia_Singh	Project Coordinator, Community Manager	54 Hrs	PENDING	Approve Now
A Aayush Mishra Qayush_Mishra	Public Speaker, Community Manager	163 Hrs	PENDING	Approve Now

© 2026 NGO Management System • Secure Blockchain Records

Blockchain_ledger.html:

Back to Control Panel

Inter-NGO Fund Transfer

Securely transfer funds between registered NGO entities.

Cancel

NGO Ascicraft created successfully!

SOURCE WALLET (SENDER)

HelpAge India (Available: ₹552290.00)

↓

DESTINATION WALLET (RECEIVER)

Greenpeace India

TRANSACTION AMOUNT

₹ 80000

Authorize & Mine Block

Blockchain Secured

Every Inter-NGO transfer is hashed and recorded as a new block in our decentralized ledger to ensure 100% transparency.

Transfer Guidelines

Ensure the sender has sufficient balance.

Double check the receiver NGO's wallet ID.

Transactions cannot be reversed once mined.

End-to-End Encrypted Financial Infrastructure

Systemsetting.html:



System Infrastructure

REAL-TIME NODE MONITORING

ALL SYSTEMS OPERATIONAL



99.9% Uptime

Database Cluster

PostgreSQL Mainnet



CONNECTED

Ledger Node #01

Syncing Block #1000



ACTIVE

AI Firewall

Fraud Detection Enabled

SYSTEM_TERMINAL.LOG

```
[18:42:01] Initializing NGO Ledger Core...
[18:42:05] Connection established with PostgreSQL Cluster.
[18:42:10] Blockchain sync complete. Last hash verified.
[18:43:55] WARN: High traffic detected from Node 04.
[18:45:12] AI Fraud Scan initiated on Block #1000.
-
```

← RETURN TO CONTROL CENTER

Smart_contract.html:



Active Smart Contracts

Automated governance rules for fund distribution

EVM Compatible



DEPLOYED

Fund Release Protocol

Funds are only released if 100% of the target amount is verified on-chain.

```
if (received >= needed) {
  status = "RELEASE_FUNDS";
  executeTransaction();
}
```

CONTRACT ID: 0x882...F21

GAS LIMIT: 21000



DEPLOYED

Volunteer Verification

Auto-verify volunteers after admin approval hash is generated.

```
mapping(address => bool) verified;
function verify(addr) {
  verified[addr] = true;
}
```

CONTRACT ID: 0x441...B09

GAS LIMIT: 45000

← CONTROL CENTER

Setting.html

Settings

Manage your NGO profile and platform preferences

General Profile

Security

Notifications

Organization Profile

NGO NAME

Ascicraft

EMAIL ADDRESS

asci@gmail.com

PUBLIC DESCRIPTION

Serving humanity through transparency and blockchain technology.

SAVE CHANGES

Danger Zone

Actions here are permanent and cannot be reversed.

DEACTIVATE NGO NODE

Payment.html:

Secure Checkouts

POWERED BY BLOCKCHAIN LEDGER

TOTAL DONATION AMOUNT

₹5,000.00

CARD HOLDER NAME

mitali sangani

CARD NUMBER

78654567

VISA

EXPIRY

98

CVV

..

Immutable Record:

This transaction will generate a SHA-256 hash and will be permanently stored on our transparent ledger.

Confirm & Sign Block

PCI-DSS COMPLIANT & AES-256 ENCRYPTED

Notificatation.html:

Notifications

Stay updated with the latest network activities

Dashboard

NGO Ascicraft successfully deployed to blockchain.

2 MINUTES AGO

SYSTEM ALERT

System Update: Block #7708 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

System Update: Block #2144 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

System Update: Block #7442 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

System Update: Block #3648 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

System Update: Block #8493 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

System Update: Block #9626 verified.

1 DAY, 7 HOURS AGO

SYSTEM ALERT

All activities are cryptographically signed and logged.

Ngo_list.html:

NGO Directory

Add New NGO

NGO Name	Location	Balance	Admin	Status
SEWA (Self Employed Women's Association)	Surat	₹296726.00		Active
Akshaya Patra Foundation	Surat	₹139756.00		Active
GIVE India	Rajkot	₹57753.00		Active
Cry Foundation	Bhavnagar	₹75196.00		Active
HelpAge India	Gandhinagar	₹552290.00		Active
Pratham Education Foundation	Vadodara	₹401649.02		Active
Sardar Patel Seva Trust	Gandhinagar	₹142714.00		Active
Ahmedabad Red Cross	Vadodara	₹172657.00		Active
Greenpeace India	Surat	₹499406.00		Active
Smile Foundation	Bhavnagar	₹571112.98		Active
Goonj Foundation	Gandhinagar	₹46736.00		Active
Deepak Foundation	Surat	₹448259.00		Active
Environmentalist Foundation of India	Vadodara	₹203175.00		Active
Agastya International Foundation	Rajkot	₹362136.00		Active
Kailash Satyarthi Children's Foundation	Vadodara	₹464786.00		Active
Bill & Melinda Gates Foundation (IN)	Vadodara	₹388569.00		Active
Care India	Surat	₹232079.00		Active
Oxfam India	Bhavnagar	₹190622.00		Active
Magic Bus India	Ahmedabad	₹446858.00		Active
Aaranyak Conservation	Bhavnagar	₹85438.00		Active
Ascicraft	mumbai	₹800000.00	prince	Active

Back to Control Panel

Create_new_ngo.html:

[← Back](#)

Register New NGO

NGO Name

Ascicraft

Location

mumbai

Initial Balance (₹)

800000

+

 Register & Save to Ledger

Role.html:

Admin Communication Hub

Manage network-wide interactions and volunteer coordination

← Dashboard

Core Communication Roles

As an Administrator, you act as the central node for all organizational updates.



Broadcast System

Instantly notify all registered volunteers about new campaigns, urgent requirements, or emergency relief efforts.



Volunteer Onboarding

Review and approve new volunteer applications. Verify credentials before granting access to the NGO node.



Resource Allocation

Assign tasks to specific teams based on geographical data and expertise level to ensure efficient ground operations.



Security & Privacy

Monitor interactions to maintain a professional environment and protect donor and beneficiary data privacy.



Note: Real-time messaging features require a secured WebSocket connection.

OPEN ADMIN CONSOLE

Blockchian_ledger:



Blockchain Ledger

Decentralized & Immutable Transaction Records

[Back to Control Panel](#)

Mainnet Active

BLOCK #1027

SENDER NGO

HelpAge India
0x2877...abc

₹80000.00

SUPPORT TRANSFER

RECEIVER NGO

Greenpeace India
0x3222...xyz

BLOCK HASH: sha256(77a...27...092b)

PREV HASH: sha256(000...x926...110z)

[RETURN TO FINANCE](#)

Report.html:

Impact Reports

Measuring the change we bring to the community

Export PDF



0+

LIVES TOUCHED



20

ACTIVE HEROES



100%

TRANSPARENCY



Success Stories (Funded Projects)

No completed projects to report yet.

[Back to Control Panel](#)



AI Integrity Scanner

Scanning blockchain nodes for suspicious activities...

SCANNING STATUS

100% Real-time Enabled



Transaction #27

From: HelpAge India To: Greenpeace India

High amount transaction detected

Perfect round figure (Potential manually forced)

₹80000.00

DANGER

Score: 50%



AI Prediction Engine

Predicting future donation trends using historical node data

CONFIDENCE SCORE

92%

CURRENT TOTAL FUNDS

₹0

Verified on Blockchain Ledger

PREDICTED TARGET (NEXT QTR)

₹0.0

+15.2% Est. Community Growth



Funding Trajectory Analysis

Actual Donations

AI Forecast

30,000

20,000

10,000

Jan

Feb

Mar

Apr

May

Jun

←

BACK TO CONTROL CENTER

LIVE AI IMPACT

74%

CHAIN BLOCKS

10

NGO NETWORK

42

Smart Operations



Mint Block



Fraud Scan



Verify Chain



AI Forecast

+ NGO Onboarding

Organization Name

↔ Inter-NGO P2P Transfer

Select Receiver NGO...

ECONEXUS

Guide Manual

Urmi1

Create Genesis Block

Active NGO Entities

NGO Helping 19	Gandhinagar
NGO Green 20	Vadodara
mission india	surat
education impact	surat

Active Volunteers

Volunteer_17_907	Rank #18	Active
Volunteer_18_195	Rank #19	Active
Volunteer_19_131	Rank #20	Active

Secure Admin Messenger

EcoNexus Node Verified

Broadcast message...



Blockchain Transaction Feed

Hash: 0089f5f3cbffc995e1a8f695c67fd3aeacc414883291f86e8bd...

₹2568.98 06:59

Hash: cb470e55f6f9652121309731c02378503e42538c1efd9c25bbb...

₹2416.48 06:59

Hash: fc88e984ca615bf87456fc0ac19b1968b70c0aa0ee4275a7998...

2.DONOR

User choose to donor role:

START YOUR LEGACY

Plant a Seed of Hope.

"Success is not just what you accomplish, but what you inspire others to do."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

saroj1

saroj@gmail.com

Getting better!

Perfect match!
YOUR MISSION ROLE

🔥 Donor

👤 Volunteer

🏠 NGO

🏢 Corporate

🚀 LAUNCH MY IMPACT

Already registered? [Sign In](#)

Donor.html

EcoNexus

Donor Universe

16 Ways to Drive Social Change in Gujarat

1. **Explore Projects**
Browse AI-vetted NGO campaigns across various sectors.

2. **Trust Ledger**
Immutable blockchain records of every donation's path.

3. **Impact Wallet**
Load funds and manage recurring micro-donations easily.

4. **Donor Roles**
Identify as a Financier, Mentor, or Ground Volunteer.

5. **AI Predictor**
Predict the long-term impact of your donations today.

6. **Live Field Feed**
Watch real-time status updates from the ground team.

7. **Skill Donation**
Donate your technical or professional expertise to NGOs.

8. **80G Tax Hub**
Instantly download 80G receipts for tax exemptions.

9. **Gift a Cause**
Donate on behalf of a loved one for special occasions.

10. **Impact NFTs**
Collect unique digital assets as rewards for contributions.

11. **Donor Voting**
Vote on which new social projects should get platform funding.

12. **Eco Footprint**
Track the environmental impact of your green donations.

13. **AI Legitimacy Audit**
Check AI-generated trust scores for every NGO on board.

14. **Private Giving**
Donate anonymously using Zero-Knowledge security.

15. **Milestone Tracker**
See exactly when a project goal is reached step-by-step.

16. **Auto-Pledge**
Automate your monthly kindness through smart pledges.

Live Campaigns

Search by NGO or Cause...

All Categories

Michael CeraVe

A multi-phase campaign that used a fake conspiracy theory suggesting actor Michael Cera was the ...

Raised: ₹4000.00 Goal: ₹80000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Ocean Plastic DNA

Tracking plastic source to stop ocean pollution.

Raised: ₹30507.00 Goal: ₹492734.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Zero Waste Surat

Community composting and plastic recycling awareness program.

Raised: ₹0.00 Goal: ₹60000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Mobile Health Clinic

Van-based primary healthcare for construction workers and their families.

Raised: ₹0.00 Goal: ₹70000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Skill Up Gujarat

Vocational training in Python and Tally for underprivileged youth.

Raised: ₹0.00 Goal: ₹30000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Adopt a Primary School

Renovation of building and library setup for city primary schools.

Raised: ₹0.00 Goal: ₹70000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Pure Flow Water Project

Installing RO water plants in public parks and government schools.

Raised: ₹60000.00 Goal: ₹50000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Project Pankh

Supporting local artisans to sell their handloom products online.

Raised: ₹80000.00 Goal: ₹80000.00

1 Supporters Active Now

1 people donated Verified NGO

[Donate Now](#)

Gau Seva & Nutrition

Sustainable fodder management for local animal shelters.

Raised: ₹0.00 Goal: ₹70000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Smart Anganwadi Upgrade

Transforming local anganwadis into digital learning centers.

Raised: ₹0.00 Goal: ₹90000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Emergency Response Fund

Support for rapid disaster relief and community safety kits.

Raised: ₹0.00 Goal: ₹50000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Green Roof Surat

Promoting terrace gardening and rainwater harvesting in societies.

Raised: ₹0.00 Goal: ₹30000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Bridge the Gap Literacy

Evening classes for children living in urban slums.

Raised: ₹0.00 Goal: ₹50000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Fit India CSR Drive

Building open-air gyms in public gardens of Adajan and Vesu.

Raised: ₹0.00 Goal: ₹60000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Renewable Energy Hub

Subsidizing solar panels for low-income households.

Raised: ₹0.00 Goal: ₹90000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Save the Sparrow

Distributing bird nests and water feeders across the city.

Raised: ₹0.00 Goal: ₹80000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Clean Coast Navsari

Beach cleanup and marine life protection awareness drive.

Raised: ₹0.00 Goal: ₹70000.00

0 Supporters Active Now

0 people donated Verified NGO

[Donate Now](#)

Green Today, Better Tomorrow

Inspiring eco-friendly habits for a sustainable future.

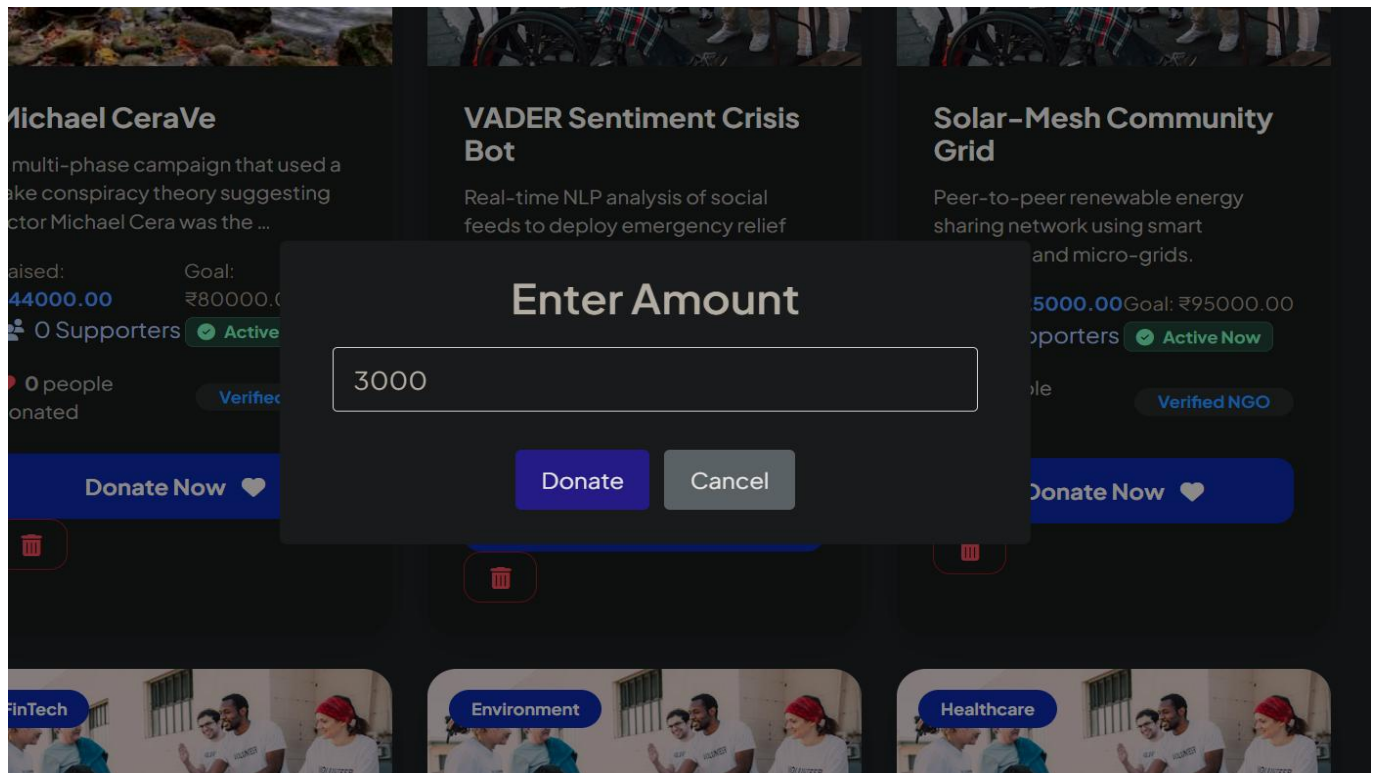
Raised: ₹7000.00 Goal: ₹400000.00

0 Supporters Active Now

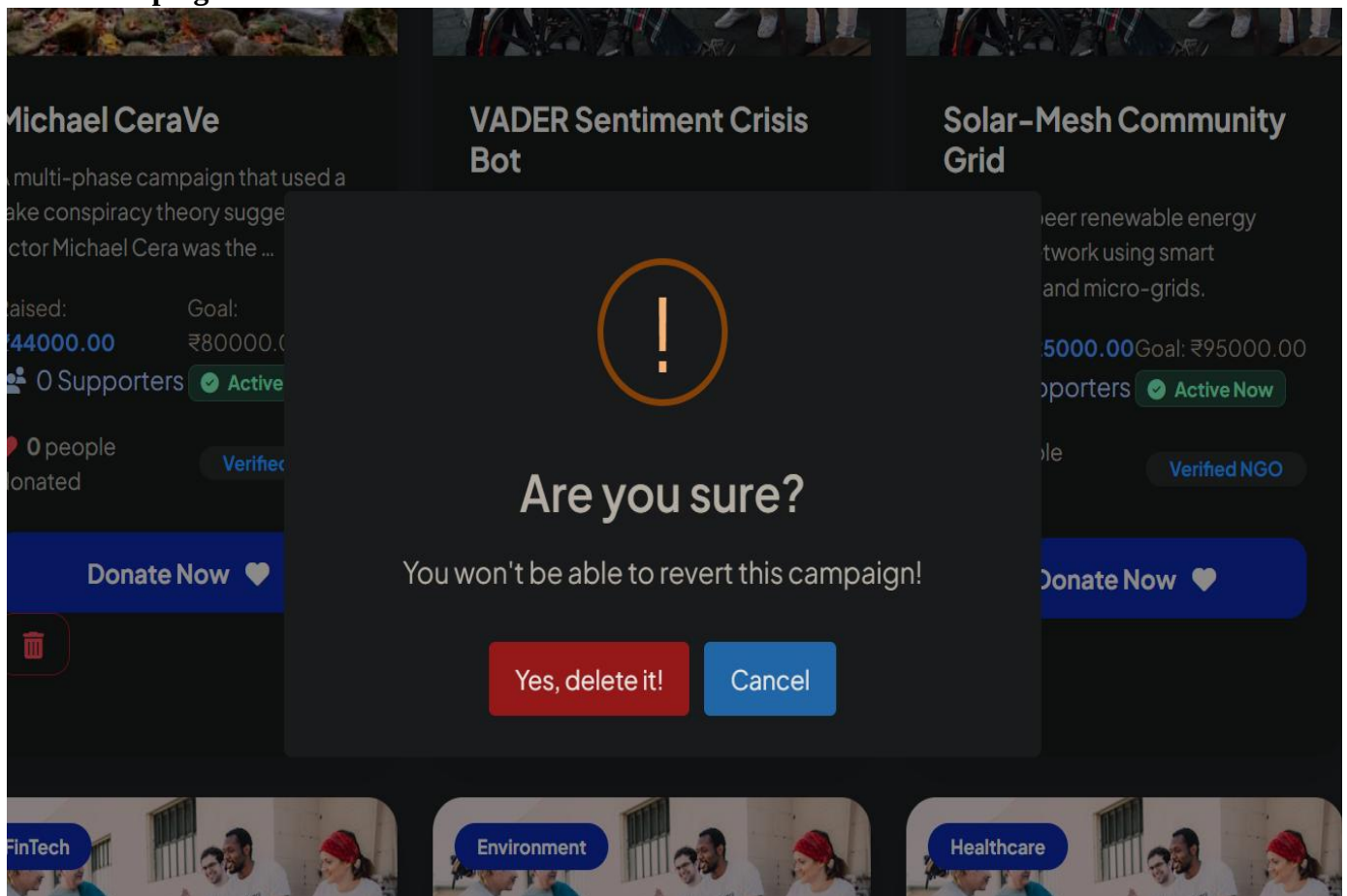
0 people donated Verified NGO

[Donate Now](#)

Donor donation add :



Delete campaign:



Ledger.html

EcoNexus Trust Ledger Explore TRUST LEDGER Verified Financial Transparency for CSR Impact.						
#	TRANSACTION ID	PROJECT NAME	DONOR ENTITY	AMOUNT	VERIFIED AT	STATUS
1	TXN-57CBA7A7FF	Michael CeraVe	Saroji	₹44090.00	14 Apr 2025, 13:05	SECURED
2	TXN-D8ADDD8B	Fit India CSR Drive	Surat Diamond Assoc.	₹16773.00	08 Apr 2025, 13:24	SECURED
3	TXN-C8D2AC4D	Skil Up Gujarat	Reliance Industries	₹36900.00	08 Apr 2025, 13:24	SECURED
4	TXN-28C628BC	Zero Waste Surat	Surat Diamond Assoc.	₹6692.00	08 Apr 2025, 13:24	SECURED
5	TXN-C8D19D9C	Renewable Energy Hub	HDFC Bank CSR	₹42759.00	08 Apr 2025, 13:24	SECURED
6	TXN-B8D1AD0E	Green Roof Surat	HDFC Bank CSR	₹8087.00	08 Apr 2025, 13:24	SECURED
7	TXN-ED8E6E9B	Pure Flow Water Project	Adani Foundation	₹46870.00	08 Apr 2025, 13:24	SECURED
8	TXN-3F4F28ED	Save the Sparrow	Surat Diamond Assoc.	₹36277.00	08 Apr 2025, 13:24	SECURED
9	TXN-B8D273ED	Sau Seva Nutrition	Adani Foundation	₹5923.00	08 Apr 2025, 13:24	SECURED
10	TXN-88F1208E	Emergency Response Fund	HDFC Bank CSR	₹21545.00	08 Apr 2025, 13:24	SECURED
11	TXN-41E79AED	Emergency Response Fund	Reliance Industries	₹47261.00	08 Apr 2025, 13:24	SECURED
12	TXN-3D3551F6	Green Roof Surat	HDFC Bank CSR	₹23859.00	08 Apr 2025, 13:24	SECURED
13	TXN-E8E837A0	Pure Flow Water Project	Adani Foundation	₹42172.00	08 Apr 2025, 13:24	SECURED
14	TXN-C8D19D9B	Adopt a Primary School	Tata Group	₹31460.00	08 Apr 2025, 13:24	SECURED
15	TXN-7F3A1FCC	Save the Sparrow	HDFC Bank CSR	₹18715.00	08 Apr 2025, 13:24	SECURED
16	TXN-B8D273ED	Save the Sparrow	HDFC Bank CSR	₹38927.00	08 Apr 2025, 13:24	SECURED
17	TXN-A18D9D41	Pure Flow Water Project	HDFC Bank CSR	₹35532.00	08 Apr 2025, 13:24	SECURED
18	TXN-D8E68E3D	Renewable Energy Hub	Tata Group	₹11283.00	08 Apr 2025, 13:24	SECURED
19	TXN-2D6C8D8B	Fit India CSR Drive	HDFC Bank CSR	₹43816.00	08 Apr 2025, 13:24	SECURED
20	TXN-AE23433C	Mobile Health Clinic	Surat Diamond Assoc.	₹18729.00	08 Apr 2025, 13:24	SECURED
21	TXN-22081A8E	Save the Sparrow	HDFC Bank CSR	₹42333.00	08 Apr 2025, 13:24	SECURED

Impact_wallet.html

ECONEXUS

Impact Wallet

Dashboard

Impact Wallet

TOTAL FUNDED

₹1922734.0

Direct contribution to community growth.

PROJECTS

18

Active initiatives in Surat & Gujarat.

IMPACT

19227+

Estimated lives positively affected.

Want to see the deep-ledger?

Every rupee is tracked on our transparent trust ledger.

VIEW FULL LEDGER

SECURE ECOSYSTEM

CHOOSE YOUR **IMPACT.**

Select how you want to contribute to the EcoNexus transparency network.



INDIVIDUAL

Support local causes, track your small-scale donations, and earn impact points for your contribution to the community.

CONTINUE AS DONOR →



CORPORATE

Manage CSR budgets, view high-impact ledgers, and download verified transparency certificates for your organization.

ACCESS CSR PORTAL 

 | VERIFIED BY ECONEXUS

NEURAL IMPACT ENGINE

AI **PREDICTOR**

AI CONFIDENCE **98%** 

EDUCATION **High IMPACT**

PREDICTED SUCCESS 92%

Based on historical data and current NGO performance, donations in this sector show optimized transparency.

ENVIRONMENT **Medium IMPACT**

PREDICTED SUCCESS 85%

Based on historical data and current NGO performance, donations in this sector show optimized transparency.

HEALTHCARE **High IMPACT**

PREDICTED SUCCESS 78%

Based on historical data and current NGO performance, donations in this sector show optimized transparency.

SMART ALLOCATION SUGGESTION

Our AI suggests diversifying your contribution across **Education** and **Environment** sectors for maximum transparency and long-term community impact.

APPLY AI ALLOCATION

Live_update.html

LIVE ACTIVITY FEED

Real-time transparency logs from EcoNeasus Network.

LIVE SYSTEM STATUS

	SAROJI Contributed ₹44000.00 to the Michael CeraVe Initiative. DONOR: TRENDSHAWK - VERIFIED GUY ASSOCIATE	23 MINUTES AGO
	SURAT DIAMOND ASSOC. Contributed ₹16775.00 to The #1 India CSR Drive Initiative. DONOR: THE DADDERS - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	RELIANCE INDUSTRIES Contributed ₹36900.00 to The Skill Up Gujarat Initiative. DONOR: TRENDINGZILLI - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	SURAT DIAMOND ASSOC. Contributed ₹80000.00 to The Zero Waste Surat Initiative. DONOR: THE SCSBORG - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	HDFC BANK CSR Contributed ₹42799.00 to the Renewable Energy Hub Initiative. DONOR: THE CHORDSIC - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	HDFC BANK CSR Contributed ₹8067.00 to the Green Roof Surat Initiative. DONOR: THE BODDZOO - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	ADANI FOUNDATION Contributed ₹46870.00 to the Pure Flow Water Project Initiative. DONOR: THE POKESION - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	SURAT DIAMOND ASSOC. Contributed ₹36277.00 to the Save the Sparrow Initiative. DONOR: THE NALINDU - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	ADANI FOUNDATION Contributed ₹30624.00 to the Gut Savvy & Nutrition Initiative. DONOR: THE BASSZED - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO
	HDFC BANK CSR Contributed ₹21545.00 to the Emergency Response Fund Initiative. DONOR: THE OFFSEER - VERIFIED GUY ASSOCIATE	4 DAYS, 19 HOURS AGO

- * AKAASHAKRISHN MANDHEJIAN, Mumbai, India
- * SURVESH, who's Source Registered Services
- * VIKASRAJENDR KUMAR KATHIRAVELAN, Chennai, India
- * IYENGAR, Renu Monica, who's Source Registered Services
- * Dr. Prashant, who's Source Registered Services
- * Anandraj, who's Source Registered Services

Skill_donatation.html



SKILL DONATION

Contributing expertise to:

YOUR NAME

EXPERTISE

HOURS PER WEEK

BRIEF MESSAGE

CONFIRM SKILL CONTRIBUTION

VERIFIED BY ECONEXUS IMPACT ENGINE

Tax_benefits.html

TAX SAVINGS
Section 80G - Tax Exemption Dashboard

PAN STATUS
• **Verified**


ELIGIBLE DONATION
₹44000


ESTIMATED TAX SAVING
₹22000.0

EN **ECONEXUS DIGITAL**

PROVISIONAL RECEIPT

This is a computer-generated summary of your contributions eligible for deduction under Section 80G of the Income Tax Act, 1961.

 **DOWNLOAD 80G CERTIFICATE**

VIEW HISTORY



Gift_cause.html

SPREAD THE LOVE

GIFT A CAUSE
Transform a special moment into a lasting impact. Send a personalized donation gift card to your loved ones.

RECIPIENT NAME

Who is this gift for?

OCCASION

Birthday

AMOUNT (₹)

500

CHOOSE A MISSION

Green Today, Better Tomorrow



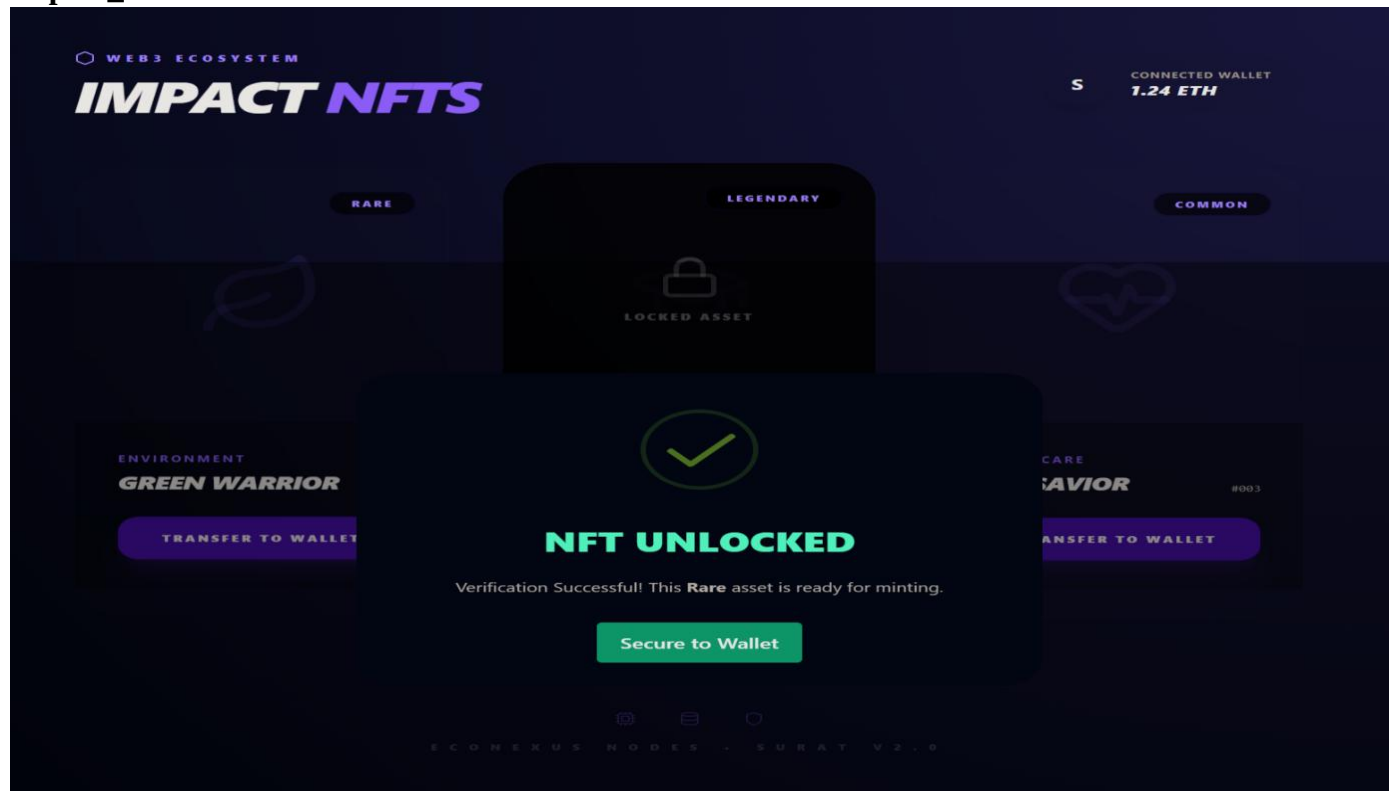
A GIFT OF HOPE
"Mitali, your contribution will be presented as a beautiful digital certificate, proving that the best gifts are the ones that change lives."

OFFICIAL ECONEXUS CERTIFICATE

DIGITAL ID: EN-GIFT-2026-AUTO

 **GENERATE GIFT CERTIFICATE**

Impact_nfts.html



Dao_voting.html



3.VOLUNTEER

User choose volunteer role:

START YOUR LEGACY

Plant a Seed of Hope.

"Your life is your canvas, and you are the masterpiece. Paint it with kindness."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

hari2

hari@gmail.com

Getting better!

Perfect match!

YOUR MISSION ROLE

Donor

Volunteer

NGO

Corporate

LAUNCH MY IMPACT

Already registered? [Sign In](#)

Volunteer_dashboard.html:

EcoNexus

VOLUNTEER

TOTAL IMPACT

HRS

Verified Change-maker Status

BADGES EARNED

Log hours to unlock

TRUST RANKING

Top 5%

VERIFIED

VOLUNTEER SERVICES

Edit Profile

Update your bio, skills, and education details.

Smart Campaigns

AI-matched projects based on your expertise.

Log Hours

Submit your work hours and share experiences.

Leaderboard

Check your rank among the top volunteers.

Impact Resume

Download your AI-verified service certificate.

Support NGOs

View and fulfill NGO resource requirements.

volunteer_profile_view

How to use this model in EcoNexus.

© 2026 EcoNexus Social Impact Tracking System

Profile_setting.html:



[← Dashboard](#)



hari2

VERIFIED VOLUNTEER
BLOCKCHAIN IDENTITY

BIO / MISSION

Data Science student and Python Developer passionate about using technology for social good. Specializing in technical education for underprivileged children and environment-driven data analysis

SKILLS INVENTORY

#full
#atack
#with
#python

ECONEXUS SYSTEM ARCHITECTURE

1

AI Matchmaking Algorithm

The system analyzes your *"Skills Inventory"* against CSR Campaign requirements using NLP to find the perfect social match.

2

Verified Proof of Work

When you log hours, our *"Sentiment AI"* analyzes your feedback to ensure genuine contribution and project impact.

3

Immutable Ledger Locking

Verified work is converted into a *"Blockchain Hash"* and stored in the public ledger, making your impact permanent and unchangeable.

THEORY OF IMPACT

The Problem: Traditional volunteering lacks transparency. It's hard to prove the hours served and the quality of work performed.

The EcoNexus Solution: By integrating **Artificial Intelligence** for verification and **Blockchain Ledger** for storage, we create a "Trustless Environment."

- ✓ Data Integrity through Hashing
- ✓ Automated Skill-based Recommendation
- ✓ Real-time Impact Analytics

ECONEXUS - POWERED BY AI & DISTRIBUTED LEDGER TECH - 2026

Ledger.html:

Blockchain Impact Ledger				
Immutable record of all social contributions and financial transactions.				
System Status: Decentralized				
Financial Transparency (CSR)				
Timestamp	Campaign	Amount	Blockchain Hash	Status
2026-04-13 04:00	Smart Anganwadi Upgrade	₹8000.00	04a7605f319d4a93a92f...	Verified
2026-04-10 05:12	Zero Waste Surat	₹45000.00	0b7f8422e8aef0403ba39d...	Verified
EcoNexus Ledger VOLUNTEER IMPACT LEDGER ← Back to Dashboard IMMERS. LAMBER				
Date	Volunteer	Contribution	Sentiment	
April 16, 2026	hari2	33 Hrs	Verified Log	
April 16, 2026	mit4	22 Hrs	Verified Log	
April 16, 2026	volunteer_user_20	6 Hrs	Verified Log	
April 16, 2026	volunteer_user_19	5 Hrs	Verified Log	
April 16, 2026	volunteer_user_18	3 Hrs	Verified Log	
April 16, 2026	volunteer_user_17	8 Hrs	Verified Log	
April 16, 2026	volunteer_user_16	7 Hrs	Verified Log	
April 16, 2026	volunteer_user_15	5 Hrs	Verified Log	
April 16, 2026	volunteer_user_14	4 Hrs	Verified Log	
April 16, 2026	volunteer_user_13	8 Hrs	Verified Log	
April 16, 2026	volunteer_user_12	3 Hrs	Verified Log	
April 16, 2026	volunteer_user_11	6 Hrs	Verified Log	
April 16, 2026	volunteer_user_10	8 Hrs	Verified Log	
April 16, 2026	volunteer_user_9	2 Hrs	Verified Log	
April 16, 2026	volunteer_user_8	3 Hrs	Verified Log	
April 16, 2026	volunteer_user_7	4 Hrs	Verified Log	
April 16, 2026	volunteer_user_6	3 Hrs	Verified Log	
April 16, 2026	volunteer_user_5	8 Hrs	Verified Log	
April 16, 2026	volunteer_user_4	4 Hrs	Verified Log	
April 16, 2026	volunteer_user_3	4 Hrs	Verified Log	
April 16, 2026	volunteer_user_2	4 Hrs	Verified Log	
April 16, 2026	volunteer_user_1	8 Hrs	Verified Log	
April 16, 2026	mitall_volunteer	10 Hrs	Verified Log	

Volunteer_dashboard.html:

Volunteer Profile Setup

Tell us about your skills so our AI can match you with the best campaigns.

 **Personal Branding**

Short Bio

Data Science student and Python Developer passionate about using technology for social good. Specializing in technical education for underprivileged children and environment-driven data analysis

 **EcoNexus**

[← Back to Dashboard](#)

 **Skills & Expertise**

COMMA SEPARATED (E.G. TEACHING, HEALTH, PLANNING)

full stack with python

 **Education**

HIGHEST DEGREE / CURRENT STATUS

B.TECH

 **Previous NGO Experience**

ISKON TRUST

BACKEND DEVELOPER

* AI WILL VERIFY THESE DETAILS WITH THE NGO FOR YOUR TRUST SCORE.

SAVE & SYNC PROFILE 

Hour_feedback.html:

Build Your Legacy

"Every act of kindness is a dream coming true for someone else."



AI Wellness Guardian

You have poured **18 hours** into campaigns this week. Take tonight to rest. Rest Recommended

Time & Energy

Log your contribution to the world.

When did you serve?

How many hours?

LOG MY IMPACT

"Time is not measured by clocks, but by moments of change."

Cloud Sync Enabled

Impact Journal

Share the emotions and results of your work.

How did today feel?

Your Energy Level after Campaign:

Replenished 🌟

Tired 😴

SUBMIT TO HISTORY

YOUR PATH TO 100 HOURS OF CHANGE

72 hours completed. You are making a real difference!

"Every act of kindness is a dream coming true for someone else."



AI Wellness Guardian

Time & Energy

LOG MV

Impact Logged!

Your time has been added to your legacy.

OK

Tired 🥱

SUBMIT TO HISTORY

YOUR PATH TO 100 HOURS OF CHANGE

72 hours completed. You are making a real difference!

Campaign.html:

Direct Impact Funding

Direct Impact Funding
Fund specific NGO requirements and track your contribution on the blockchain.

Your Contribution History

Ngo_dashboard.html(volunteer join the peoject):

Corporate Social Responsibility

Active campaigns from our corporate partners. Join and make an impact.

ACTIVE



MITALI Ocean Cleanup 2026

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Solar Power for Schools

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Clean Water Initiative

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Urban Jungle Project

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Reforestation Drive

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Green Earth 2026

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI EcoNexus

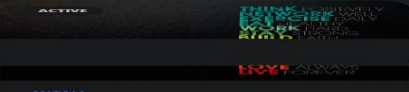
MITALI Clean Coast Navsari

Beach cleanup and marine life protection awareness drive.

FUNDING PROGRESS 0%

SUCCESS ✓

ACTIVE



MITALI Save the Sparrow

Distributing bird nests and water feeders across the city.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Renewable Energy Hub

Subsidizing solar panels for low-income households.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Fit India CSR Drive

Building open-air gyms in public gardens of Adajan and Vestu.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Bridge the Gap Literacy

Evening classes for children living in urban slums.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Green Roof Surat

Promoting terrace gardening and rainwater harvesting in societies.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Emergency Response Fund

Support for rapid disaster relief and community safety kits.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Smart Anganwadi Upgrade

Transforming local anganwadis into digital learning centers.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Gau Seva & Nutrition

Sustainable fodder management for local animal shelters.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Project Pankh

Supporting local artisans to sell their handloom products online.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Pure Flow Water Project

Installing RO water plants in public parks and government schools.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Adopt a Primary School

Renovation of building and library setup for city primary schools.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Skill Up Gujarat

Vocational training in Python and Tally for underprivileged youth.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Mobile Health Clinic

Van-based primary healthcare for construction workers and their families.

FUNDING PROGRESS 0%

JOIN PROJECT →

ACTIVE



MITALI Zero Waste Surat

Community composting and plastic recycling awareness program.

FUNDING PROGRESS 75%

JOIN PROJECT →

ACTIVE



MITALI Green Today, Better Tomorrow

Inspiring eco-friendly habits for a sustainable future.

FUNDING PROGRESS 0%

JOIN PROJECT →

4.CORPORATE(CSR)

User choose csr role:

START YOUR LEGACY

Plant a Seed of Hope.

"Do not wait for extraordinary circumstances to do good; try to use ordinary situations."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

Getting better!

Perfect match!
YOUR MISSION ROLE

 Donor

 Volunteer

 NGO

 Corporate

 LAUNCH MY IMPACT

Already registered? [Sign In](#)

Corporate.html

THE STEWARDSHIP PORTAL

Create a Legacy, Not Just a Business.

Dashboard

"Business is not just about making money, it is about making meaning."

Logged in as mitu1 — Architect of Change



ACTIVE WEALTH

Campaign Funding

Turn profit into purpose. Direct your resources to the souls who need it most and track the flow of kindness.

Direct Funds



INTEGRITY LAYER

The Trust Shield

Ensuring every promise is kept. Manage your statutory responsibilities with transparency that inspires trust.

Audit Integrity



HUMAN CONNECTION

Soul Volunteering

Empower your team to find meaning beyond their desks. Track hours of service that change local lives.

Inspire Teams



FUTURE VISION

ESG Intelligence

Using technology to protect the Earth. AI-driven scoring that balances your social and environmental footprint.

Evolving Soon...



TRUTH ENGINE

Blockchain Ledger

An immutable record of your goodness. Every cent you spend on the world is etched in time forever.

Verify Ledger



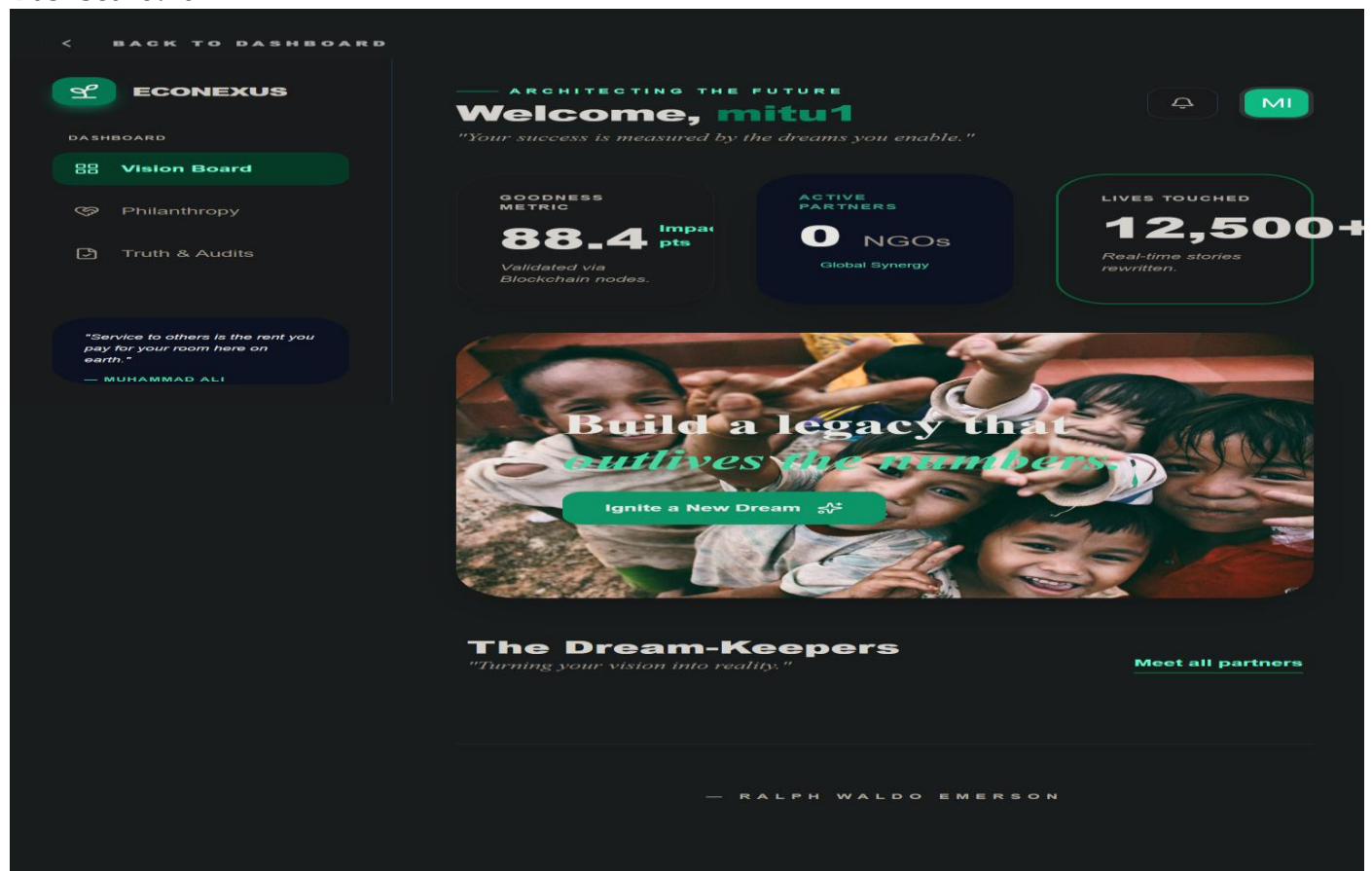
HONOR SYSTEM

Honor Certificates

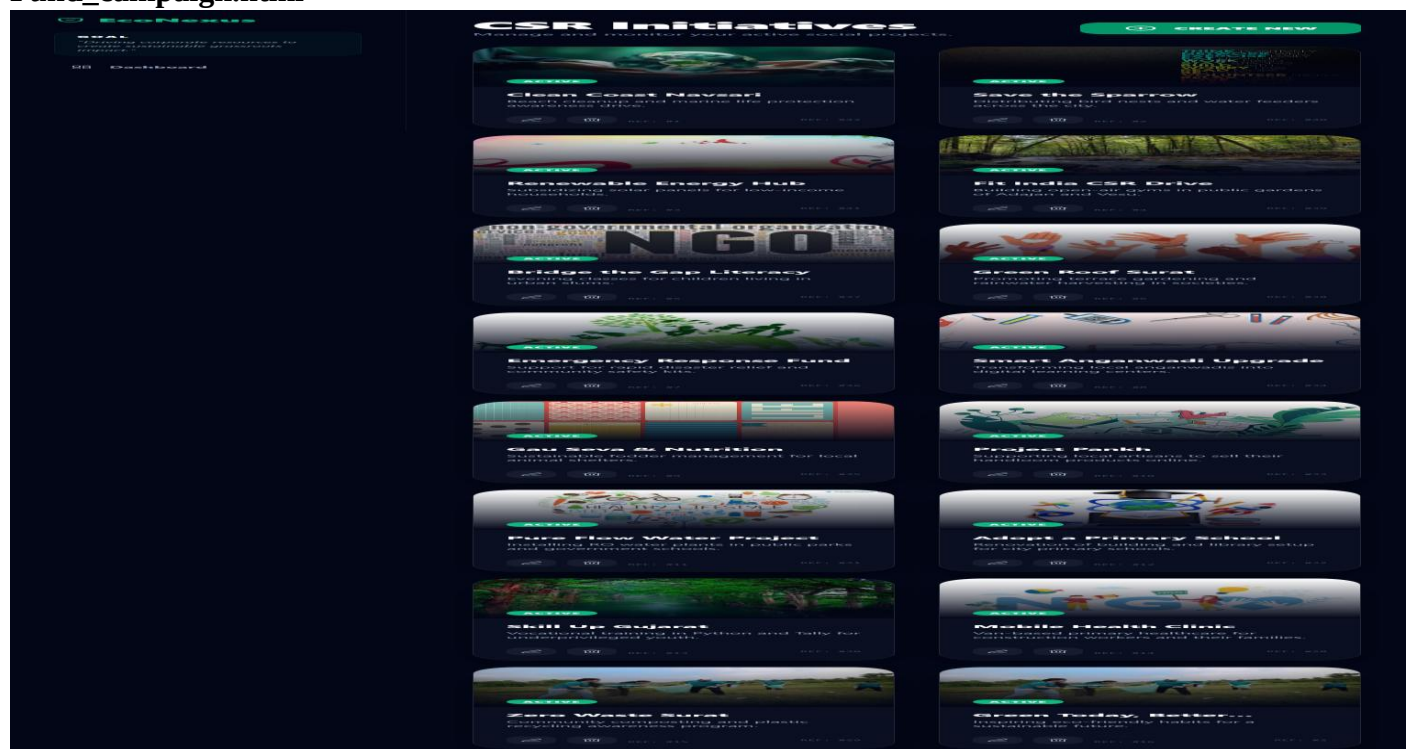
Validate your impact. Blockchain-backed certificates that serve as a digital badge for your company's honor.

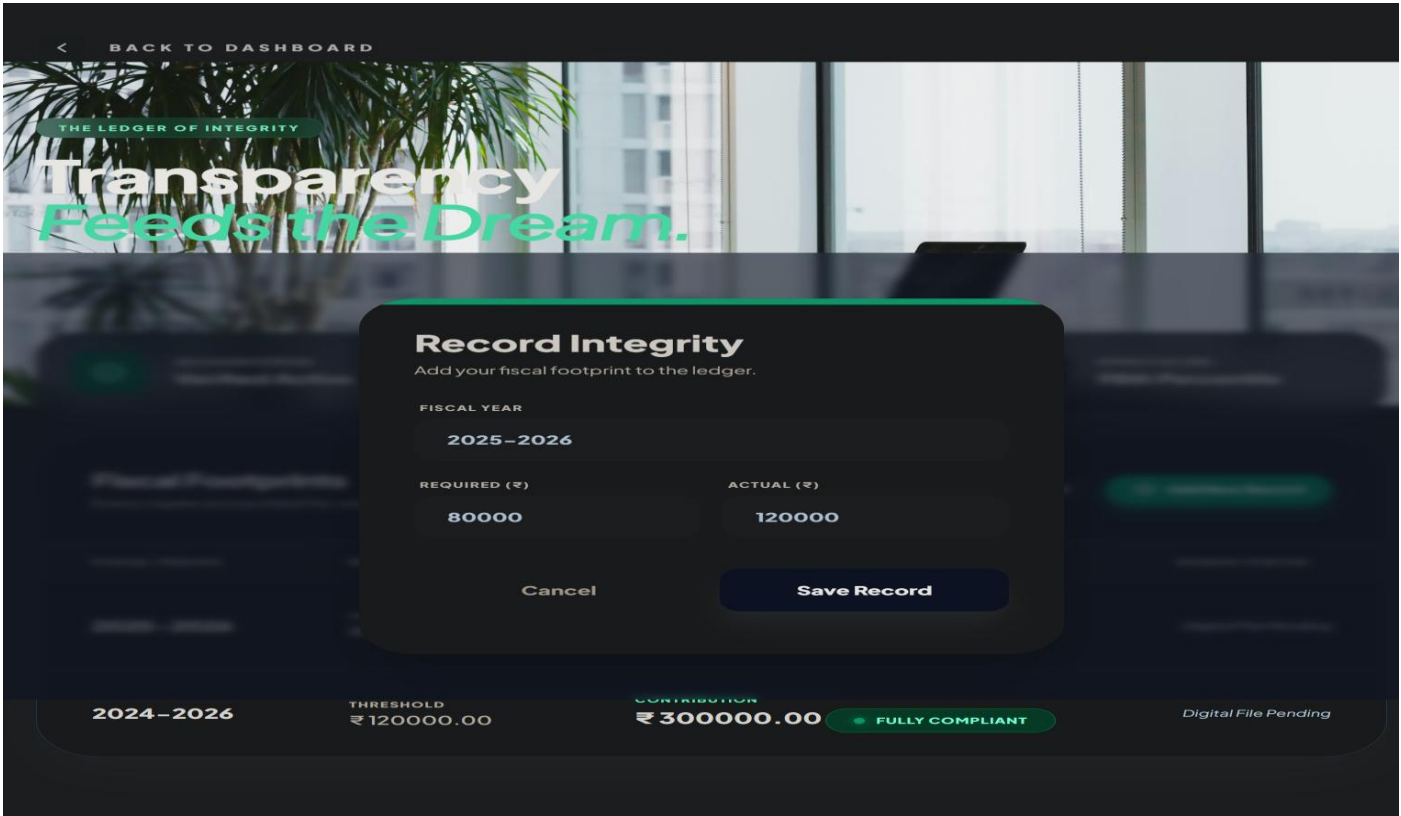
Claim Honor

Dashboard.html

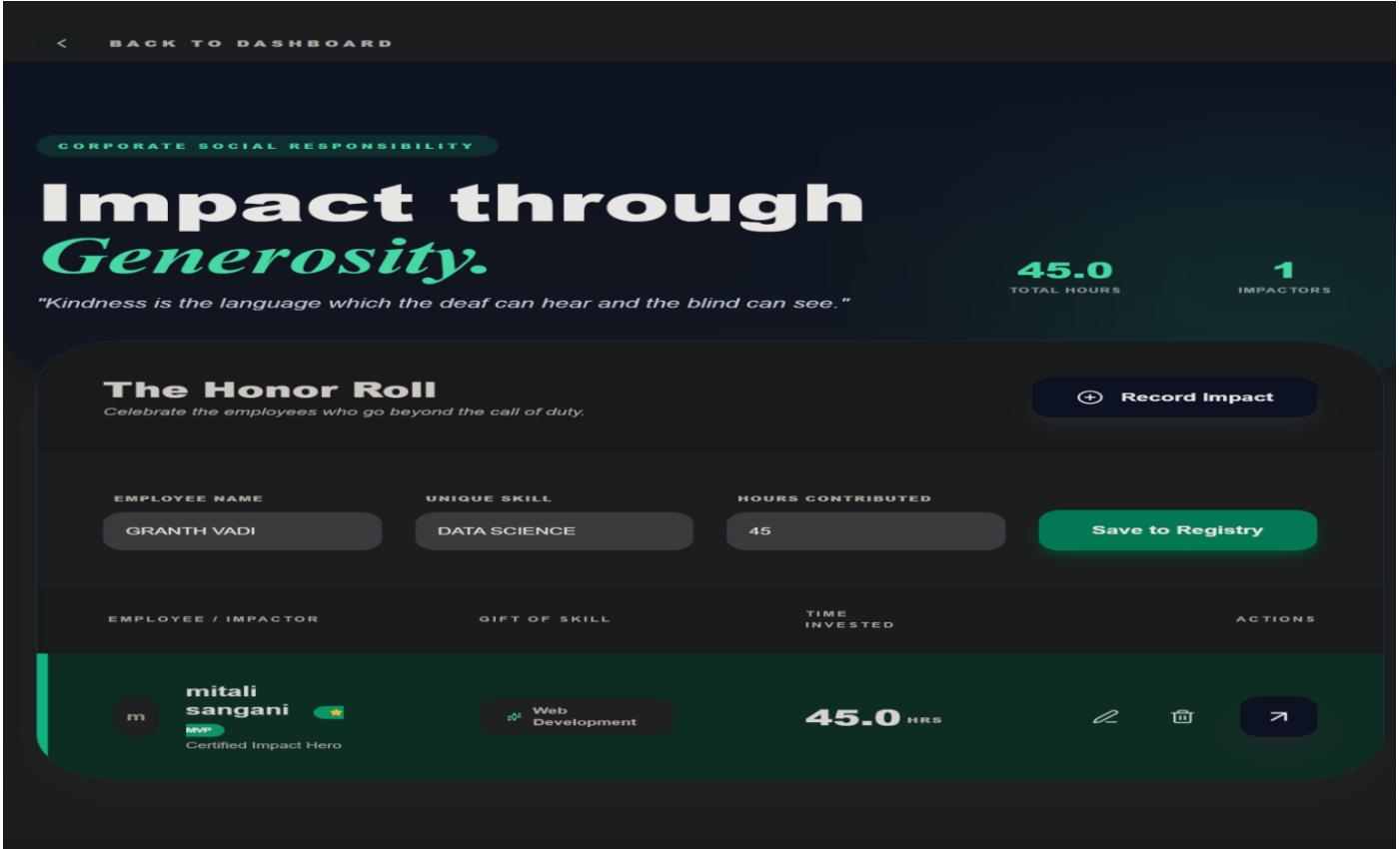


Fund_campaign.html





Employee_volunteer.html



[BACK TO DASHBOARD](#)

CORPORATE SOCIAL RESPONSIBILITY

Impact through Generosity.

"Everyone can be great because anyone can serve."

90.0
TOTAL HOURS

2
IMPACTORS

The Honor Roll

Celebrate the employees who go beyond the call of duty.

Record Impact

EMPLOYEE / IMPACTOR	GIFT OF SKILL	TIME INVESTED	ACTIONS
m mitali sangani MVP Certified Impact Hero	Web Development	45.0 HRS	
G GRANTH VADI Certified Impact Hero	DATA SCIENCE	45.0 HRS	

Ledger.html

[RETURN TO NEXUS](#)

BLOCKCHAIN VERIFIED LEDGER

Truth is Immutability.

"Transparency is the only proof of a dream realized. Every block in this ledger represents a promise kept to the world."

Generate New Block

TOTAL CIRCULATING IMPACT FUNDS

₹45000

/ Live On-Chain

AES-256 Encrypted

1 Verified Blocks

99

"Transparency is the currency of the future."

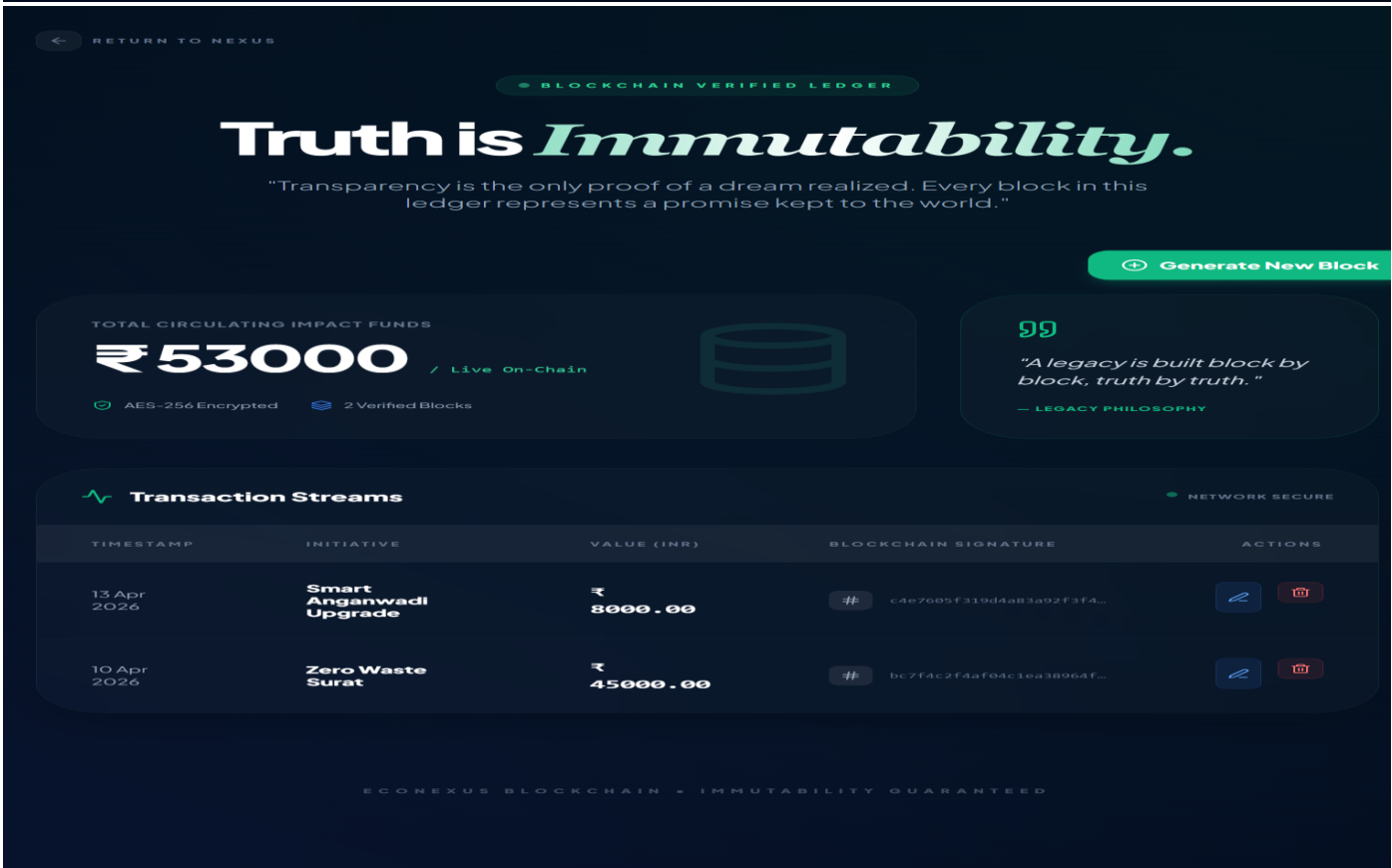
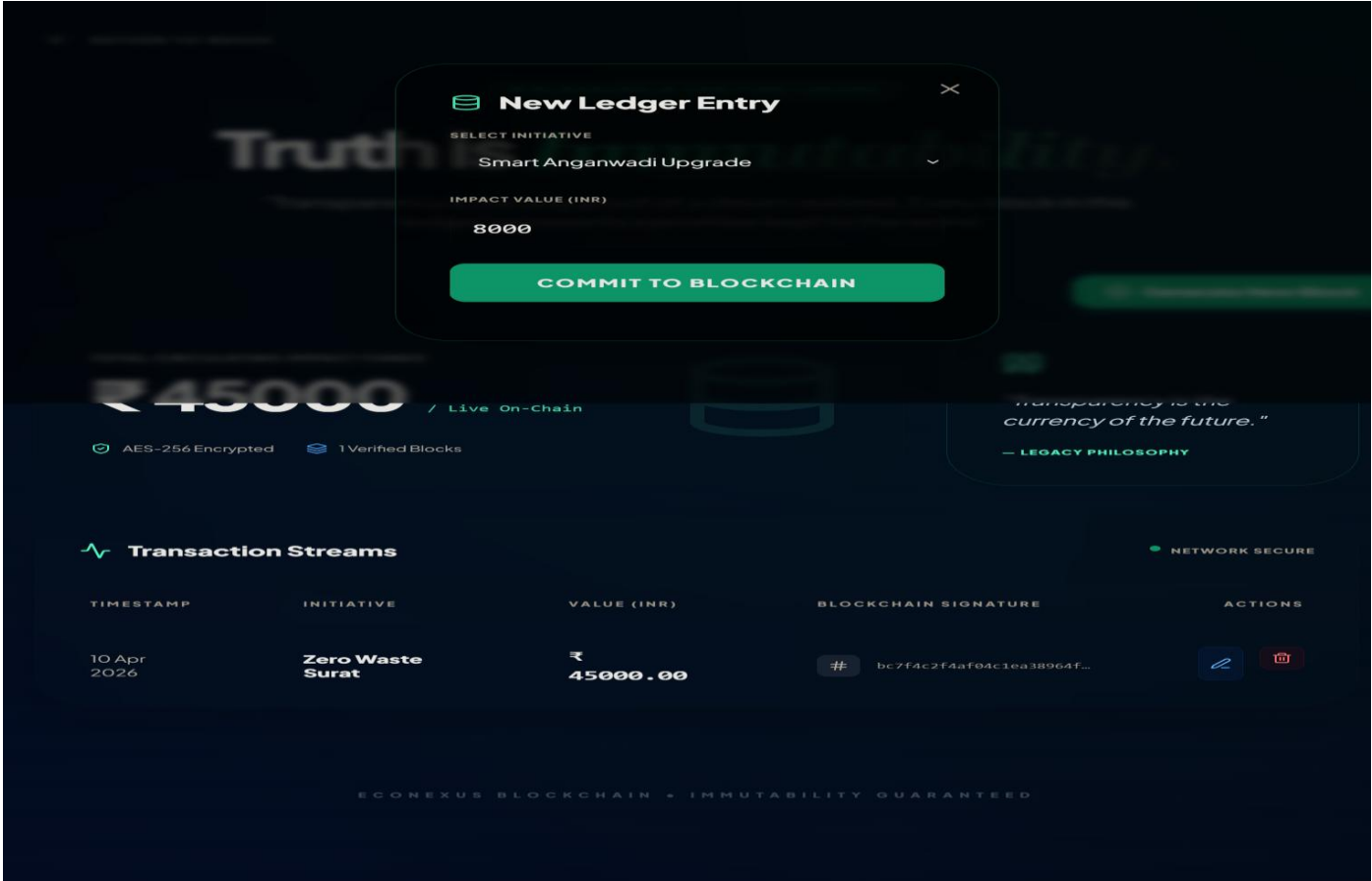
LEGACY PHILOSOPHY

Transaction Streams

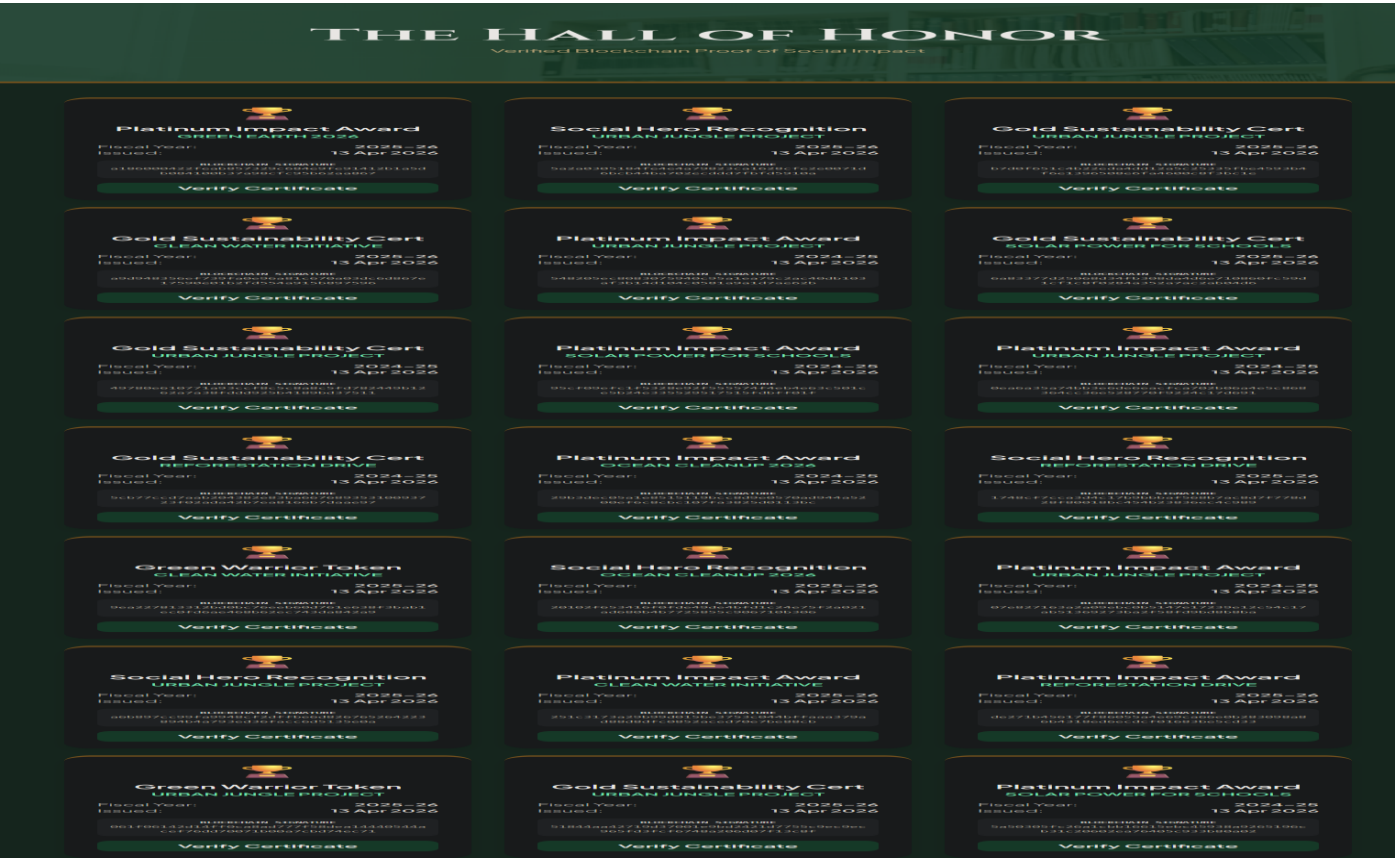
NETWORK SECURE

TIMESTAMP	INITIATIVE	VALUE (INR)	BLOCKCHAIN SIGNATURE	ACTIONS
10 Apr 2026	Zero Waste Surat	₹45000.00	# bc7f4c2f4af04c1ea38964f...	

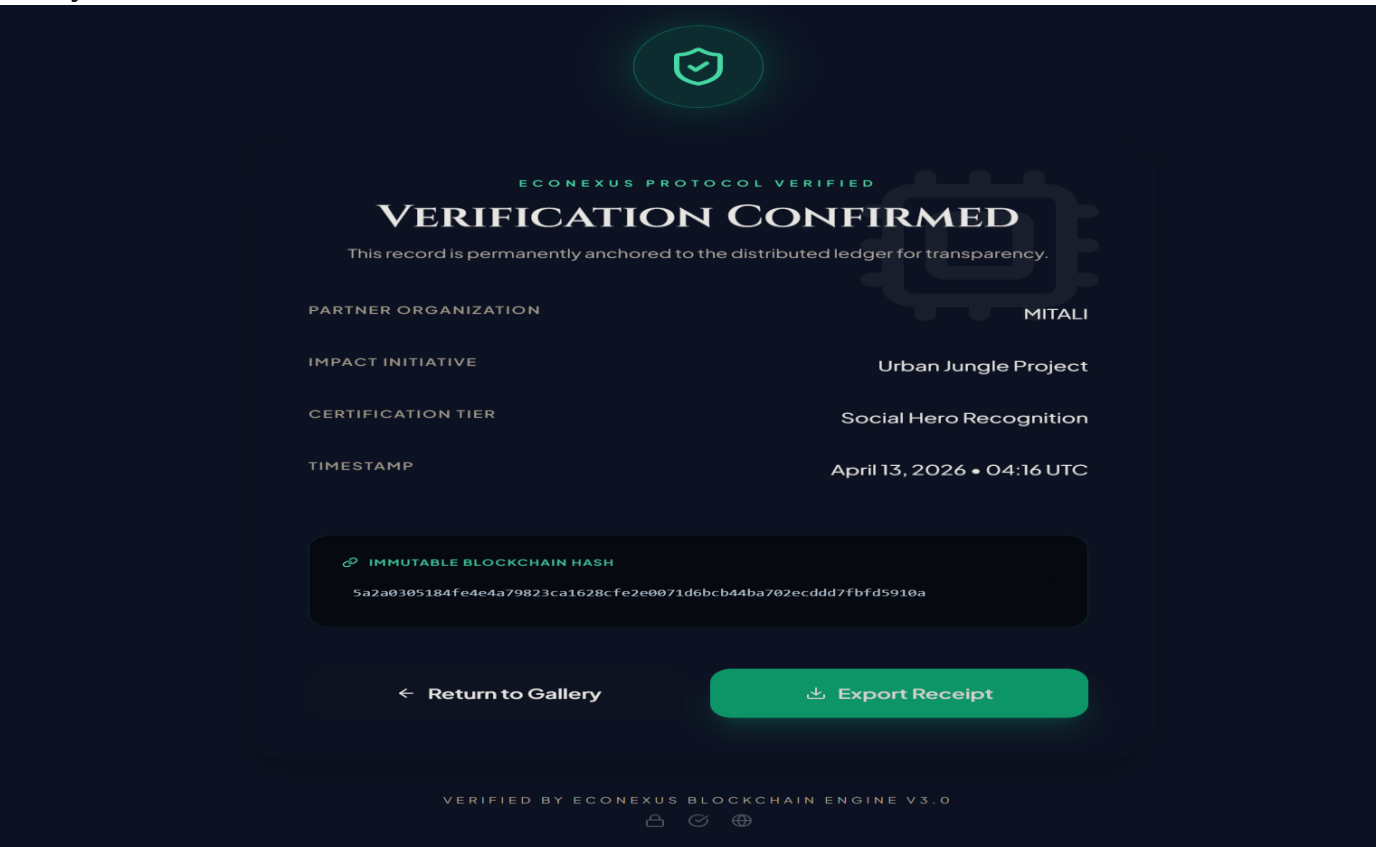
ECONEXUS BLOCKCHAIN • IMMUTABILITY GUARANTEED



Certificate.html



Verify_success.html



5.DEFULT USER
User choose not role:

START YOUR LEGACY

Plant a Seed of Hope.

"Success is not just what you accomplish, but what you inspire others to do."

Join 5,000+ others who have turned their dreams into measurable impact.

Begin Your Journey

saroj

saroj@gmail.com

Getting better!

Perfect match!

YOUR MISSION ROLE

Donor

Volunteer

NGO

Corporate

LAUNCH MY IMPACT

Already registered? [Sign In](#)

Save Our Environment.

Empowering the social sector with AI-driven matching, blockchain transparency, and real-time impact verification.

VOLUNTEERS

Access verified opportunities and build a professionally signed "Impact Resume" for your career.

NGOs

Streamline volunteer management with automated onboarding and geo-verified attendance tracking.

DONORS

Track fund tracking from donation to ground-level delivery via our proprietary trust protocol.

CORPORATES

Automated CSR & ESG compliance reports ready for government audits with one-click exports.

New Ecosystem Features

Proof of Trust Score
A dynamic reputation engine for users based on verified actions, rates, and community feedback.

Live Impact Heatmap
Visualize real-time social impact density and resource allocation.

AI Mentor - Matching
Connecting under-volunteers with mentors and providing guidance via the app during social service.

Operational Workflow

STEP 01
Intelligent Onboarding
All users undergo KYC (KYB) ensuring only legitimate entities enter the trust circle.

STEP 02
Predictive Matching
AI-driven algorithm identifies need, and urgency to match the right resources to the right causes.

STEP 03
Smart Audit Logs
Every action is recorded in a tamper-proof system, ensuring a clear audit trail for stakeholders.

The Autonomous Intelligence Layer
ENSO solves the "Trust Deficit" in the social sector using a proprietary AI stack.

Fraud Prevention
Behavioral pattern analysis to detect suspicious activity and suggest course of action.

Anti-Fraud ENSO
Real-time detection to identify and suspend fraudulent or suspicious donation requests.

Impact Forecasting
Machine learning models to estimate the long-term social ROI of current campaigns.

98.2% Matching Accuracy

100% Audit Reliability

2.4K+ Verified NGOs

Zero Fund Leakage

The Roadmap

2026
Decentralized ID (DID)
Implementing W3C standards for true data ownership.

2027
Smart-Contract Grants
Automated disbursement of grants based on milestones.

2028
Global Impact Network
Expand to 100+ countries for direct on-ground processing.

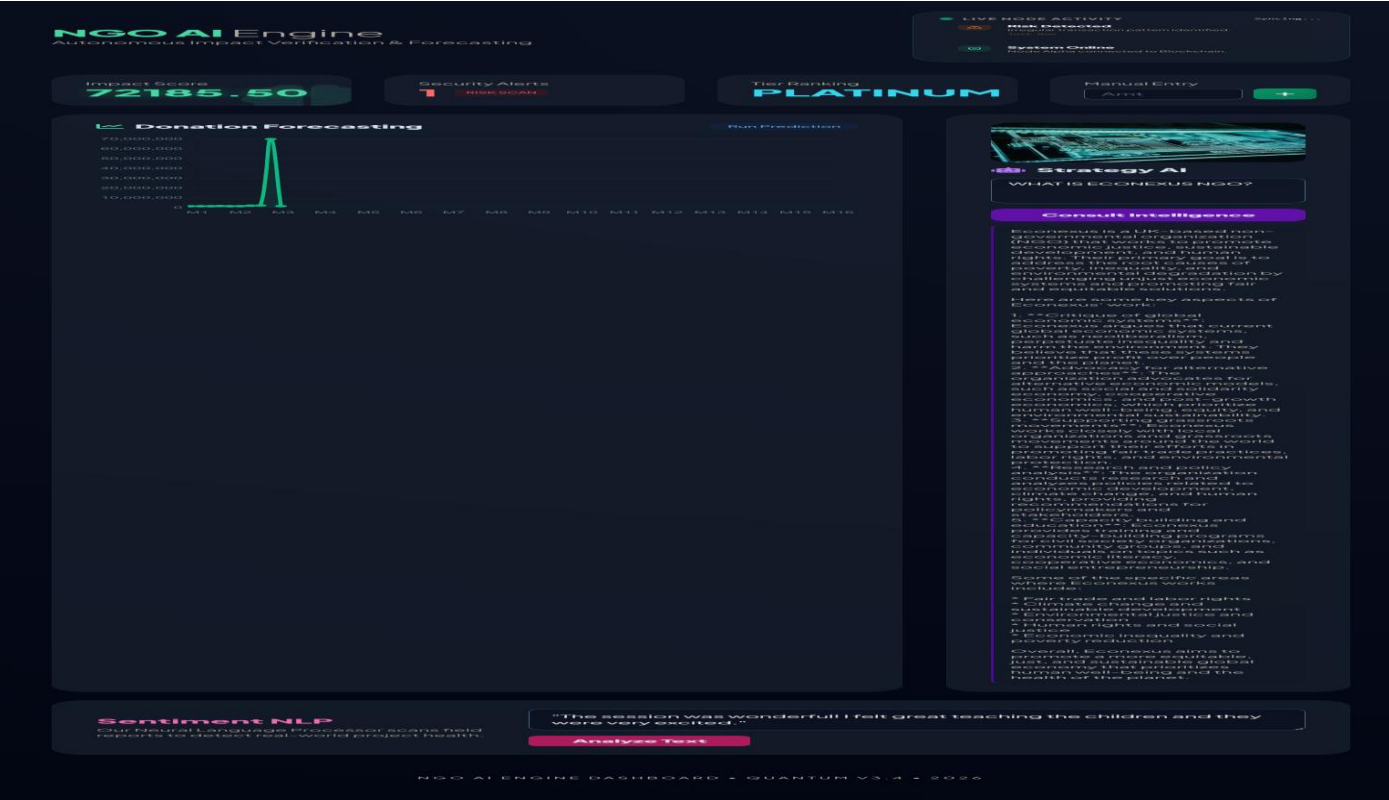
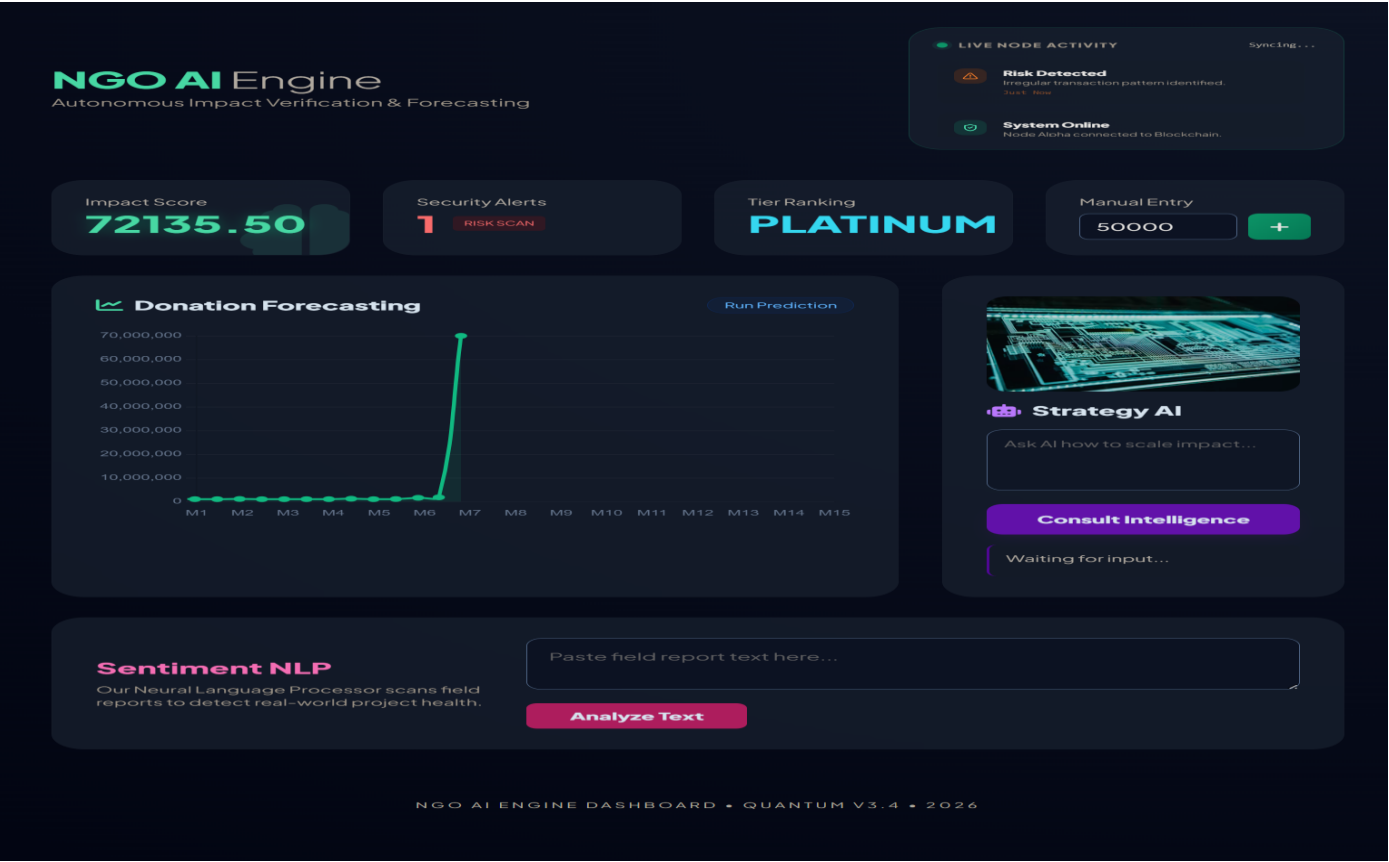
ENSO

A unified ecosystem for the social good. We leverage technology to solve humanity's most pressing challenges with transparency and intelligence.

© 2024 ENSO Global Ecosystem. All Rights Reserved.

B.Admin Side

1. AI ADMIN



2. SYSTEM ADMIN

NEXUS CORE

Dashboard

AI Command

NGO Tenants

Governance

Audit Logs

System Settings

SYSTEM STATUS

99.9% Uptime (Healthy)

Strategic Overview

REAL-TIME PREDICTIVE ANALYTICS

Avg CPU Load

34.1%

Global Donations

₹4.2M

Active Campaigns

48

AI Resource Balancer

AUTO-SCALING: ACTIVE

Seva Foundation

LOW

OPTIMIZE

EcoCare Surat

OPTIMAL

BALANCED

Goonj India

STABLE

MONITOR

NEXUS CORE

Dashboard

AI Command

NGO Tenants

Governance

Audit Logs

System Settings

SYSTEM STATUS

99.9% Uptime (Healthy)

Immutable Audit Trail

13:44:21 AI_ACTION: Added NatureGuard Vadodara from suggestions.

13:44:19 AI_ACTION: Added Lighthouse Trust (Ahmedabad) from suggestions.

13:44:12 AI_DRAFT: Campaign architect generated: Gujarat Green Drive: 10k Trees

13:44:11 AI_DRAFT: Campaign architect generated: Gujarat Green Drive: 10k Trees

13:44:10 AI_DRAFT: Campaign architect generated: Gujarat Green Drive: 10k Trees

13:44:07 AI_DRAFT: Campaign architect generated: Empower 100 Children in Surat Slums

NGO Management

15 ACTIVE TENANTS

Provision New Tenant

NGO Full Name

Sub-domain/Schema Name

PROVISION TENANT

NGO NAME	DOMAIN/SCHEMA	STATUS	ACTIONS
Seva Foundation	seva_org	● Online	<button>Open</button> <button>Suspend</button>
EcoCare Surat	ecocare_surat	● Online	<button>Open</button> <button>Suspend</button>
Goonj India	goonj_main	● Online	<button>Open</button> <button>Suspend</button>
Smile Foundation	smile_edu	● Online	<button>Open</button> <button>Suspend</button>
Akshaya Patra	akshaya_meals	● Online	<button>Open</button> <button>Suspend</button>

AI Strategic Command

V2.4.0 PREDICTIVE ENGINE

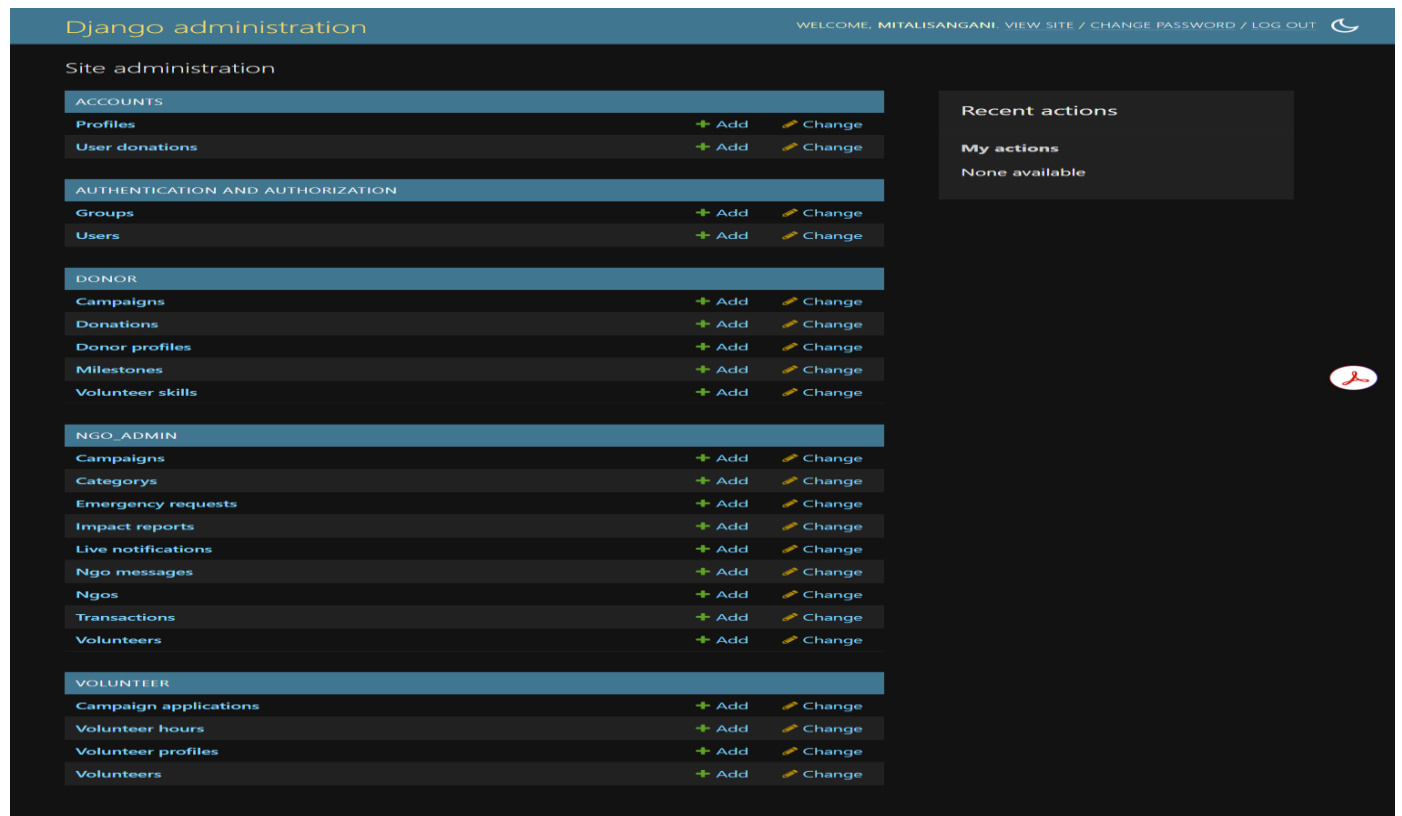
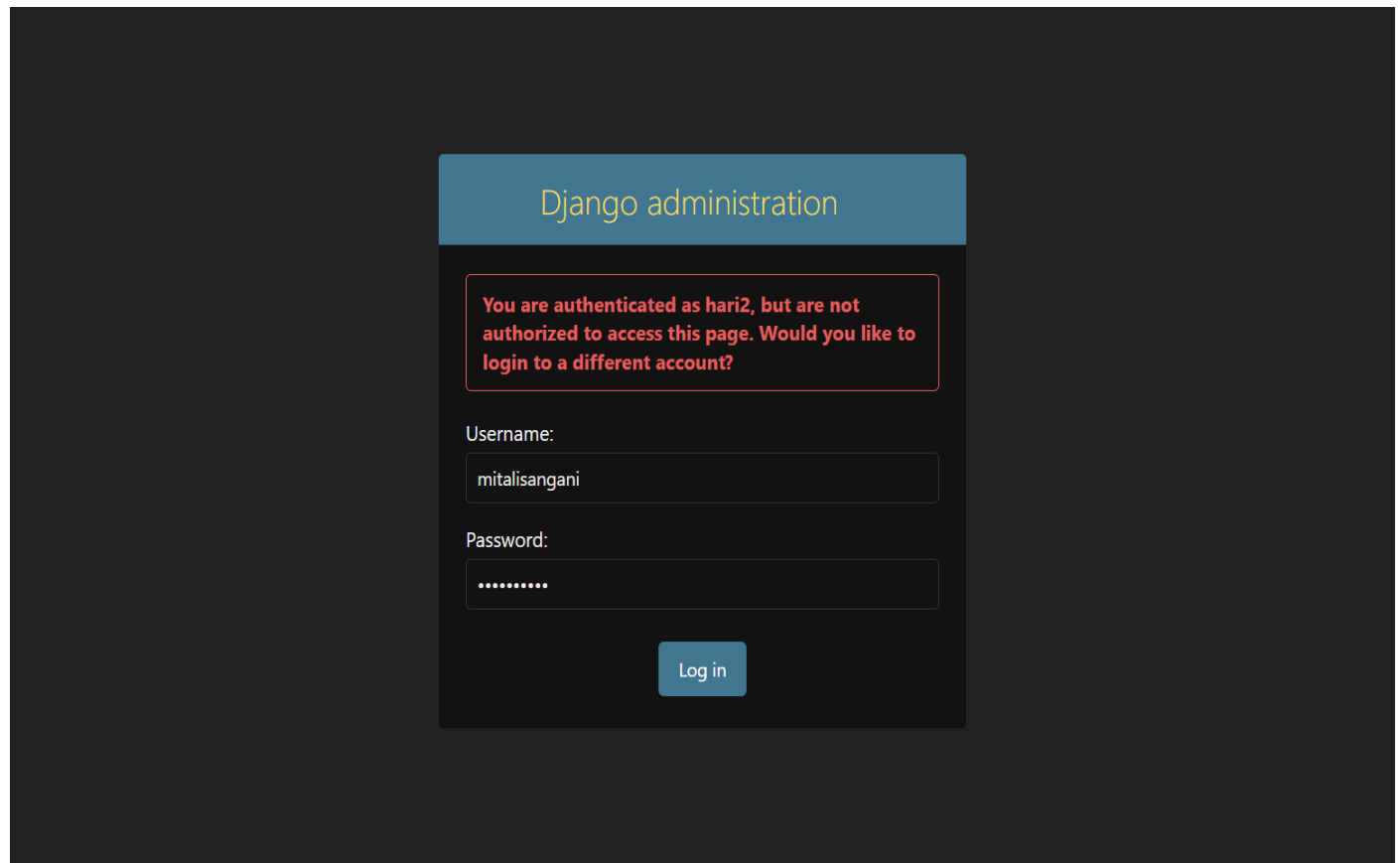
Impact Heatmap



Smart Suggestions REFRESH

Lighthouse Trust (Ahmedabad)	<button>Add</button>
Animal Relief Vapi	<button>Add</button>
Surat Clean Earth Initiative	<button>Add</button>
NatureGuard Vadodara	<button>Add</button>
Vidya Guj Foundation	<button>Add</button>

3. DJANGO ADMIN



5.4 Database Screenshots

Vs code db.sqlite3 (SQLite viewer):

id	role	verified	user_id
1	ngo		176
2	donor		202
3	donor		203
4	ngo		204
5	volunteer		205
6	volunteer		206

```
Microsoft Windows [Version 10.0.26200.8117]
(c) Microsoft Corporation. All rights reserved.

C:\sqlite-tools-win-x64-3510200>sqlite3.exe "C:\ENGO APP\db.sqlite3"
SQLite version 3.51.2 2026-01-09 17:27:48
Enter ".help" for usage hints.
sqlite> .tables
accounts_profile          ngo_admin_category
accounts_userdonation     ngo_admin_emergencyrequest
ai_engine_donation        ngo_admin_impactreport
ai_engine_impactreport    ngo_admin_livenotification
ai_engine_ngo             ngo_admin_ngo
auth_group                ngo_admin_ngomessage
auth_group_permissions    ngo_admin_transaction
auth_permission           ngo_admin_volunteer
auth_user                 system_admin_anomalyevent
auth_user_groups          system_admin_auditlog
auth_user_user_permissions system_admin_ngo
csr_corporatecsrprofile   system_admin_role
csr_csrcampaign           system_admin_systemmetric
csr_csrcertificate        system_admin_transaction
csr_csrcompliance         system_admin_userprofile
csr_csrtransaction        volunteer_campaign
csr_employeevolunteer     volunteer_campaignapplication
django_admin_log          volunteer_chatmessage
django_content_type       volunteer_csr
django_migrations         volunteer_donation
django_session            volunteer_feedback
donor_campaign            volunteer_ngo
donor_donation            volunteer_ngorequirement
```



```

sqlite> .schema
CREATE TABLE IF NOT EXISTS "django_migrations" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "app" varchar(255) NOT NULL, "name" varchar(255) NOT NULL, "applied" datetime NOT NULL);
CREATE TABLE sqlite_sequence(name,seq);
CREATE TABLE IF NOT EXISTS "auth_group_permissions" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "group_id" integer NOT NULL REFERENCES "auth_group" ("id") DEFERRABLE INITIALLY DEFERRED, "permission_id" integer NOT NULL REFERENCES "auth_permission" ("id") DEFERRABLE INITIALLY DEFERRED);
CREATE TABLE IF NOT EXISTS "auth_user_groups" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "user_id" integer NOT NULL REFERENCES "auth_user" ("id") DEFERRABLE INITIALLY DEFERRED, "group_id" integer NOT NULL REFERENCES "auth_group" ("id") DEFERRABLE INITIALLY DEFERRED);
CREATE TABLE IF NOT EXISTS "auth_user_user_permissions" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "user_id" integer NOT NULL REFERENCES "auth_user" ("id") DEFERRABLE INITIALLY DEFERRED, "permission_id" integer NOT NULL REFERENCES "auth_permission" ("id") DEFERRABLE INITIALLY DEFERRED);
CREATE UNIQUE INDEX "auth_group_permissions_group_id_permission_id_0cd325b0_uniq" ON "auth_group_permissions" ("group_id", "permission_id");
CREATE INDEX "auth_group_permissions_group_id_b120cbf9" ON "auth_group_permissions" ("group_id");
CREATE INDEX "auth_group_permissions_permission_id_84c5c92e" ON "auth_group_permissions" ("permission_id");
CREATE UNIQUE INDEX "auth_user_groups_user_id_group_id_94350c0c_uniq" ON "auth_user_groups" ("user_id", "group_id");
CREATE INDEX "auth_user_groups_user_id_6a12ed8b" ON "auth_user_groups" ("user_id");
CREATE INDEX "auth_user_groups_group_id_97559544" ON "auth_user_groups" ("group_id");
CREATE UNIQUE INDEX "auth_user_user_permissions_user_id_permission_id_14a6b632_uniq" ON "auth_user_user_permissions" ("user_id", "permission_id");
CREATE INDEX "auth_user_user_permissions_user_id_a95ead1b" ON "auth_user_user_permissions" ("user_id");
CREATE INDEX "auth_user_user_permissions_permission_id_1fbb5f2c" ON "auth_user_user_permissions" ("permission_id");
CREATE TABLE IF NOT EXISTS "accounts_profile" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "role" varchar(20) NOT NULL, "verified" bool NOT NULL, "user_id" integer NOT NULL UNIQUE REFERENCES "auth_user" ("id") DEFERRABLE INITIALLY DEFERRED);
CREATE TABLE IF NOT EXISTS "accounts_userdonation" ("id" integer NOT NULL PRIMARY KEY AUTOINCREMENT, "name" varchar(100) NOT NULL, "email" varchar(254) NOT NULL, "donation_type" varchar(20) NOT NULL, "amount" integer unsigned NULL CHECK ("amount" >= 0), "description" text NOT NULL, "created_at" datetime NOT NULL, "is_verified" bool NOT NULL, "user_id" integer NOT NULL REFERENCES "auth_user" ("id") DEFERRABLE INITIALLY DEFERRED);

```

File Edit Selection View Go Run ... ENGO APP

EXPLORER

- OPEN EDITORS
- ENGO APP
 - volunteer
 - templates\volunteers
 - hours_feedback.html
 - ledger.html
 - profile_settings.html
 - volunteer_dashboard.html
 - volunteer_details.html
 - _init_.py
 - admin.py
 - ai.py
 - apps.py
 - models.py
 - tests.py
 - urls.py
 - utils.py
 - views.py
 - ~\$path.doc
 - db.sqlite3
 - manage.py
 - path.doc
 - requirements.txt
 - OUTLINE
 - TIMELINE

db.sqlite3

Filter 58 Rows: 232 Filter 232 rows... Upgrade to PRO

TABLES

- accounts_profile
- accounts_userdonation
- ai_engine_donation
- ai_engine_impact
- ai_engine_ngo
- auth_group
- auth_group_permissions
- auth_permission
- auth_user
- auth_user_groups
- auth_user_user_permissions
- csr_corporatecsr
- csr_csccampaign
- csr_csccertificate
- csr_csccompliance
- csr_csctransaction
- csr_employeeevaluation
- django_admin_log
- django_content_type
- django_migrations
- django_session
- donor_campaign

id	content_type_id	codename	name
1	1	add_logentry	Can add log entry
2	2	change_logentry	Can change log entry
3	3	delete_logentry	Can delete log entry
4	4	view_logentry	Can view log entry
5	5	add_permission	Can add permission
6	6	change_permission	Can change permission
7	7	delete_permission	Can delete permission
8	2	view_permission	Can view permission
9	9	add_group	Can add group
10	10	change_group	Can change group
11	11	delete_group	Can delete group
12	12	view_group	Can view group
13	13	add_user	Can add user
14	14	change_user	Can change user
15	15	delete_user	Can delete user
16	16	view_user	Can view user
17	17	add_contenttype	Can add content type
18	18	change_contenttype	Can change content type

SQLITE VIEW... v25.12.2 Page 1 / 3 Copy as Excel

5.5 Important Code Snippets

1. Ollama use:

```
@csrf_exempt
def ask_gpt(request):
    if request.method == "POST":
        try:
            data = json.loads(request.body)
            user_message = data.get("message", "")

            (function) chat: Any

            response = ollama.chat(model='llama3', messages=[
                {
                    'role': 'user',
                    'content': user_message,
                },
            ])

            return JsonResponse({'reply': response['message']['content']})

        except Exception as e:
            return JsonResponse({'reply': f"Llama Error: {str(e)}"}, status=500)
```

2. Blockchainhash:

views.py volunteer 1 views.py donor ● blockchain.py X hours_feedback.html

donor > blockchain.py > ...

```
1 import hashlib
2
3 def generate_blockchain_hash(donation):
4     raw = f"{donation.donor_id}{donation.amount}{donation.created_at}"
5     return hashlib.sha256(raw.encode()).hexdigest()
6
```

3. csr/models.py:

```
File Edit Selection View Go ... ENGO APP

EXPLORER
  OPEN EDITORS 1 unsaved
  ENGO APP
    al_engine
    campaigns
    core
    csr
      __pycache__
      migrations
      templates
      __init__.py
      admin.py
      al.py
      apps.py
      blockchain.py
      forms.py
      models.py
      tests.py
      urls.py
      views.py
    engo_project
    media
    ngo_admin
    static
    system_admin
    templates
    venv
    volunteer
      __pycache__
      migrations
      templates \ volunteers
      campaigns.html
      donations.html
      hours_feedback.html
      ledger.html
      profile_settings.html
      volunteer_dashboard.html
      volunteer_details.html
      __init__.py
      admin.py
      al.py
      apps.py

csr > models.py > ...
1 import uuid
2 from django.db import models
3 from django.contrib.auth.models import User
4 from django.utils import timezone
5 from django.db.models.signals import post_save
6 from django.dispatch import receiver
7 from decimal import Decimal
8
9 # 1. Corporate Profile: Stores company-specific details linked to a User
10 class CorporateCSRProfile(models.Model):
11     user = models.OneToOneField(User, on_delete=models.CASCADE)
12     company_name = models.CharField(max_length=255)
13     csr_budget = models.DecimalField(max_digits=12, decimal_places=2, default=0)
14     industry = models.CharField(max_length=150, blank=True)
15     created_at = models.DateTimeField(auto_now_add=True)
16
17     def __str__(self):
18         return self.company_name
19
20 # 2. CSR Campaign: Represents specific social initiatives created by a Corporate
21 class CSRCampaign(models.Model):
22     STATUS_CHOICES = (
23         ('Active', 'Active'),
24         ('Completed', 'Completed'),
25         ('Paused', 'Paused'),
26     )
27     corporate = models.ForeignKey(CorporateCSRProfile, on_delete=models.CASCADE, related_name="campaigns")
28     title = models.CharField(max_length=255)
29     description = models.TextField()
30     image = models.ImageField(upload_to='campaigns/', null=True, blank=True)
31     target_amount = models.DecimalField(max_digits=12, decimal_places=2)
32     funded_amount = models.DecimalField(max_digits=12, decimal_places=2, default=0)
33     status = models.CharField(max_length=20, choices=STATUS_CHOICES, default='Active')
34     blockchain_hash = models.CharField(max_length=255, blank=True, null=True)
35     created_at = models.DateTimeField(auto_now_add=True)
36
37     # Calculates the funding progress as a percentage
38     def progress_percentage(self):
39         if self.target_amount > 0:
40             return int((self.funded_amount / self.target_amount) * 100)
41         return 0
42
43     def __str__(self):
44         return self.title
45
46 # 3. CSR Transaction: Records financial movement and generates a unique Blockchain Hash
47 class CSRTransaction(models.Model):
48     campaign = models.ForeignKey(CSRCampaign, on_delete=models.CASCADE, related_name="transactions")
49     amount = models.DecimalField(max_digits=12, decimal_places=2)
50     created_at = models.DateTimeField(auto_now_add=True) # Used for ledger timestamp
51     blockchain_hash = models.CharField(max_length=64, unique=True, editable=False, blank=True)
52
```

4. accounts/models.py:

```
File Edit Selection View Go ... ENGO APP

EXPLORER
  OPEN EDITORS 2 unsaved
  ENGO APP
    accounts
      __pycache__
      migrations
      templates
      admin.py
      apps.py
      CustomUserCreationForm.py
      decorators.py
      forms.py
      models.py
      signals.py
      tests.py
      urls.py
      utils.py
      views.py
    al_engine
    campaigns
    core
    csr
    donor
    engo_project
    media
    ngo_admin
    static
    system_admin
    templates
    venv
    volunteer
      __pycache__
      migrations
      templates \ volunteers
      campaigns.html
      donations.html
      hours_feedback.html
      ledger.html
      profile_settings.html
      volunteer_dashboard.html
      volunteer_details.html
      __init__.py
      admin.py
      al.py
      apps.py

accounts > models.py > ...
1 from django.db import models
2 from django.contrib.auth.models import User
3
4 class Profile(models.Model):
5     ROLE_CHOICES = (
6         ('user', 'User'),
7         ('donor', 'Donor'),
8         ('volunteer', 'Volunteer'),
9         ('ngo', 'NGO'),
10        ('corporate', 'Corporate'),
11    )
12
13     user = models.OneToOneField(User, on_delete=models.CASCADE)
14     role = models.CharField(
15         max_length=20,
16         choices=ROLE_CHOICES,
17         default='user', # IMPORTANT
18     )
19     verified = models.BooleanField(default=False)
20
21     def __str__(self):
22         return self.user.username
23
24 class UserDonation(models.Model):
25     DONATION_TYPE_CHOICES = (
26         ('money', 'Money'),
27         ('food', 'Food'),
28         ('clothes', 'Clothes'),
29         ('medicine', 'Medicine'),
30         ('other', 'Other'),
31     )
32
33     # Optional: link donation to logged-in user
34     user = models.ForeignKey(
35         User,
36         on_delete=models.SET_NULL,
37         null=True,
38         blank=True
39     )
40
41     name = models.CharField(max_length=100)
42     email = models.EmailField()
43
44     donation_type = models.CharField(
45         max_length=20,
46         choices=DONATION_TYPE_CHOICES
47     )
48
49
```

6. Future Scope

1. Mobile Application Development

- Develop dedicated mobile applications for Android and iOS platforms.
- Provide easy access to donation, volunteering, and campaign features.
- Enable push notifications for real-time updates and alerts.
- Improve user engagement and convenience.
- Support offline data access and synchronization.

2. Advanced Artificial Intelligence (AI) Integration

- Enhance campaign recommendation systems based on user preferences.
- Implement predictive analytics for donation trends and campaign success.
- Analyze donor behavior for better engagement strategies.
- AI-based volunteer skill matching for efficient resource utilization.
- Generate automated impact reports showing real-world outcomes.

3. Blockchain Technology Implementation

- Implement full blockchain integration for secure donation tracking.
- Generate unique hash values for each transaction.
- Enable smart contracts for automatic fund release based on milestones.
- Provide public verification of fund usage.
- Ensure transparency and build trust among donors.

4. Global Platform Expansion

- Expand the platform to support international NGOs and users.
- Implement multi-language support for global accessibility.
- Enable multi-currency transactions for global donations.
- Promote collaboration between NGOs across different countries.
- Increase the reach and impact of social initiatives.

5. Enhanced Security Features

- Implement OTP-based login and verification systems.
- Add biometric authentication such as fingerprint or facial recognition.
- Use advanced encryption techniques to secure sensitive data.
- Protect against cyber threats like hacking and data breaches.
- Ensure safe and secure transactions for all users.

6. Gamification Features

- Introduce badges, rewards, and achievement systems.
- Create leaderboards for donors and volunteers.
- Motivate users to participate actively in campaigns.

- Enhance user engagement through interactive elements.
- Encourage long-term involvement in social activities.

7. Geo-Based Services

- Provide location-based campaign suggestions.
- Help volunteers find nearby opportunities.
- Integrate map-based tracking for campaigns and services.
- Improve local engagement and participation.
- Enable real-time location tracking for events.

8. Government & CSR Integration

- Integrate with government systems for NGO verification.
- Automate CSR compliance and reporting processes.
- Generate tax-related documents for donors and corporates.
- Ensure legal compliance and transparency.
- Strengthen trust between organizations and users.

9. Advanced Analytics & Dashboards

- Provide detailed insights into donations and campaign performance.
- Use AI for predictive decision-making.
- Real-time data visualization for better understanding.
- Generate customized reports for NGOs and corporates.
- Improve strategic planning and management.

10. Decentralized Governance (DAO)

- Implement blockchain-based voting systems.
- Allow stakeholders to participate in decision-making.
- Ensure transparency in governance processes.
- Promote community-driven management.
- Increase accountability and trust in the system.

11. Integration with External Platforms

- Connect with social media platforms for campaign promotion.
- Enable easy sharing and awareness generation.
- Integrate with third-party APIs for payment, messaging, and analytics.
- Expand system capabilities through external services.

12. Offline & Hybrid Functionality

- Allow offline data entry for remote areas.
- Synchronize data when internet connectivity is available.
- Support field volunteers working in low-network regions.
- Improve usability in rural and underdeveloped areas.

7. Conclusion

The ENGO Web Application (Eco Nexus NGO) is a modern and efficient platform designed to simplify and enhance NGO operations through digital technology. The system successfully integrates various functionalities such as donation management, volunteer coordination, campaign handling, and CSR activities into a single unified platform.

The project demonstrates how technology can be used to improve transparency, efficiency, and accessibility in the NGO sector. By providing features like real-time updates, secure donation processing, and user-friendly dashboards, the system ensures a smooth and engaging experience for all users including donors, volunteers, NGOs, and corporate partners.

The implementation of modular architecture allows the system to be scalable and easy to maintain. Advanced features such as Artificial Intelligence (AI) for smart recommendations and Blockchain for secure and transparent transactions further enhance the reliability and future readiness of the platform.

The system has been successfully tested and validated, ensuring that all modules function correctly and efficiently. It reduces manual work, improves communication, and enables better decision-making through data-driven insights.

Overall, the ENGO Web Application provides a comprehensive solution for managing NGO activities in a digital environment. It not only addresses current challenges but also offers flexibility for future enhancements such as mobile applications, global expansion, and advanced analytics.

In conclusion, this project highlights the importance of integrating modern technologies in social development systems and serves as a strong foundation for building a transparent, scalable, and impactful NGO ecosystem.

8. Bibliography

The following resources were referred to during the development of the ENGO Web Application (Eco Nexus NGO):

Research Papers:

1. “A Study on NGO Management Systems using Web Technologies” – International Journal of Computer Applications (IJCA)
2. “Online Donation System Using Secure Payment Gateway” – IEEE Research Paper
3. “Block chain Technology for Transparent Charity Systems” – IEEE Access Journal
4. “Application of Artificial Intelligence in Social Good Platforms” – Springer Research Paper

Useful Online Platforms:

1. <https://github.com> (for project references)
2. <https://ieeexplore.ieee.org> (for research papers)
3. <https://scholar.google.com> (for academic references)

Books:

1. Vincent, W. S., “Django for Beginners”
2. Matthes, E., “Python Crash Course”
3. Silberschatz, A., “Database System Concept